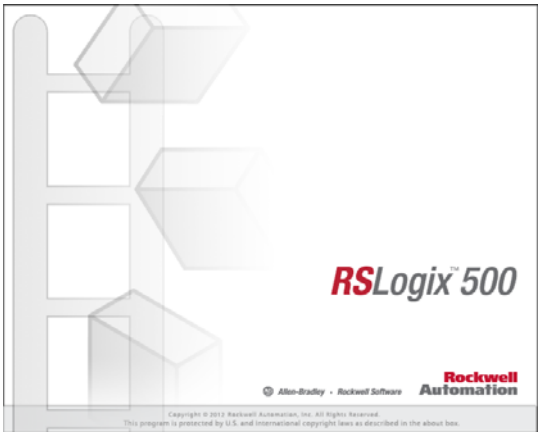
RSLogix 500 Project Report



Processor Information

Processor Type: Bul.1766 MicroLogix 1400 Series B

Processor Name: SAMPLE2

Total Memory Used: *

Total Memory Left: *

Program Files: 17

Data Files: 14

Program ID: 0

I/O Configuration

0	Bul.1766	MicroLogix 1400 Series B
1	1762-OB16	16-Output (TRANS-SRC) 10/50 VDC
2	1762-IF4	Analog 4 Chan. Input
3	1762-IF4	Analog 4 Chan. Input
4		
5		
6		
7		

PID Configuration

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

CHANNEL 1 (SYSTEM) - Driver: Ethernet
   CHANNEL 1 (SYSTEM) - Driver: Ethernet Edit Resource/Owner Timeout: 60
   CHANNEL 1 (SYSTEM) - Driver: Ethernet Passthru Link ID: 1
   CHANNEL 1 (SYSTEM) - Driver: Ethernet Write Protected: No
   CHANNEL 1 (SYSTEM) - Driver: Ethernet Comms Servicing Selection: Yes
   CHANNEL 1 (SYSTEM) - Driver: Ethernet Message Servicing Selection: Yes

   Hardware Address: 00:00:00:00:00:00
   IP Address: 192.168.125.119
   Subnet Mask: 255.255.255.0
   Gateway Address: 192.125.168.1
   Msg Connection Timeout (x 1mS): 15000
   Msg Reply Timeout (x mS): 3000
   Inactivity Timeout (x Min): 30
   Bootp Enable: No
   Dhcp Enable No
   SMTP Enable: No
   SNMP Enable: Yes
   HTTP Enable: Yes
   Auto Negotiate Enable: Yes
   DNP3 over IP Enable: No
   Modbus TCP Enable: No
   Disable EtherNet/IP Incoming Connection: No
   Disable Duplicate IP Address Detection: No
   Port Speed Enable: 10/100 Mbps Full Duplex/Half Duplex
   Contact:
   Location:

CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
   CHANNEL 2 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

Program File List

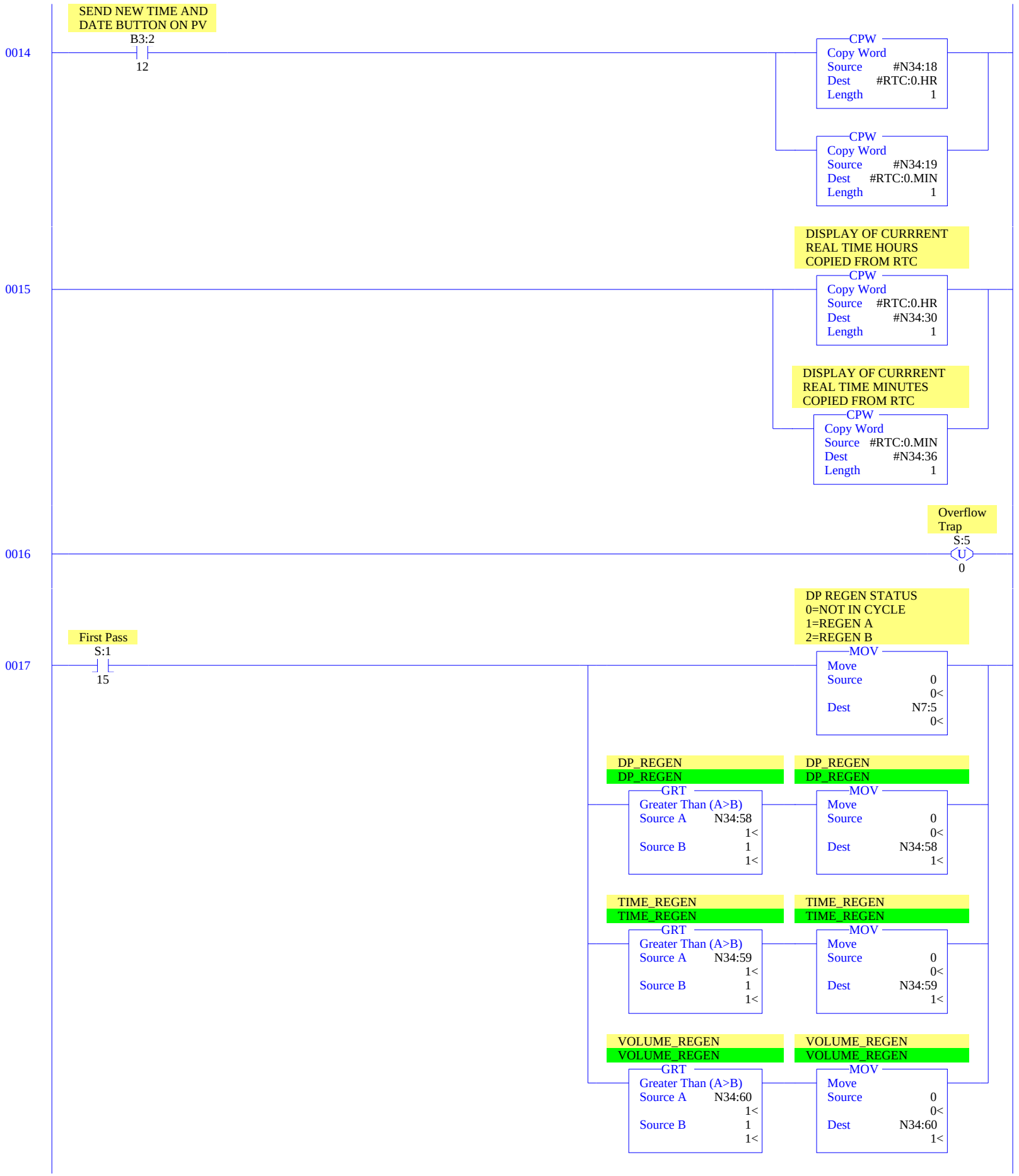
Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
MAIN	2	LADDER	21	No	379
REGEN	3	LADDER	29	No	1295
FLOW AB	4	LADDER	43	No	1759
ALARMS	5	LADDER	22	No	827
TIME TRIG	6	LADDER	17	No	680
A VESSEL	7	LADDER	38	No	1872
ANALOG IN	8	LADDER	6	No	198
PRESSURE	9	LADDER	11	No	466
B VESSEL	10	LADDER	38	No	1865
DIGITAL IN	11	LADDER	4	No	51
WELL PUMP	12	LADDER	4	No	63
DIGITL OUT	16	LADDER	15	No	297
REGEN A	17	LADDER	17	No	679
REGEN B	18	LADDER	17	No	679
RECYCLE	19	LADDER	17	No	574

Data File List

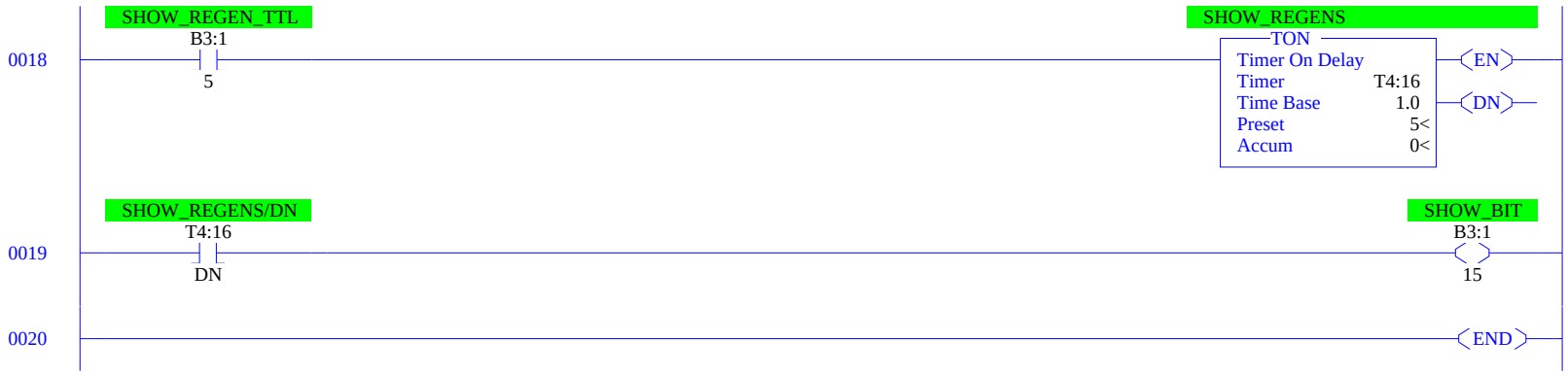
Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	21	7	O:6
INPUT	1	I	Global	No	66	22	I:21
STATUS	2	S	Global	No	0	66	S:65
COMMON	3	B	Global	No	14	14	B3:13
TIMER	4	T	Global	No	168	56	T4:55
COUNTER	5	C	Global	No	39	13	C5:12
CONTROL	6	R	Global	No	3	1	R6:0
	7	N	Global	No	52	52	N7:51
FLOAT	8	F	Global	No	244	122	F8:121
GENCTRL	9	B	Global	No	3	3	B9:2
	10	C	Global	No	15	5	C10:4
A BITS	21	B	Global	No	7	7	B21:6
B BITS	22	B	Global	No	7	7	B22:6
	34	N	Global	No	92	92	N34:91

LAD 2 - MAIN --- Total Rungs in File = 21

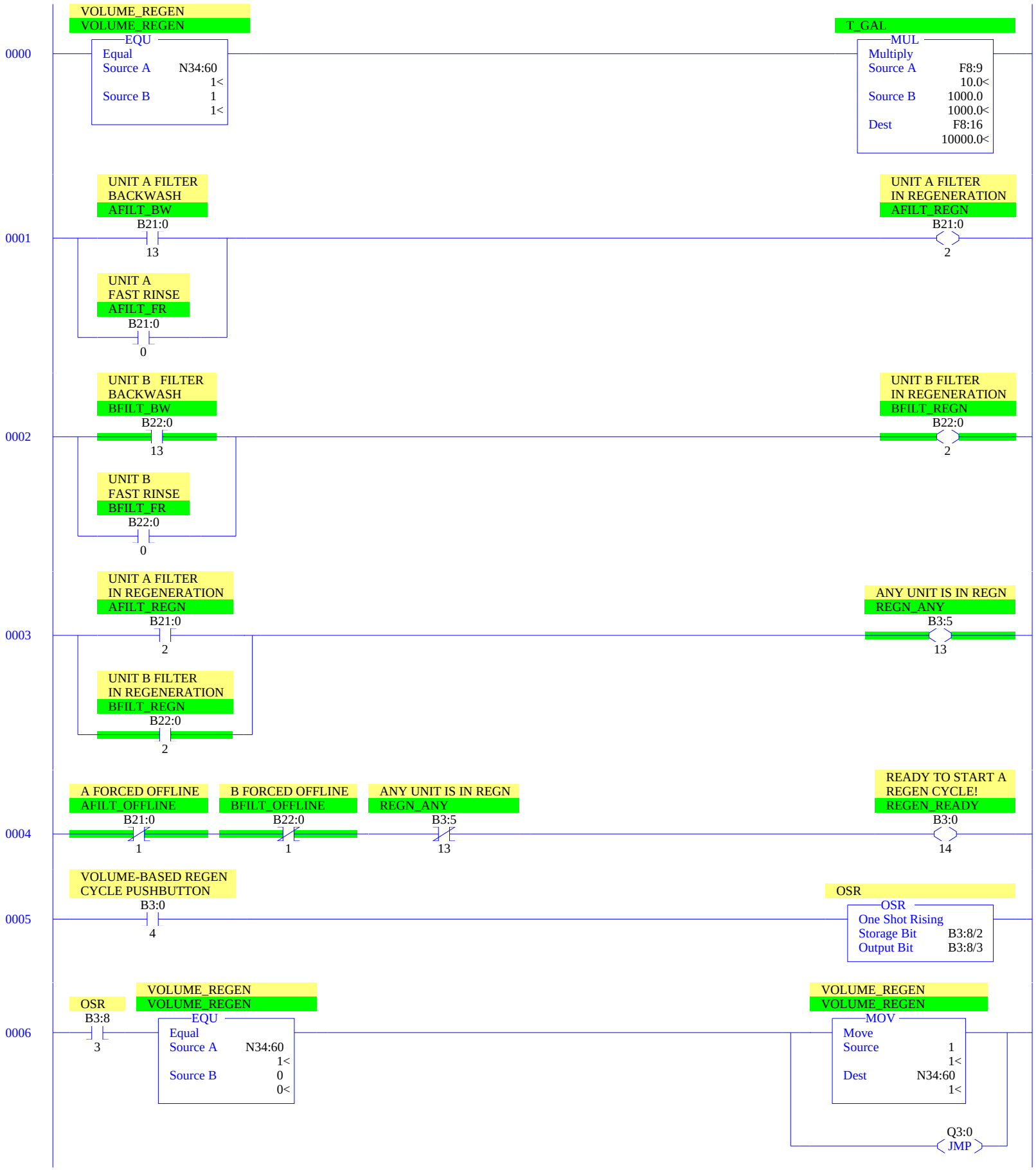



SEND NEW TIME AND DATE BUTTON ON PV
B3:2
12
0014
CPW
Copy Word
Source #N34:18
Dest #RTC:0.HR
Length 1
CPW
Copy Word
Source #N34:19
Dest #RTC:0.MIN
Length 1
DISPLAY OF CURRRENT REAL TIME HOURS COPIED FROM RTC
0015
CPW
Copy Word
Source #RTC:0.HR
Dest #N34:30
Length 1
DISPLAY OF CURRRENT REAL TIME MINUTES COPIED FROM RTC
CPW
Copy Word
Source #RTC:0.MIN
Dest #N34:36
Length 1
Overflow Trap
S:5
U
0
0016
DP REGEN STATUS
0=NOT IN CYCLE
1=REGEN A
2=REGEN B
First Pass
S:1
15
0017
MOV
Move
Source 0
0<
Dest N7:5
0<
DP_REGEN
DP_REGEN
GRT
Greater Than (A>B)
Source A N34:58
1<
Source B 1
1<
DP_REGEN
DP_REGEN
MOV
Move
Source 0
0<
Dest N34:58
1<
TIME_REGEN
TIME_REGEN
GRT
Greater Than (A>B)
Source A N34:59
1<
Source B 1
1<
TIME_REGEN
TIME_REGEN
MOV
Move
Source 0
0<
Dest N34:59
1<
VOLUME_REGEN
VOLUME_REGEN
GRT
Greater Than (A>B)
Source A N34:60
1<
Source B 1
1<
VOLUME_REGEN
VOLUME_REGEN
MOV
Move
Source 0
0<
Dest N34:60
1<

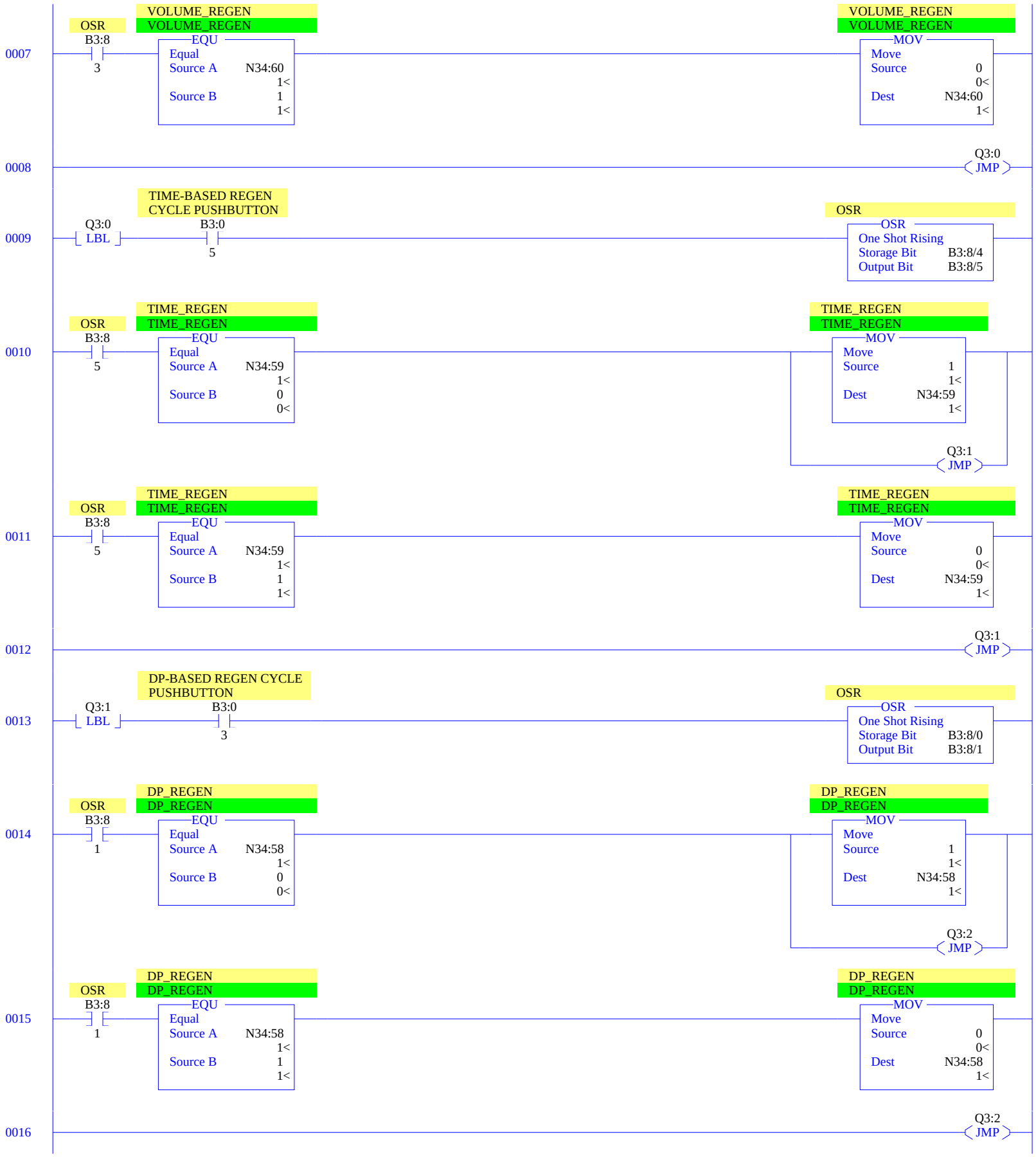
LAD 2 - MAIN --- Total Rungs in File = 21



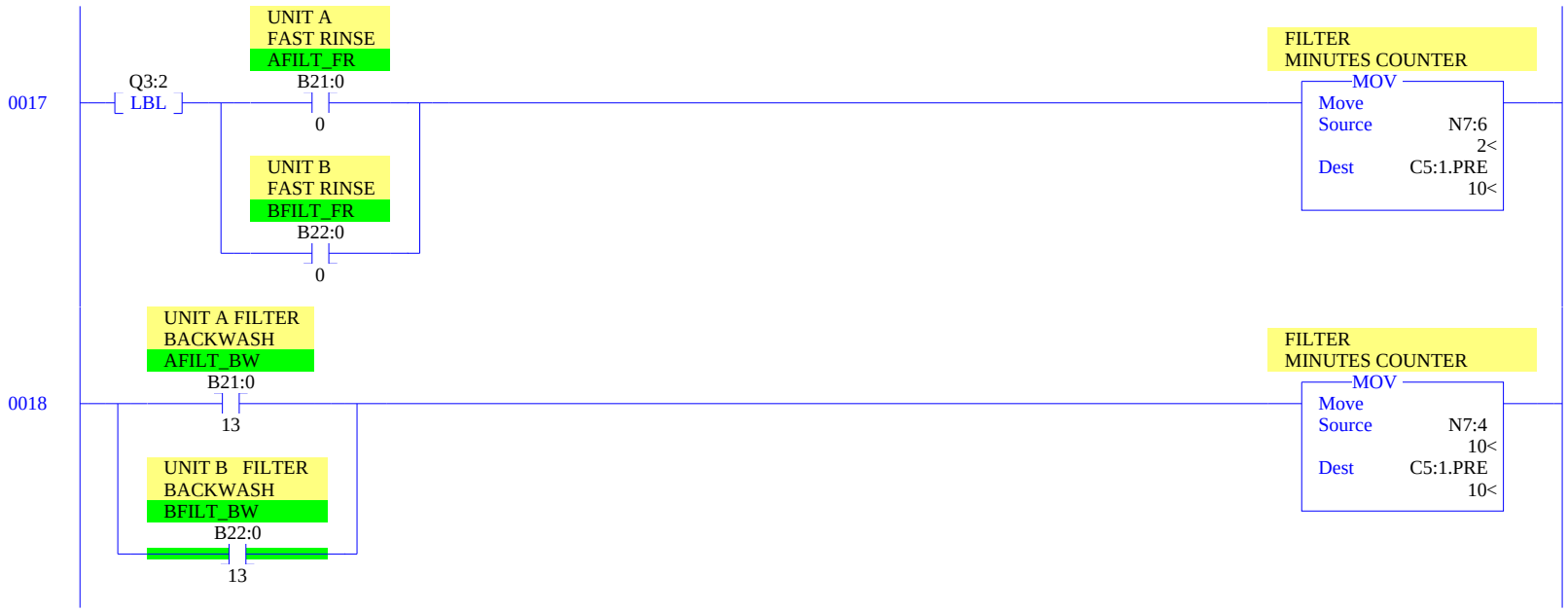
LAD 3 - REGEN --- Total Rungs in File = 29

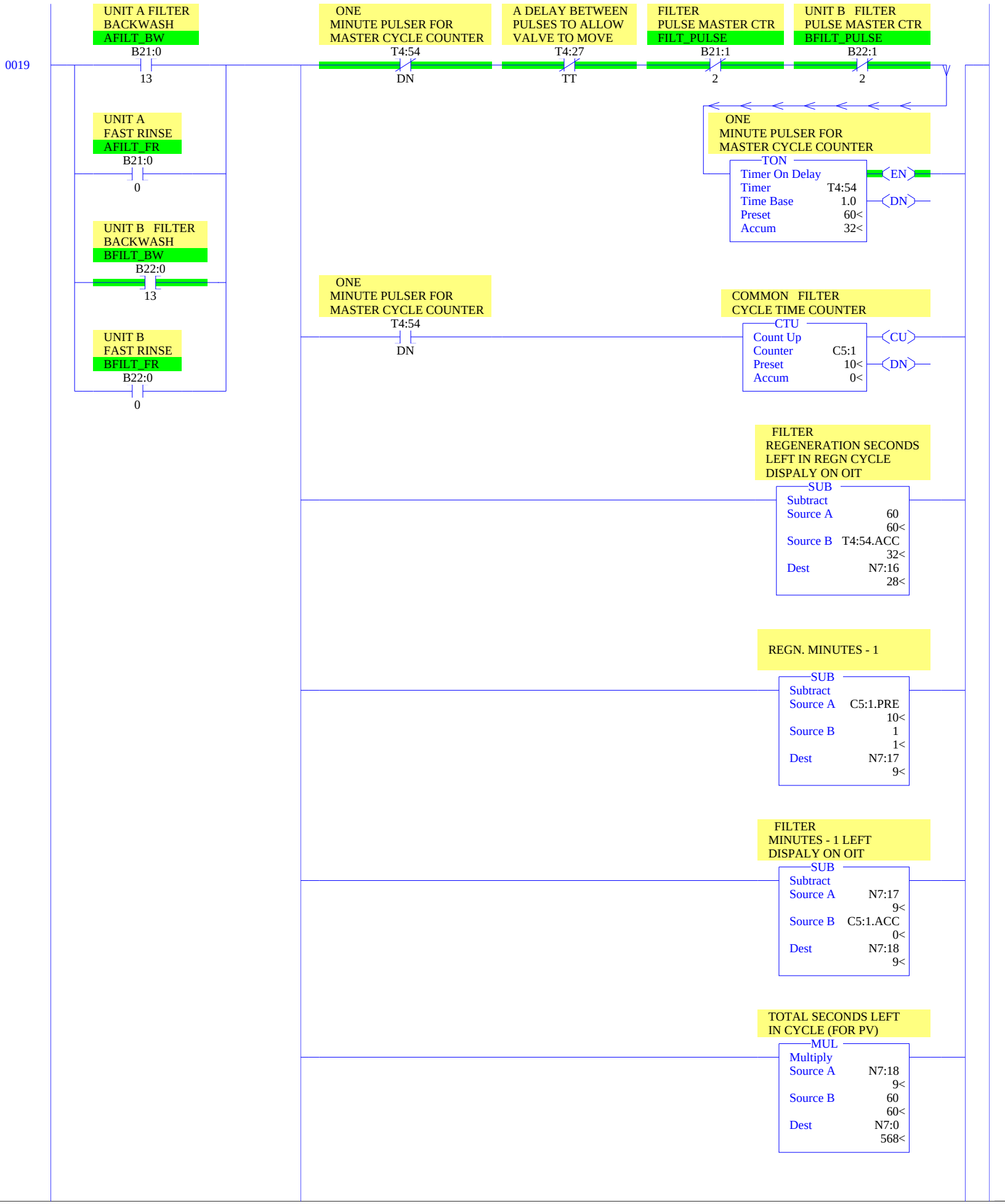


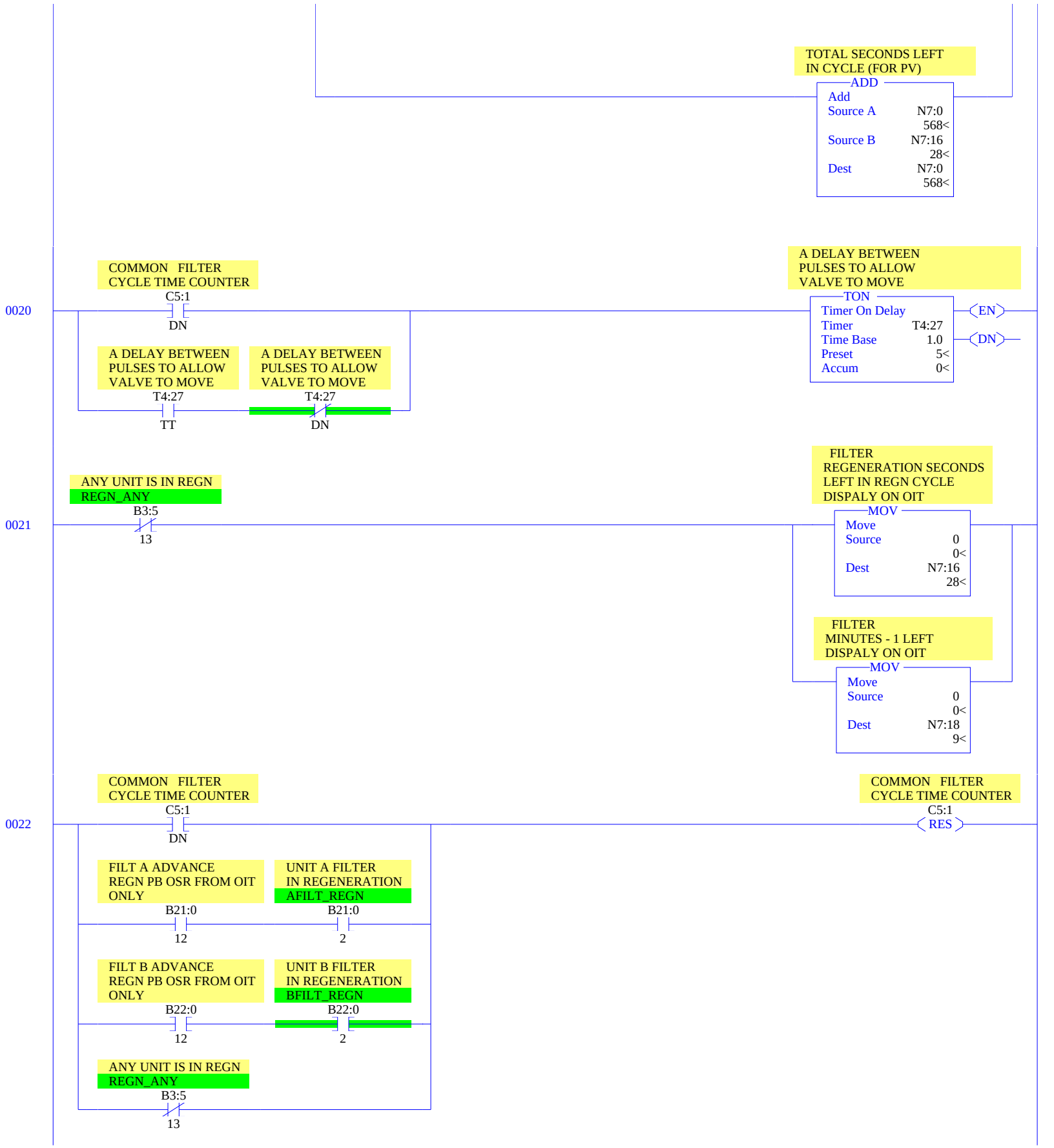
LAD 3 - REGEN --- Total Rungs in File = 29



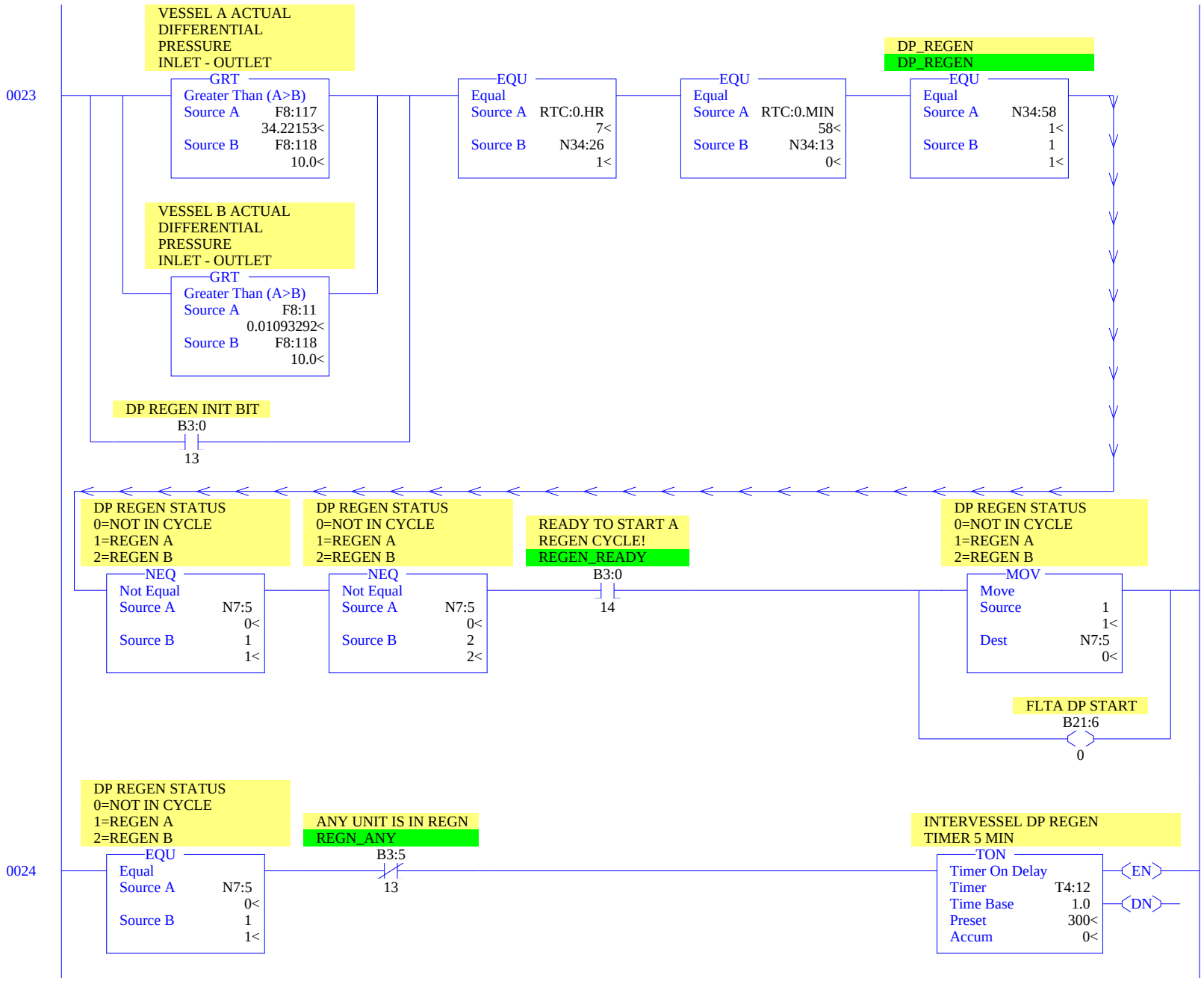
LAD 3 - REGEN --- Total Rungs in File = 29

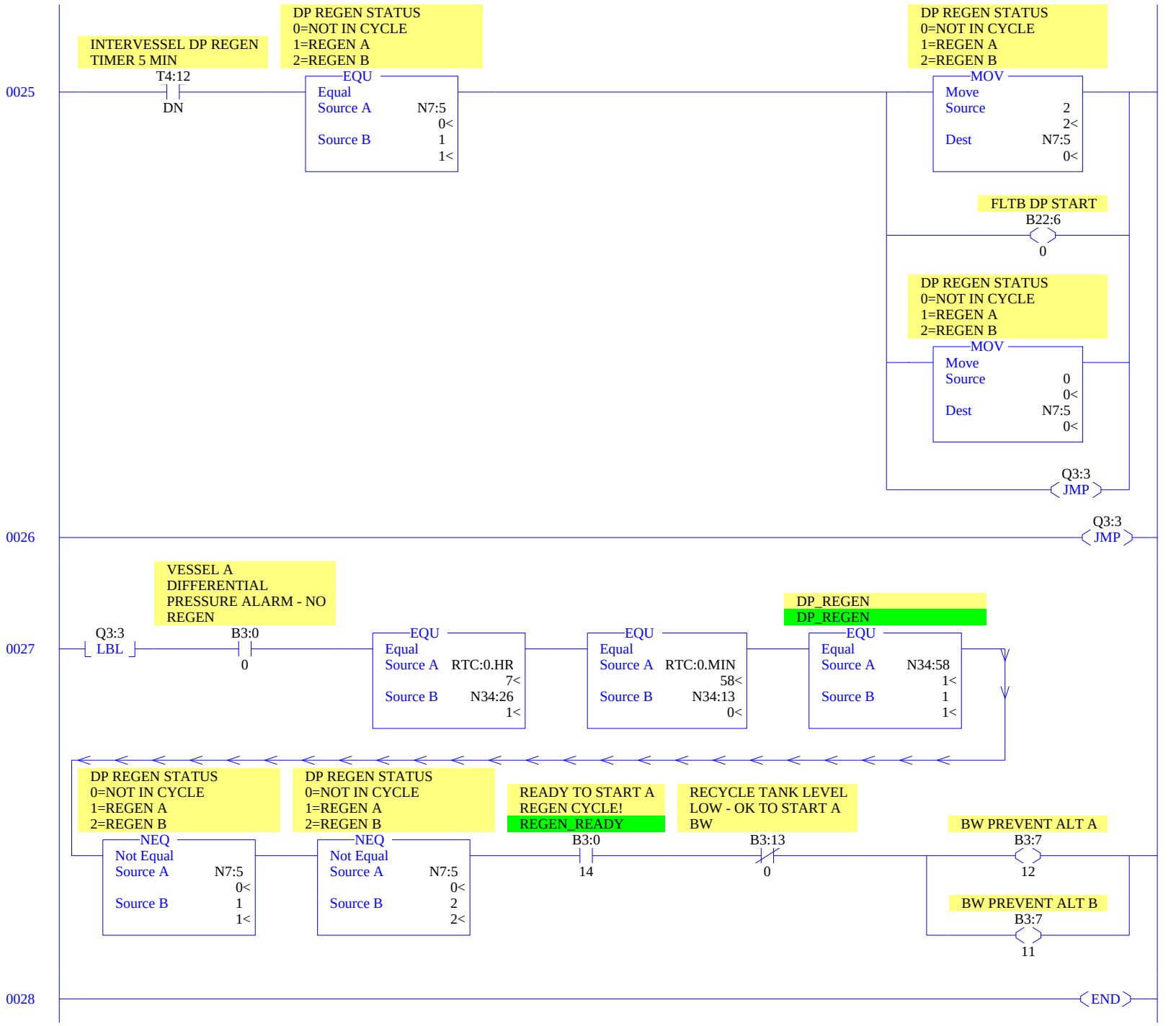



0019
UNIT A FILTER BACKWASH
AFILT_BW
B21:0
13
UNIT A FAST RINSE
AFILT_FR
B21:0
0
UNIT B FILTER BACKWASH
BFILT_BW
B22:0
13
UNIT B FAST RINSE
BFILT_FR
B22:0
0
ONE MINUTE PULSER FOR MASTER CYCLE COUNTER
T4:54
DN
A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:27
TT
FILTER PULSE MASTER CTR
FILT_PULSE
B21:1
2
UNIT B FILTER PULSE MASTER CTR
BFILT_PULSE
B22:1
2
ONE MINUTE PULSER FOR MASTER CYCLE COUNTER
TON
Timer On Delay
Timer T4:54
Time Base 1.0
Preset 60<
Accum 32<
EN
DN
ONE MINUTE PULSER FOR MASTER CYCLE COUNTER
T4:54
DN
COMMON FILTER CYCLE TIME COUNTER
CTU
Count Up
Counter C5:1
Preset 10<
Accum 0<
CU
DN
FILTER REGENERATION SECONDS LEFT IN REGN CYCLE DISPALY ON OIT
SUB
Subtract
Source A 60
60<
Source B T4:54.ACC
32<
Dest N7:16
28<
REGN. MINUTES - 1
SUB
Subtract
Source A C5:1.PRE
10<
Source B 1
1<
Dest N7:17
9<
FILTER MINUTES - 1 LEFT DISPALY ON OIT
SUB
Subtract
Source A N7:17
9<
Source B C5:1.ACC
0<
Dest N7:18
9<
TOTAL SECONDS LEFT IN CYCLE (FOR PV)
MUL
Multiply
Source A N7:18
9<
Source B 60
60<
Dest N7:0
568<

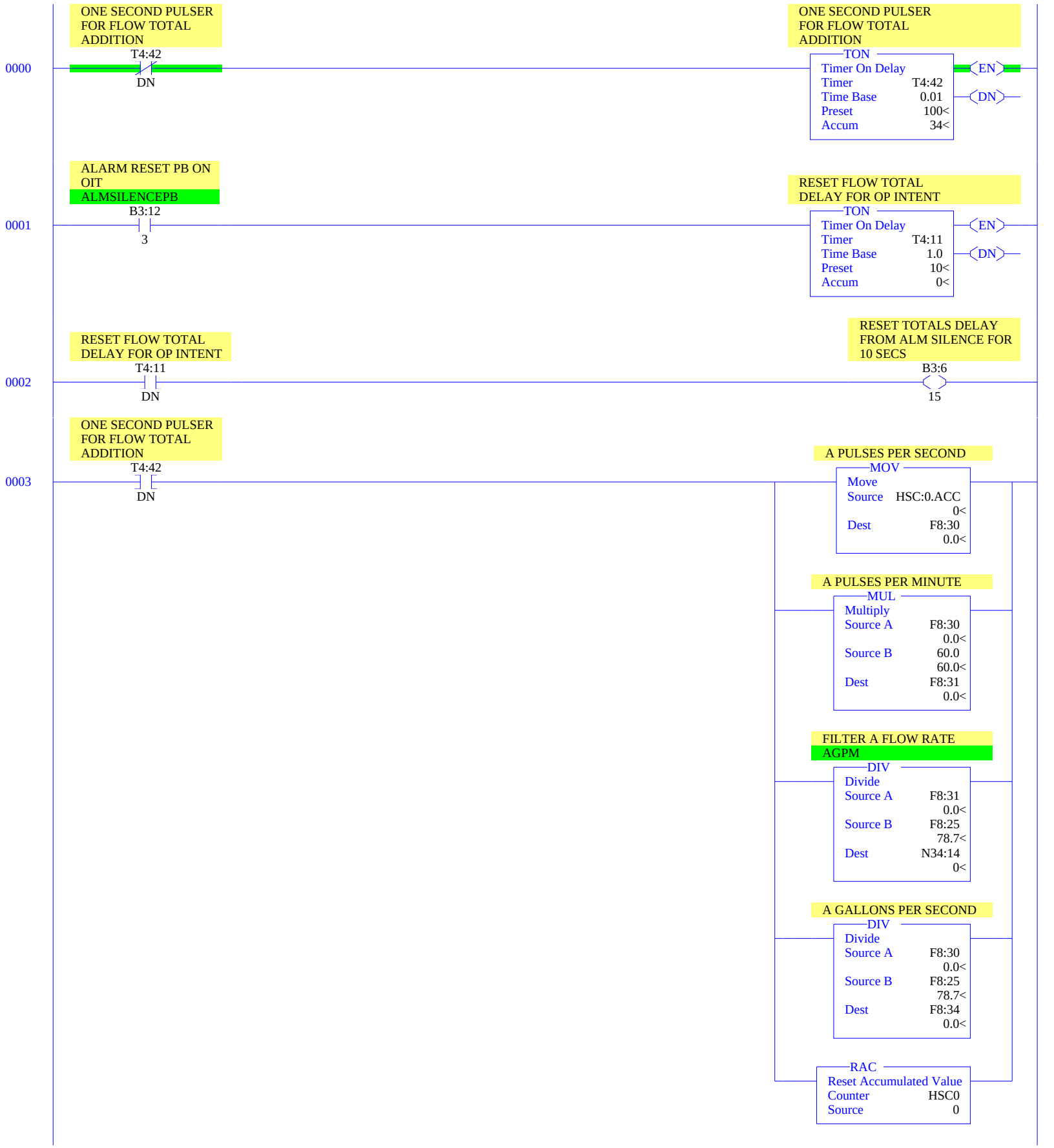

TOTAL SECONDS LEFT IN CYCLE (FOR PV)
ADD
Add
Source A N7:0 568<
Source B N7:16 28<
Dest N7:0 568<
0020
COMMON FILTER CYCLE TIME COUNTER
C5:1
DN
A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:27
TT
A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:27
DN
A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
TON
Timer On Delay
Timer T4:27
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
0021
ANY UNIT IS IN REGN
REGN_ANY
B3:5
13
FILTER REGENERATION SECONDS LEFT IN REGN CYCLE DISPALY ON OIT
MOV
Move
Source 0 0<
Dest N7:16 28<
FILTER MINUTES - 1 LEFT DISPALY ON OIT
MOV
Move
Source 0 0<
Dest N7:18 9<
0022
COMMON FILTER CYCLE TIME COUNTER
C5:1
DN
FILT A ADVANCE REGN PB OSR FROM OIT ONLY
B21:0
12
UNIT A FILTER IN REGENERATION
AFILT_REGN
B21:0
2
FILT B ADVANCE REGN PB OSR FROM OIT ONLY
B22:0
12
UNIT B FILTER IN REGENERATION
BFILT_REGN
B22:0
2
ANY UNIT IS IN REGN
REGN_ANY
B3:5
13
COMMON FILTER CYCLE TIME COUNTER
C5:1
RES

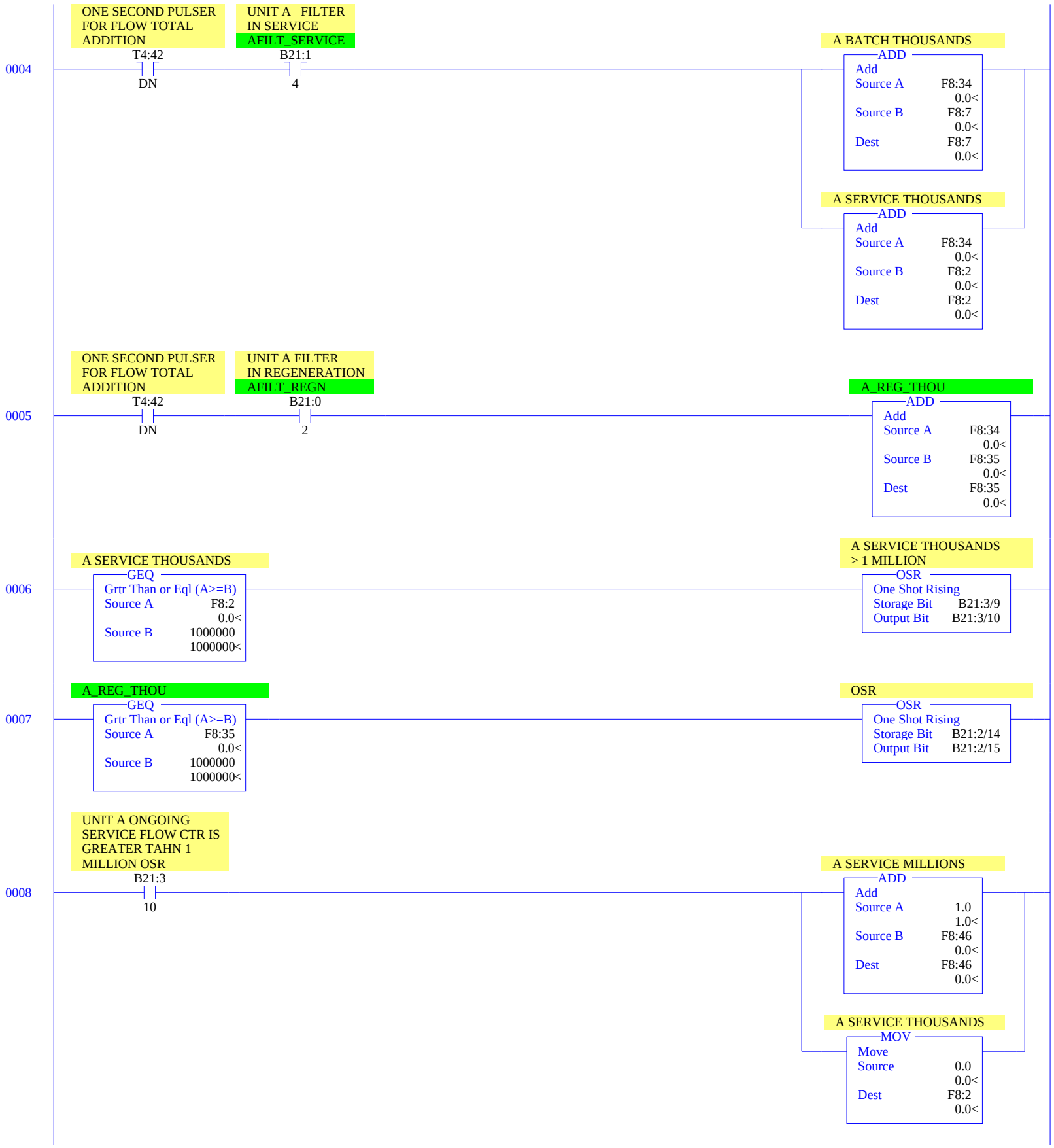
LAD 3 - REGEN --- Total Rungs in File = 29

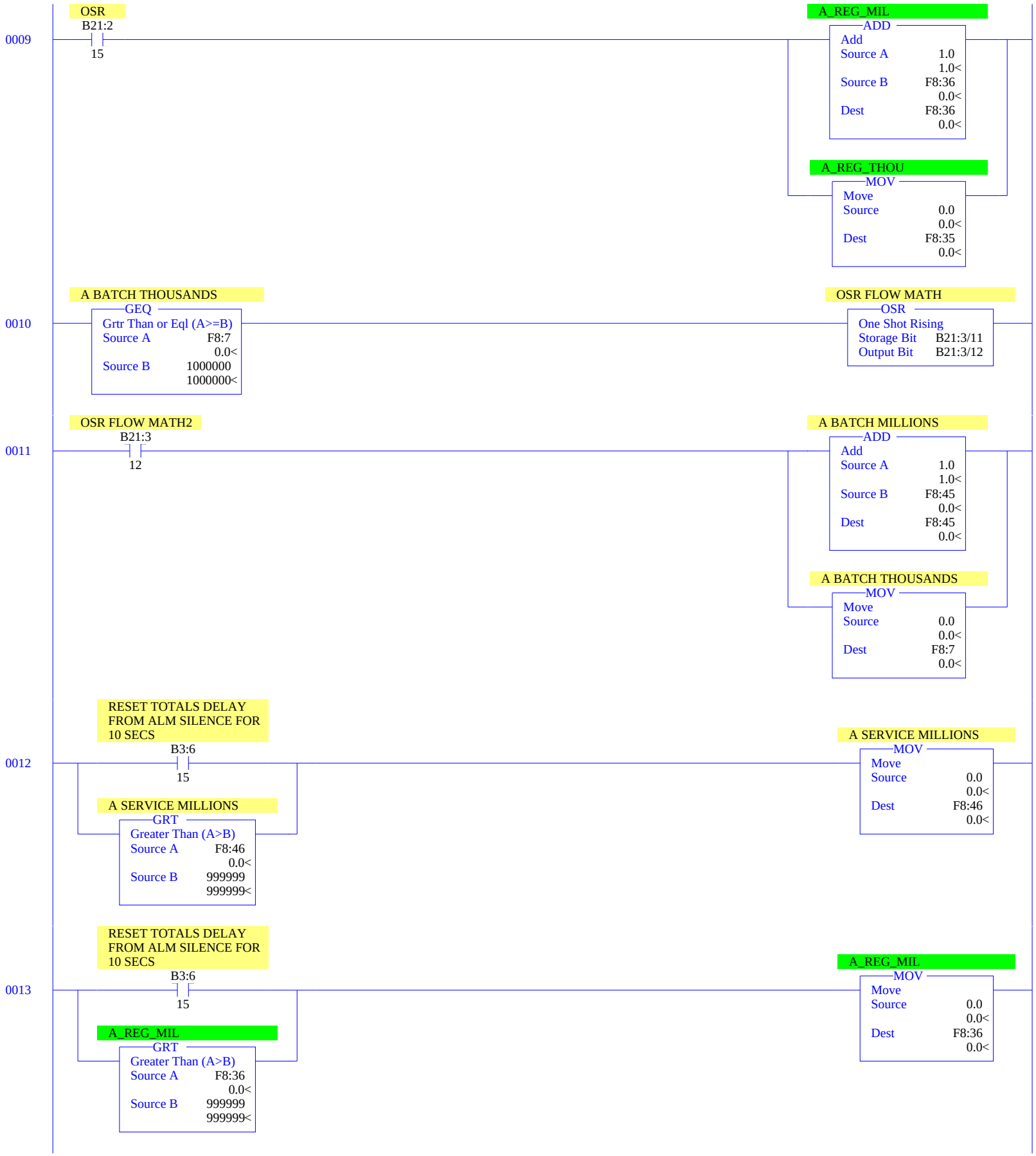


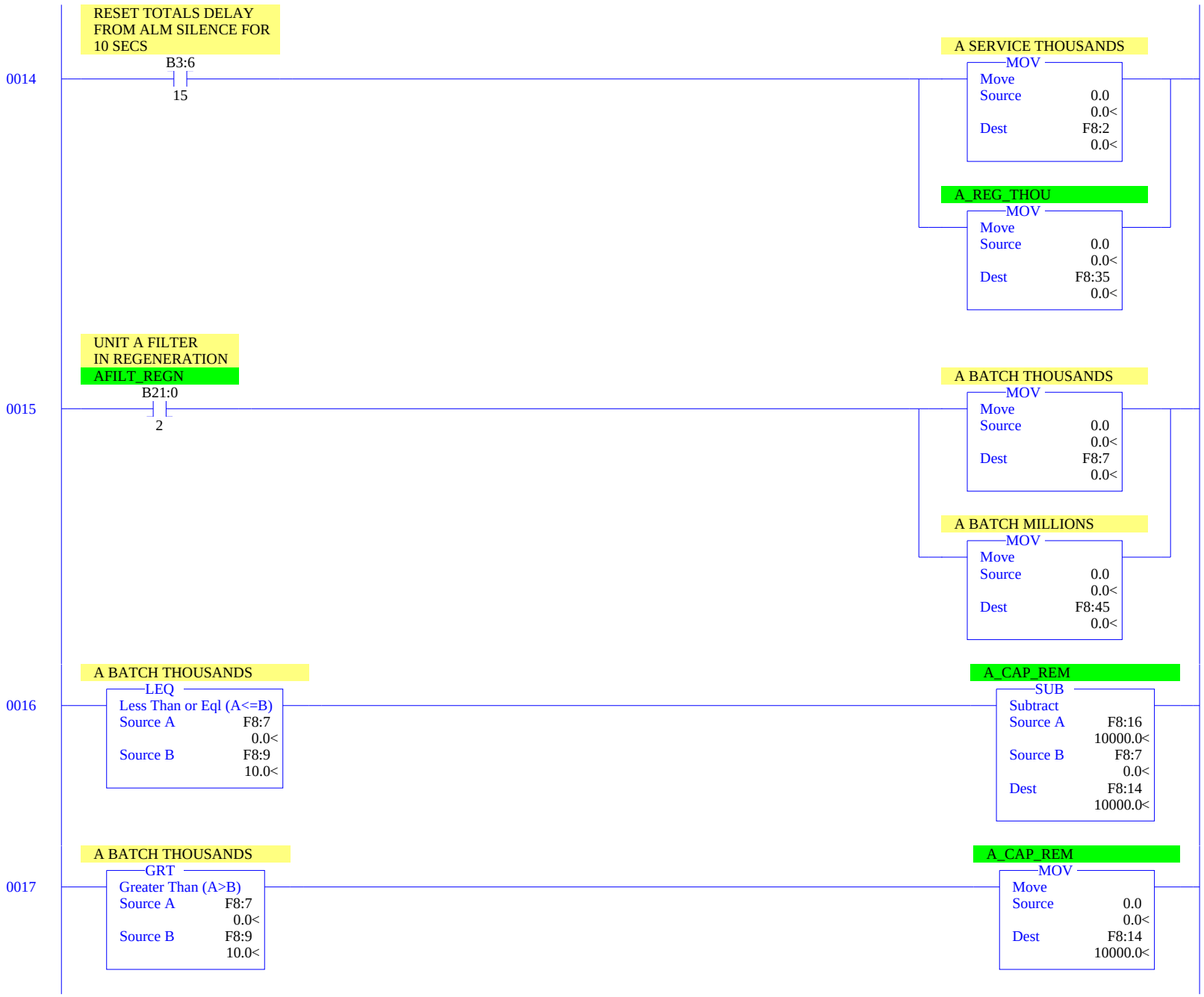

0025
INTERVESSEL DP REGEN TIMER 5 MIN
T4:12
DN
DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
EQU
Equal
Source A N7:5 0<
Source B 1 1<
DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
MOV
Move
Source 2 2<
Dest N7:5 0<
FLTB DP START
B22:6
0
DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
MOV
Move
Source 0 0<
Dest N7:5 0<
Q3:3
JMP
0026
Q3:3
JMP
0027
VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
Q3:3
LBL
B3:0
0
EQU
Equal
Source A RTC:0.HR 7<
Source B N34:26 1<
EQU
Equal
Source A RTC:0.MIN 58<
Source B N34:13 0<
DP_REGEN
DP_REGEN
EQU
Equal
Source A N34:58 1<
Source B 1 1<
DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
NEQ
Not Equal
Source A N7:5 0<
Source B 1 1<
DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
NEQ
Not Equal
Source A N7:5 0<
Source B 2 2<
READY TO START A REGEN CYCLE!
REGEN_READY
B3:0
14
RECYCLE TANK LEVEL LOW - OK TO START A BW
B3:13
0
BW PREVENT ALT A
B3:7
12
BW PREVENT ALT B
B3:7
11
0028
END

LAD 4 - FLOW AB --- Total Rungs in File = 43




0004
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
UNIT A FILTER IN SERVICE
AFILT_SERVICE
B21:1
4
A BATCH THOUSANDS
ADD
Add
Source A F8:34 0.0<
Source B F8:7 0.0<
Dest F8:7 0.0<
A SERVICE THOUSANDS
ADD
Add
Source A F8:34 0.0<
Source B F8:2 0.0<
Dest F8:2 0.0<
0005
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
UNIT A FILTER IN REGENERATION
AFILT_REGN
B21:0
2
A_REG_THOU
ADD
Add
Source A F8:34 0.0<
Source B F8:35 0.0<
Dest F8:35 0.0<
0006
A SERVICE THOUSANDS
GEQ
Grtr Than or Eql (A>=B)
Source A F8:2 0.0<
Source B 1000000 1000000<
A SERVICE THOUSANDS > 1 MILLION
OSR
One Shot Rising
Storage Bit B21:3/9
Output Bit B21:3/10
0007
A_REG_THOU
GEQ
Grtr Than or Eql (A>=B)
Source A F8:35 0.0<
Source B 1000000 1000000<
OSR
OSR
One Shot Rising
Storage Bit B21:2/14
Output Bit B21:2/15
0008
UNIT A ONGOING SERVICE FLOW CTR IS GREATER TAHN 1 MILLION OSR
B21:3
10
A SERVICE MILLIONS
ADD
Add
Source A 1.0 1.0<
Source B F8:46 0.0<
Dest F8:46 0.0<
A SERVICE THOUSANDS
MOV
Move
Source 0.0 0.0<
Dest F8:2 0.0<


0009
OSR
B21:2
15
A_REG_MIL
ADD
Add
Source A 1.0 1.0<
Source B F8:36 0.0<
Dest F8:36 0.0<
A_REG_THOU
MOV
Move
Source 0.0 0.0<
Dest F8:35 0.0<
0010
A BATCH THOUSANDS
GEQ
Grtr Than or Eql (A>=B)
Source A F8:7 0.0<
Source B 1000000 1000000<
OSR FLOW MATH
OSR
One Shot Rising
Storage Bit B21:3/11
Output Bit B21:3/12
0011
OSR FLOW MATH2
B21:3
12
A BATCH MILLIONS
ADD
Add
Source A 1.0 1.0<
Source B F8:45 0.0<
Dest F8:45 0.0<
A BATCH THOUSANDS
MOV
Move
Source 0.0 0.0<
Dest F8:7 0.0<
0012
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
A SERVICE MILLIONS
GRT
Greater Than (A>B)
Source A F8:46 0.0<
Source B 999999 999999<
A SERVICE MILLIONS
MOV
Move
Source 0.0 0.0<
Dest F8:46 0.0<
0013
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
A_REG_MIL
GRT
Greater Than (A>B)
Source A F8:36 0.0<
Source B 999999 999999<
A_REG_MIL
MOV
Move
Source 0.0 0.0<
Dest F8:36 0.0<


RESET TOTALS DELAY
FROM ALM SILENCE FOR
10 SECS
B3:6
15
0014
A SERVICE THOUSANDS
MOV
Move
Source 0.0
0.0<
Dest F8:2
0.0<
A_REG_THOU
MOV
Move
Source 0.0
0.0<
Dest F8:35
0.0<
UNIT A FILTER
IN REGENERATION
AFILT_REGN
B21:0
2
0015
A BATCH THOUSANDS
MOV
Move
Source 0.0
0.0<
Dest F8:7
0.0<
A BATCH MILLIONS
MOV
Move
Source 0.0
0.0<
Dest F8:45
0.0<
A BATCH THOUSANDS
LEQ
Less Than or Eql (A<=B)
Source A F8:7
0.0<
Source B F8:9
10.0<
0016
A_CAP_REM
SUB
Subtract
Source A F8:16
10000.0<
Source B F8:7
0.0<
Dest F8:14
10000.0<
A BATCH THOUSANDS
GRT
Greater Than (A>B)
Source A F8:7
0.0<
Source B F8:9
10.0<
0017
A_CAP_REM
MOV
Move
Source 0.0
0.0<
Dest F8:14
10000.0<

0018

ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN

B PULSES PER SECOND
MOV
Move
Source HSC:1.ACC
0<
Dest F8:32
0.0<

B PULSES PER MINUTE
MUL
Multiply
Source A F8:32
0.0<
Source B 60.0
60.0<
Dest F8:33
0.0<

UNIT B FLOW RATE
BGPM
DIV
Divide
Source A F8:33
0.0<
Source B F8:26
78.7<
Dest N34:6
0<

UNIT B GALLONS PER SECOND
DIV
Divide
Source A F8:32
0.0<
Source B F8:26
78.7<
Dest F8:10
0.0<

RAC
Reset Accumulated Value
Counter HSC1
Source 0

0019

ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN

UNIT B FILTER IN SERVICE
BFILT_SERVICE
B22:1
4

B BATCH THOUSANDS
ADD
Add
Source A F8:10
0.0<
Source B F8:6
0.0<
Dest F8:6
0.0<

B SERVICE THOUSANDS
ADD
Add
Source A F8:10
0.0<
Source B F8:12
0.0<
Dest F8:12
0.0<

0020

ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN

UNIT B FILTER IN REGENERATION
BFILT_REGN
B22:0
2

B_REG_THOU
ADD
Add
Source A F8:10 0.0<
Source B F8:37 0.0<
Dest F8:37 0.0<

0021

B SERVICE THOUSANDS
GEQ
Grtr Than or Eql (A>=B)
Source A F8:12 0.0<
Source B 1000000 1000000<

B SERVICE THOUSANDS > 1 MILLION
OSR
One Shot Rising
Storage Bit B22:3/9
Output Bit B22:3/10

0022

B_REG_THOU
GEQ
Grtr Than or Eql (A>=B)
Source A F8:37 0.0<
Source B 1000000 1000000<

OSR
OSR
One Shot Rising
Storage Bit B22:2/14
Output Bit B22:2/15

0023

UNIT B ONGOING SERVICE FLOW CTR IS GREATER THAN 1 MILLION OSR
B22:3
10

B SERVICE MILLIONS
ADD
Add
Source A 1.0 1.0<
Source B F8:47 0.0<
Dest F8:47 0.0<

B SERVICE THOUSANDS
MOV
Move
Source 0.0 0.0<
Dest F8:12 0.0<

0024

OSR
B22:2
15

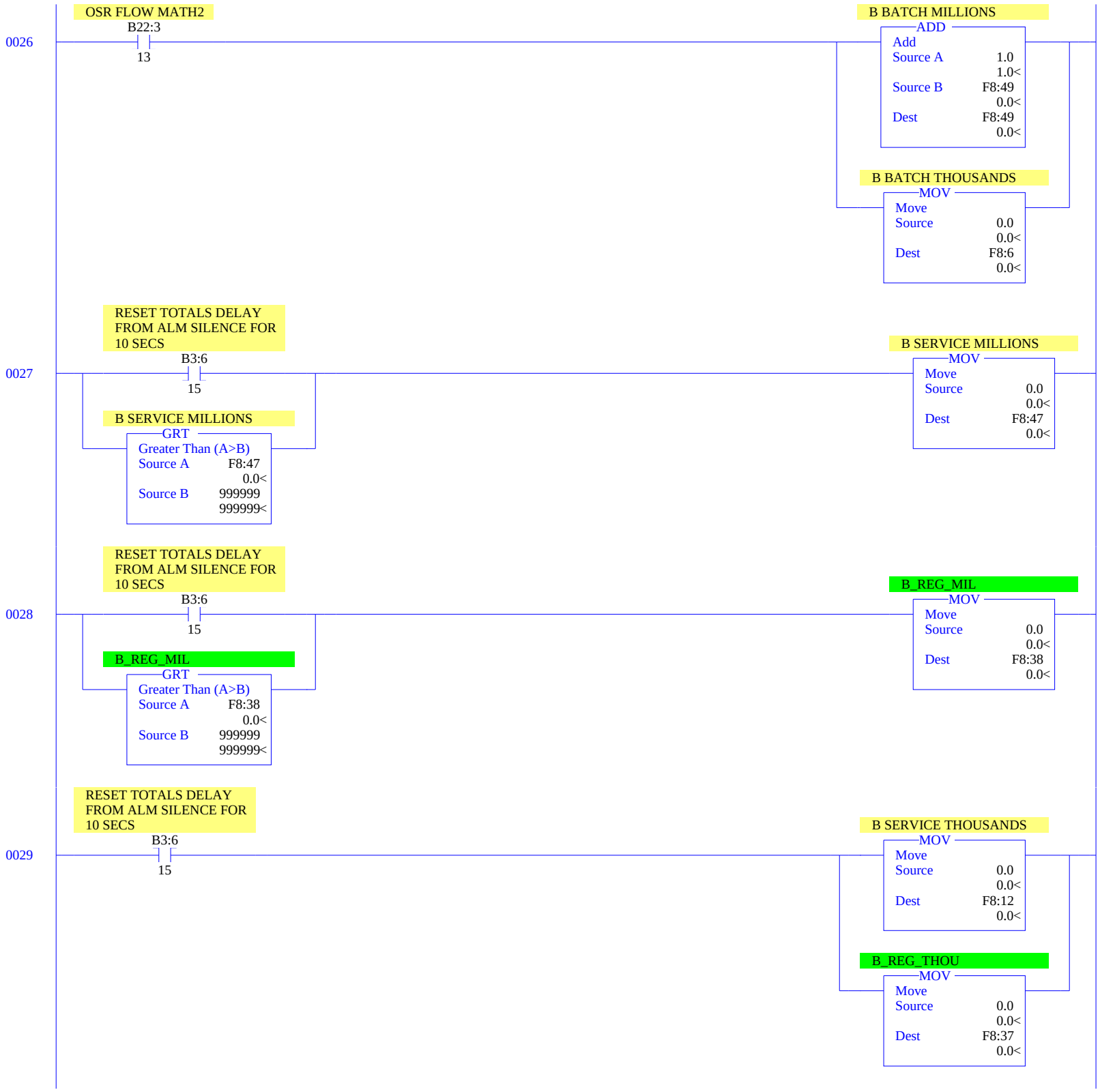
B_REG_MIL
ADD
Add
Source A 1.0 1.0<
Source B F8:38 0.0<
Dest F8:38 0.0<

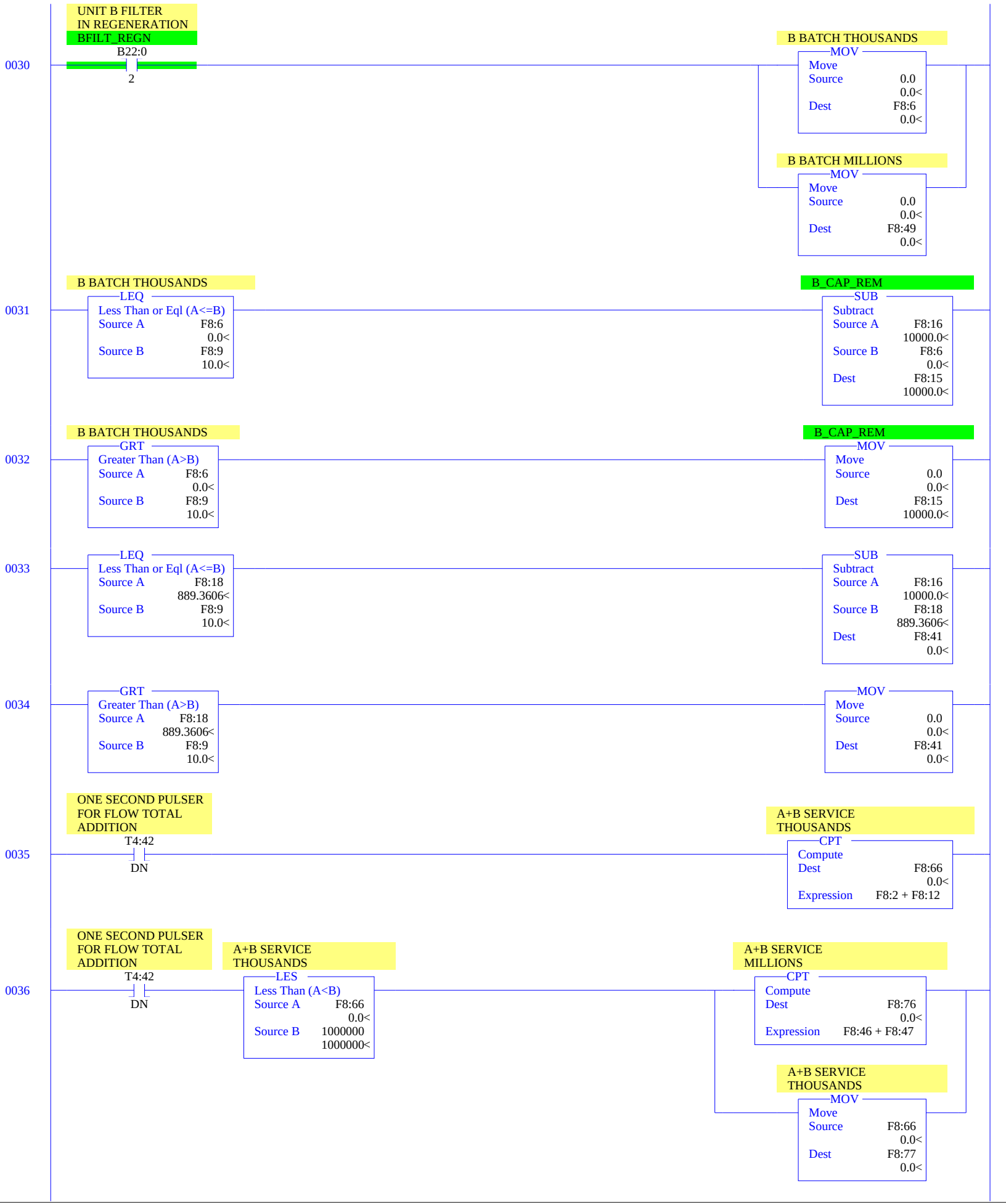
B_REG_THOU
MOV
Move
Source 0.0 0.0<
Dest F8:37 0.0<

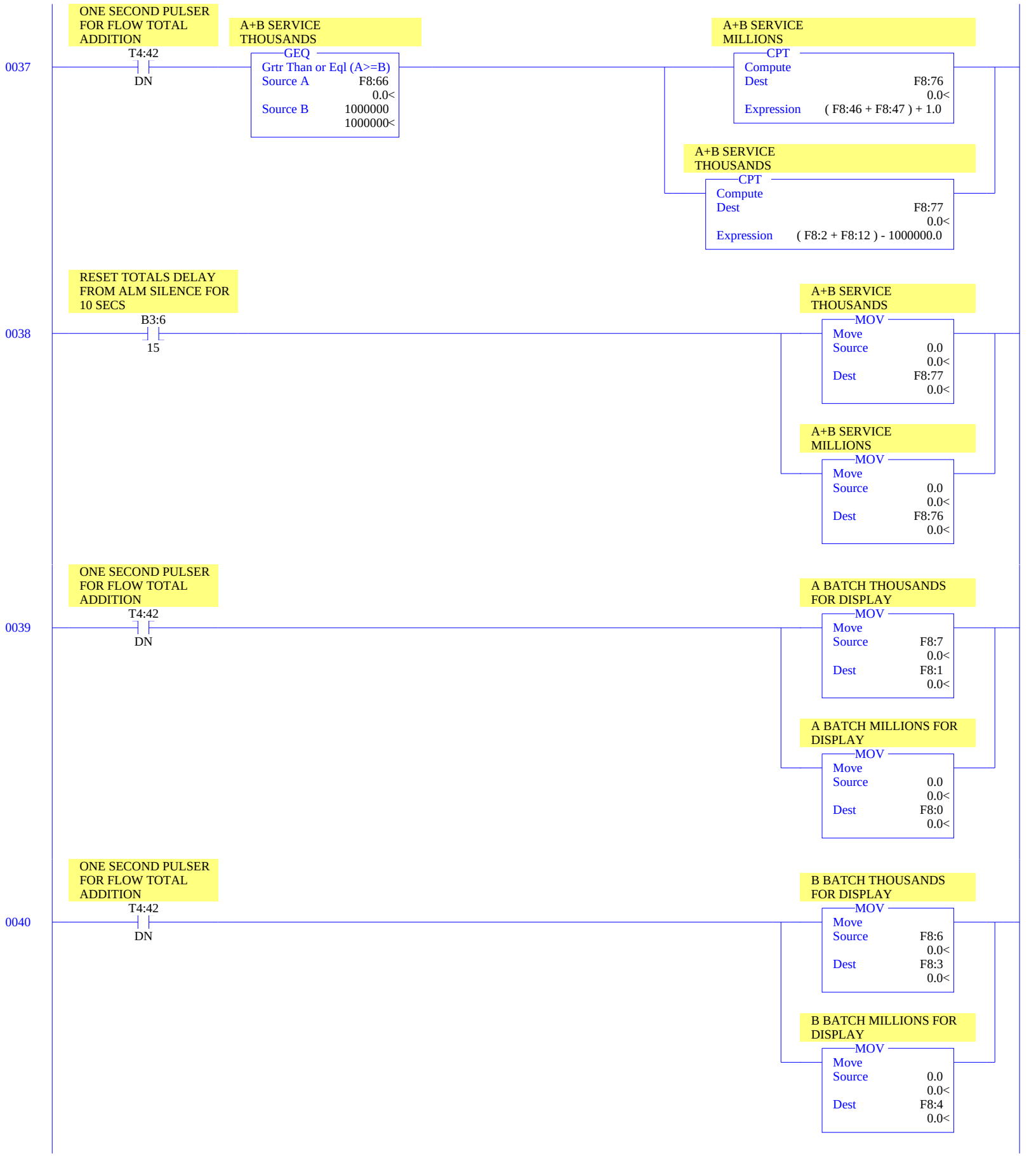
0025

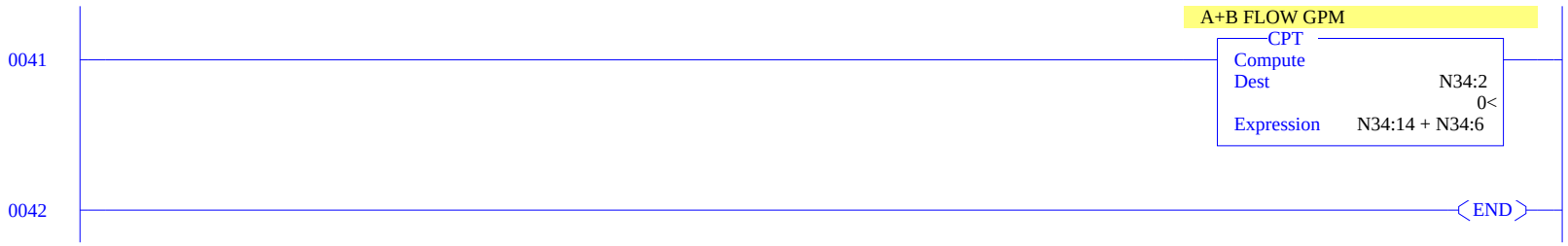
B BATCH THOUSANDS
GEQ
Grtr Than or Eql (A>=B)
Source A F8:6 0.0<
Source B 1000000 1000000<

OSR FLOW MATH
OSR
One Shot Rising
Storage Bit B22:3/12
Output Bit B22:3/13

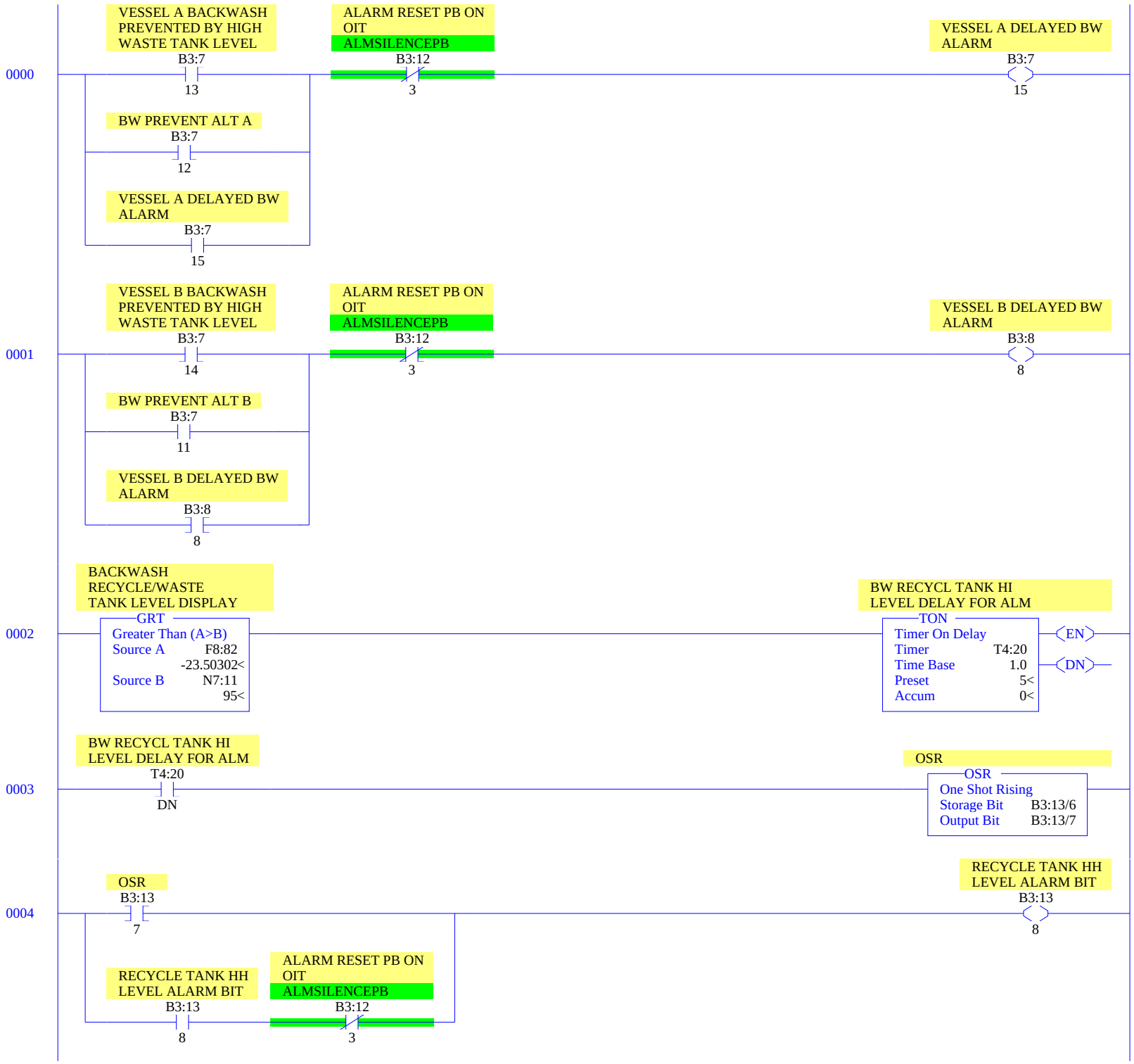

OSR FLOW MATH2
B22:3
13
0026
B BATCH MILLIONS
ADD
Add
Source A 1.0
1.0<
Source B F8:49
0.0<
Dest F8:49
0.0<
B BATCH THOUSANDS
MOV
Move
Source 0.0
0.0<
Dest F8:6
0.0<
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
0027
B SERVICE MILLIONS
GRT
Greater Than (A>B)
Source A F8:47
0.0<
Source B 999999
999999<
B SERVICE MILLIONS
MOV
Move
Source 0.0
0.0<
Dest F8:47
0.0<
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
0028
B_REG_MIL
GRT
Greater Than (A>B)
Source A F8:38
0.0<
Source B 999999
999999<
B_REG_MIL
MOV
Move
Source 0.0
0.0<
Dest F8:38
0.0<
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
0029
B SERVICE THOUSANDS
MOV
Move
Source 0.0
0.0<
Dest F8:12
0.0<
B_REG_THOU
MOV
Move
Source 0.0
0.0<
Dest F8:37
0.0<

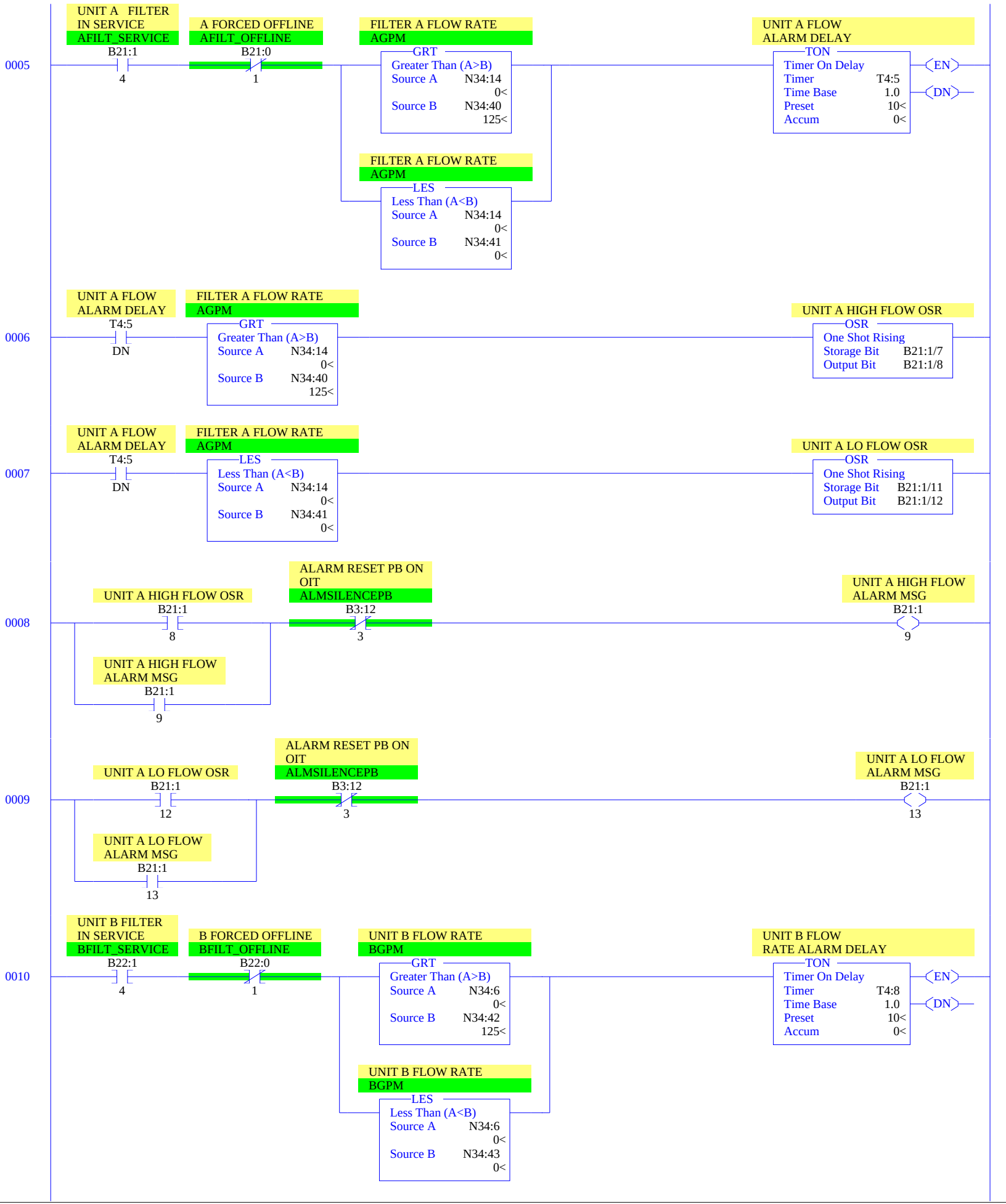

UNIT B FILTER IN REGENERATION
BFILT_REGN
B22:0
2
B BATCH THOUSANDS
MOV
Move
Source 0.0
0.0<
Dest F8:6
0.0<
B BATCH MILLIONS
MOV
Move
Source 0.0
0.0<
Dest F8:49
0.0<
0030
B BATCH THOUSANDS
LEQ
Less Than or Eql (A<=B)
Source A F8:6
0.0<
Source B F8:9
10.0<
B_CAP_REM
SUB
Subtract
Source A F8:16
10000.0<
Source B F8:6
0.0<
Dest F8:15
10000.0<
0031
B BATCH THOUSANDS
GRT
Greater Than (A>B)
Source A F8:6
0.0<
Source B F8:9
10.0<
B_CAP_REM
MOV
Move
Source 0.0
0.0<
Dest F8:15
10000.0<
0032
LEQ
Less Than or Eql (A<=B)
Source A F8:18
889.3606<
Source B F8:9
10.0<
SUB
Subtract
Source A F8:16
10000.0<
Source B F8:18
889.3606<
Dest F8:41
0.0<
0033
GRT
Greater Than (A>B)
Source A F8:18
889.3606<
Source B F8:9
10.0<
MOV
Move
Source 0.0
0.0<
Dest F8:41
0.0<
0034
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
A+B SERVICE THOUSANDS
CPT
Compute
Dest F8:66
0.0<
Expression F8:2 + F8:12
0035
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
A+B SERVICE THOUSANDS
LES
Less Than (A<B)
Source A F8:66
0.0<
Source B 1000000
1000000<
A+B SERVICE MILLIONS
CPT
Compute
Dest F8:76
0.0<
Expression F8:46 + F8:47
A+B SERVICE THOUSANDS
MOV
Move
Source F8:66
0.0<
Dest F8:77
0.0<
0036

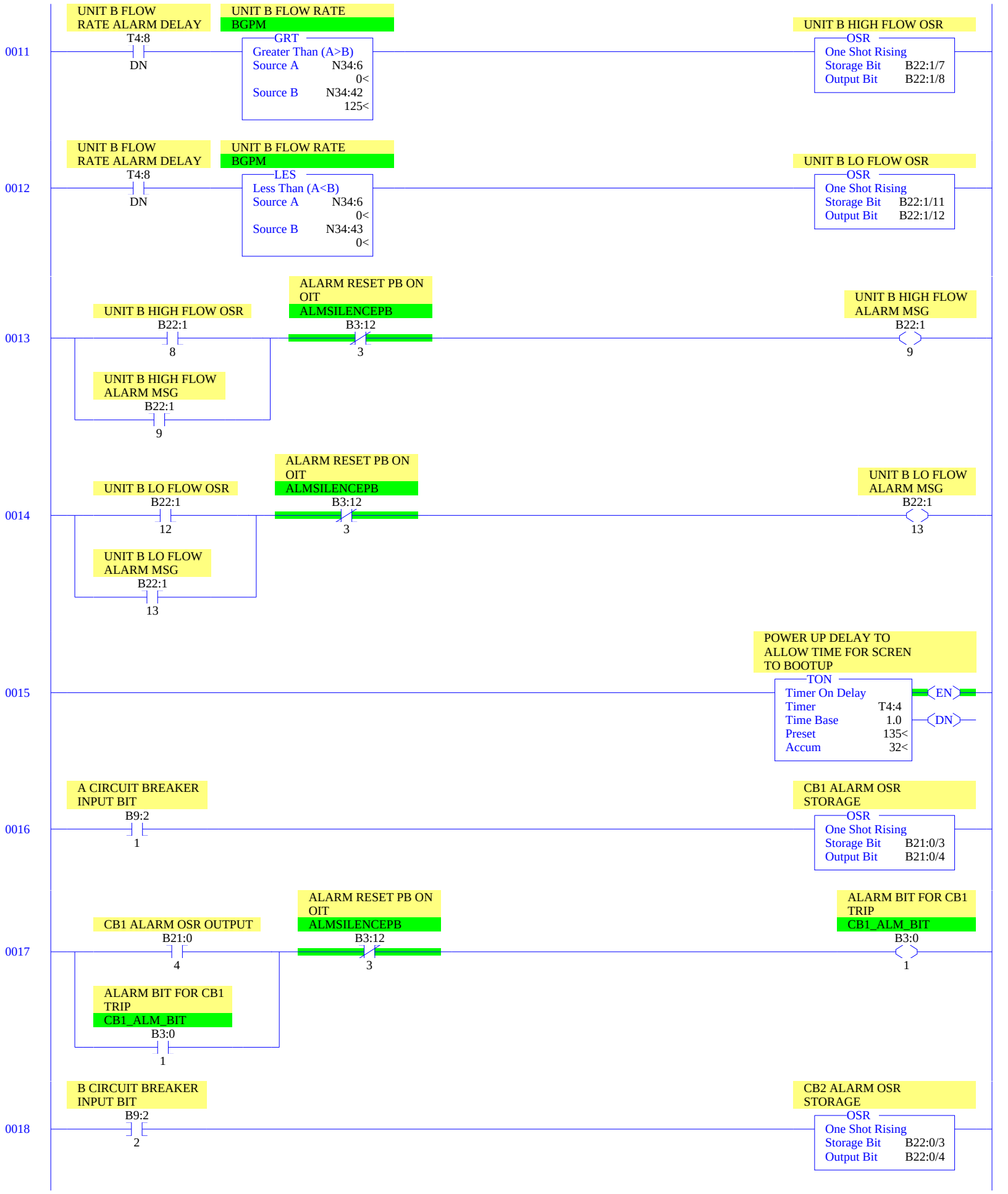

ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
0037
A+B SERVICE THOUSANDS
GEQ
Grtr Than or Eql (A>=B)
Source A F8:66 0.0<
Source B 1000000 1000000<
A+B SERVICE MILLIONS
CPT
Compute
Dest F8:76 0.0<
Expression (F8:46 + F8:47) + 1.0
A+B SERVICE THOUSANDS
CPT
Compute
Dest F8:77 0.0<
Expression (F8:2 + F8:12) - 1000000.0
RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:6
15
0038
A+B SERVICE THOUSANDS
MOV
Move
Source 0.0 0.0<
Dest F8:77 0.0<
A+B SERVICE MILLIONS
MOV
Move
Source 0.0 0.0<
Dest F8:76 0.0<
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
0039
A BATCH THOUSANDS FOR DISPLAY
MOV
Move
Source F8:7 0.0<
Dest F8:1 0.0<
A BATCH MILLIONS FOR DISPLAY
MOV
Move
Source 0.0 0.0<
Dest F8:0 0.0<
ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42
DN
0040
B BATCH THOUSANDS FOR DISPLAY
MOV
Move
Source F8:6 0.0<
Dest F8:3 0.0<
B BATCH MILLIONS FOR DISPLAY
MOV
Move
Source 0.0 0.0<
Dest F8:4 0.0<


A+B FLOW GPM
0041
CPT
Compute
Dest
N34:2
0<
Expression
N34:14 + N34:6
0042
END

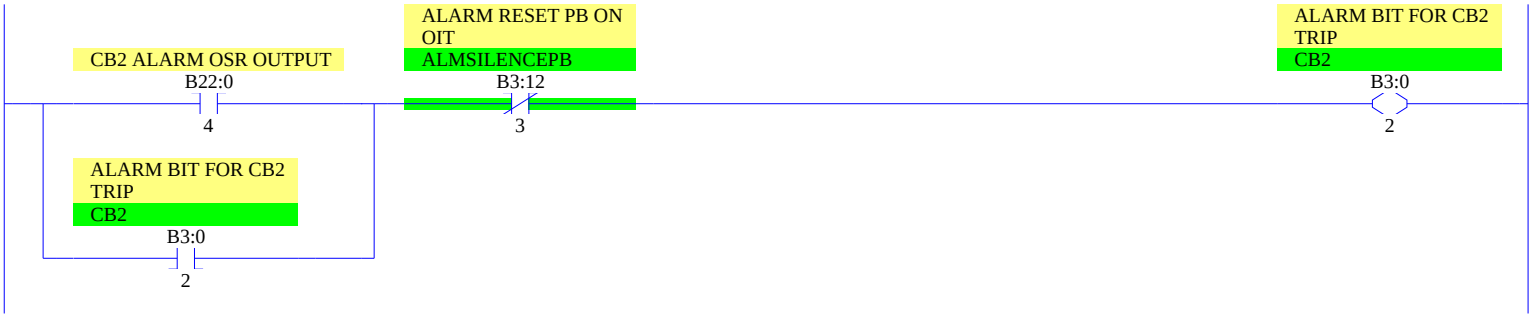
LAD 5 - ALARMS --- Total Rungs in File = 22

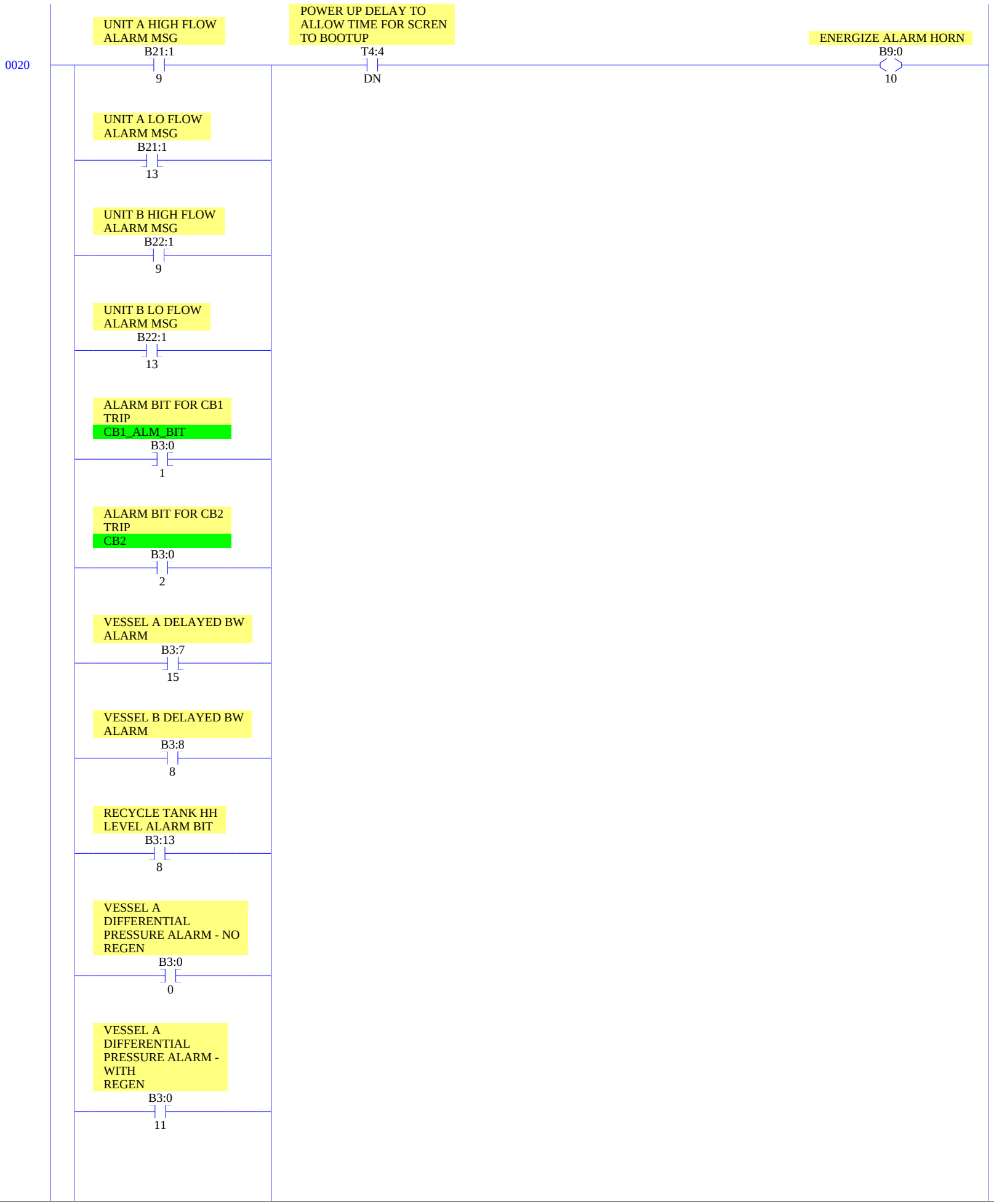



0005
UNIT A FILTER IN SERVICE
AFILT_SERVICE
B21:1
4
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
FILTER A FLOW RATE
AGPM
GRT
Greater Than (A>B)
Source A N34:14 0<
Source B N34:40 125<
FILTER A FLOW RATE
AGPM
LES
Less Than (A<B)
Source A N34:14 0<
Source B N34:41 0<
UNIT A FLOW ALARM DELAY
TON
Timer On Delay
Timer T4:5
Time Base 1.0
Preset 10<
Accum 0<
EN
DN
0006
UNIT A FLOW ALARM DELAY
T4:5
DN
FILTER A FLOW RATE
AGPM
GRT
Greater Than (A>B)
Source A N34:14 0<
Source B N34:40 125<
UNIT A HIGH FLOW OSR
OSR
One Shot Rising
Storage Bit B21:1/7
Output Bit B21:1/8
0007
UNIT A FLOW ALARM DELAY
T4:5
DN
FILTER A FLOW RATE
AGPM
LES
Less Than (A<B)
Source A N34:14 0<
Source B N34:41 0<
UNIT A LO FLOW OSR
OSR
One Shot Rising
Storage Bit B21:1/11
Output Bit B21:1/12
0008
UNIT A HIGH FLOW OSR
B21:1
8
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
UNIT A HIGH FLOW ALARM MSG
B21:1
9
UNIT A HIGH FLOW ALARM MSG
B21:1
9
0009
UNIT A LO FLOW OSR
B21:1
12
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
UNIT A LO FLOW ALARM MSG
B21:1
13
UNIT A LO FLOW ALARM MSG
B21:1
13
0010
UNIT B FILTER IN SERVICE
BFILT_SERVICE
B22:1
4
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
UNIT B FLOW RATE
BGPM
GRT
Greater Than (A>B)
Source A N34:6 0<
Source B N34:42 125<
UNIT B FLOW RATE
BGPM
LES
Less Than (A<B)
Source A N34:6 0<
Source B N34:43 0<
UNIT B FLOW RATE ALARM DELAY
TON
Timer On Delay
Timer T4:8
Time Base 1.0
Preset 10<
Accum 0<
EN
DN


0011
UNIT B FLOW RATE ALARM DELAY
T4:8
DN
UNIT B FLOW RATE
BGPM
GRT
Greater Than (A>B)
Source A N34:6 0<
Source B N34:42 125<
UNIT B HIGH FLOW OSR
OSR
One Shot Rising
Storage Bit B22:1/7
Output Bit B22:1/8
0012
UNIT B FLOW RATE ALARM DELAY
T4:8
DN
UNIT B FLOW RATE
BGPM
LES
Less Than (A<B)
Source A N34:6 0<
Source B N34:43 0<
UNIT B LO FLOW OSR
OSR
One Shot Rising
Storage Bit B22:1/11
Output Bit B22:1/12
0013
UNIT B HIGH FLOW OSR
B22:1
8
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
UNIT B HIGH FLOW ALARM MSG
B22:1
9
UNIT B HIGH FLOW ALARM MSG
B22:1
9
0014
UNIT B LO FLOW OSR
B22:1
12
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
UNIT B LO FLOW ALARM MSG
B22:1
13
UNIT B LO FLOW ALARM MSG
B22:1
13
0015
POWER UP DELAY TO ALLOW TIME FOR SCREN TO BOOTUP
TON
Timer On Delay
Timer T4:4
Time Base 1.0
Preset 135<
Accum 32<
EN
DN
0016
A CIRCUIT BREAKER INPUT BIT
B9:2
1
CB1 ALARM OSR STORAGE
OSR
One Shot Rising
Storage Bit B21:0/3
Output Bit B21:0/4
0017
CB1 ALARM OSR OUTPUT
B21:0
4
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
ALARM BIT FOR CB1 TRIP
CB1_ALM_BIT
B3:0
1
ALARM BIT FOR CB1 TRIP
CB1_ALM_BIT
B3:0
1
0018
B CIRCUIT BREAKER INPUT BIT
B9:2
2
CB2 ALARM OSR STORAGE
OSR
One Shot Rising
Storage Bit B22:0/3
Output Bit B22:0/4

0019

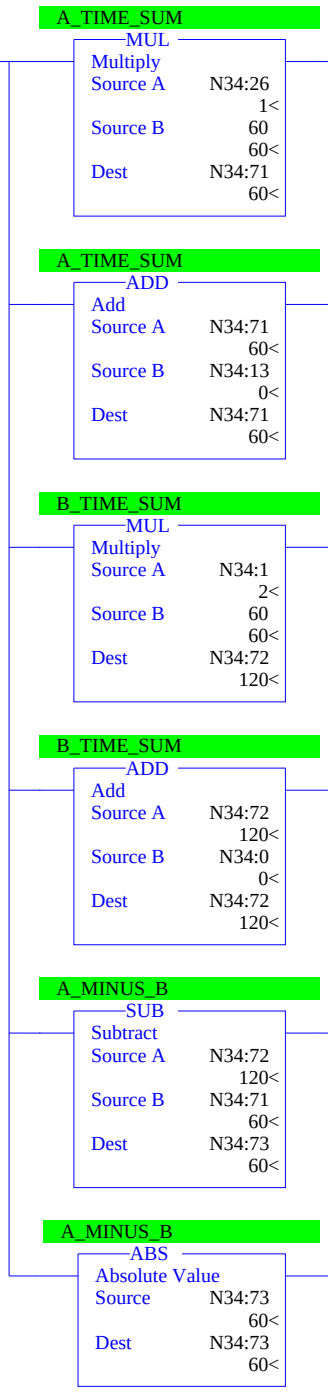

CB2 ALARM OSR OUTPUT
B22:0
4
ALARM BIT FOR CB2 TRIP
CB2
B3:0
2
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
ALARM BIT FOR CB2 TRIP
CB2
B3:0
2

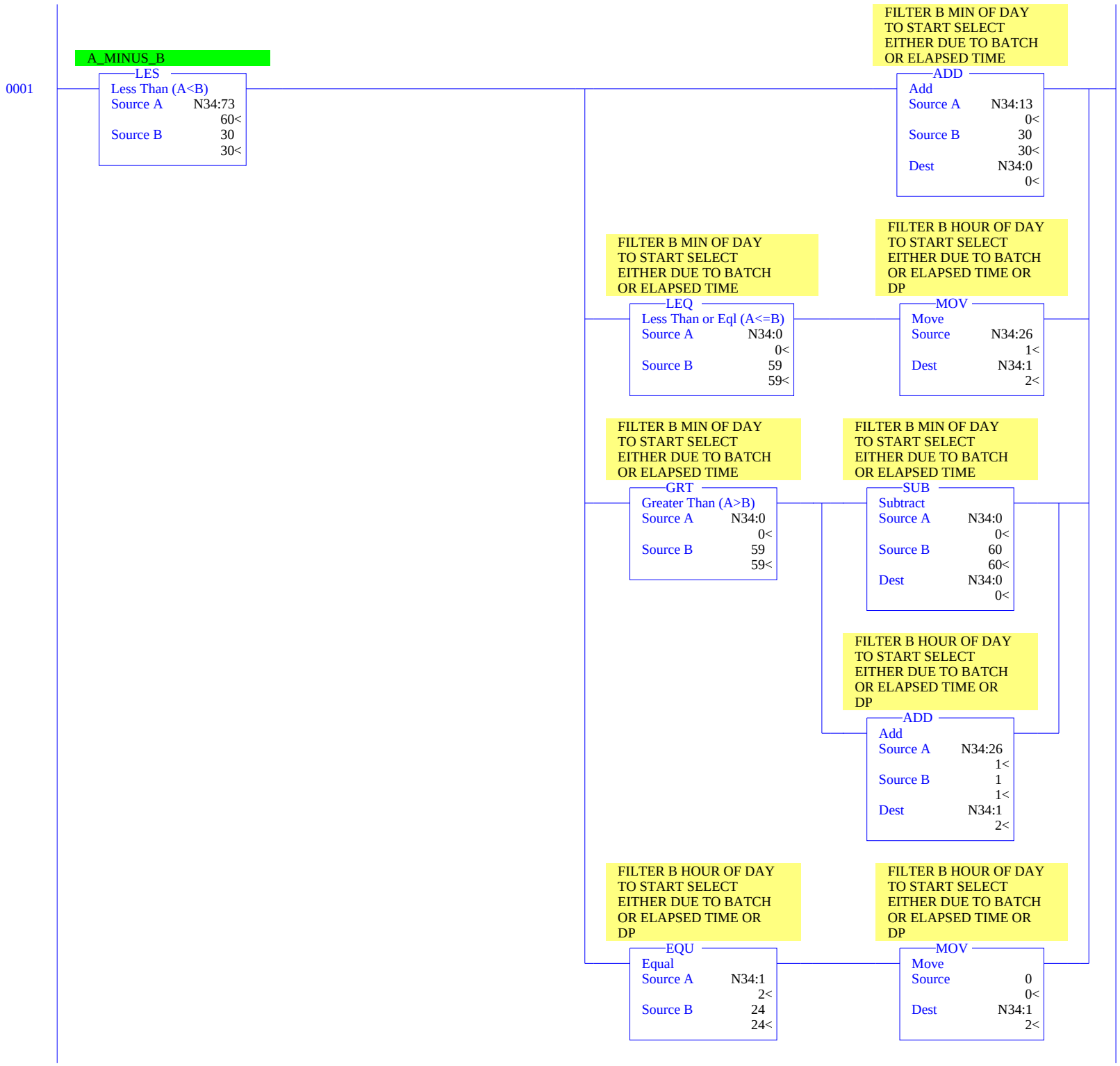

0020
UNIT A HIGH FLOW ALARM MSG
B21:1
9
POWER UP DELAY TO ALLOW TIME FOR SCREN TO BOOTUP
T4:4
DN
ENERGIZE ALARM HORN
B9:0
10
UNIT A LO FLOW ALARM MSG
B21:1
13
UNIT B HIGH FLOW ALARM MSG
B22:1
9
UNIT B LO FLOW ALARM MSG
B22:1
13
ALARM BIT FOR CB1 TRIP
CB1_ALM_BIT
B3:0
1
ALARM BIT FOR CB2 TRIP
CB2
B3:0
2
VESSEL A DELAYED BW ALARM
B3:7
15
VESSEL B DELAYED BW ALARM
B3:8
8
RECYCLE TANK HH LEVEL ALARM BIT
B3:13
8
VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
0
VESSEL A DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
11


VESSEL B DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
6
VESSEL B DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
12
0021
END

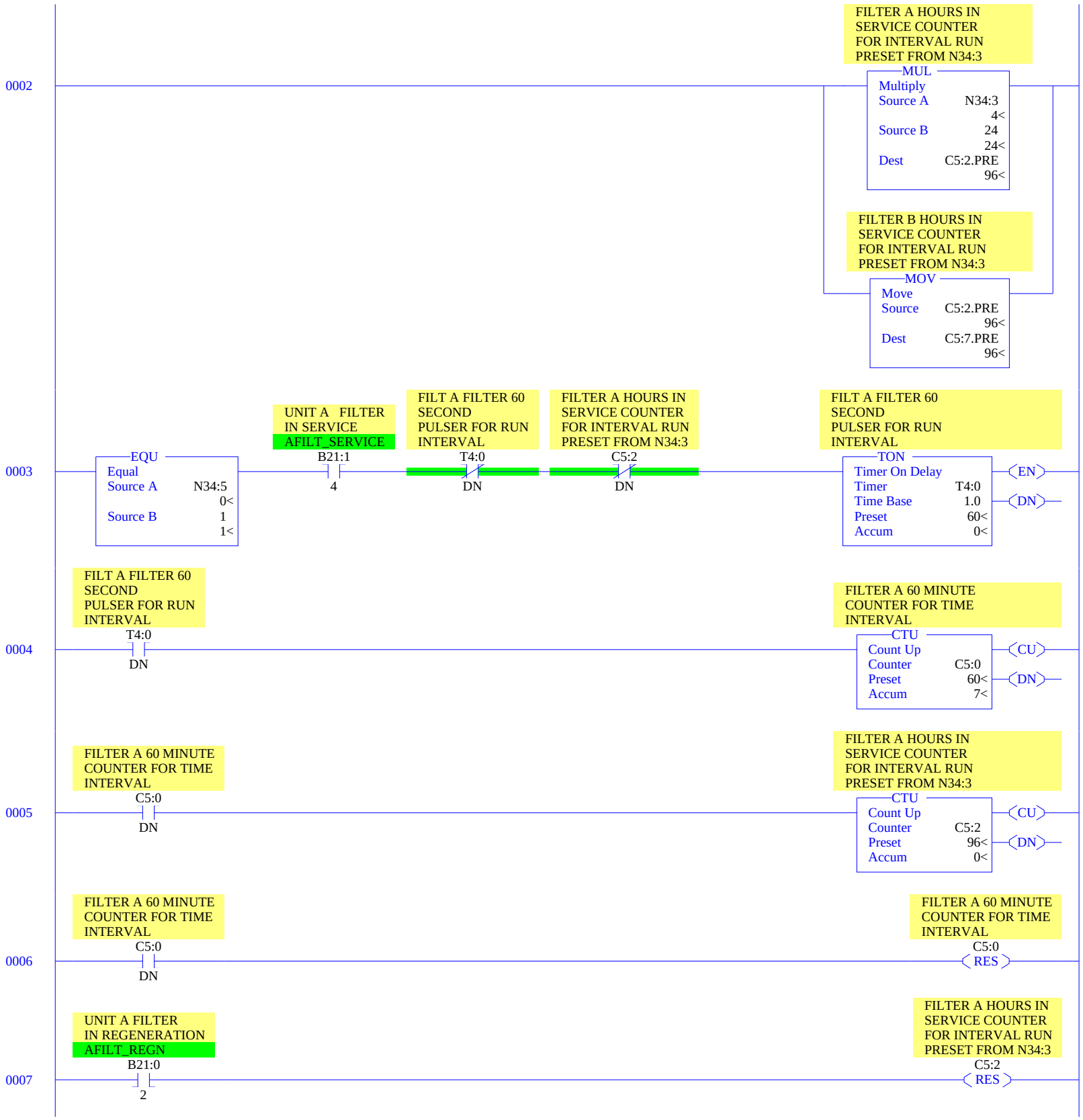
LAD 6 - TIME TRIG - TIME INTERVAL REGN INITIATION --- Total Rungs in File = 17

0000

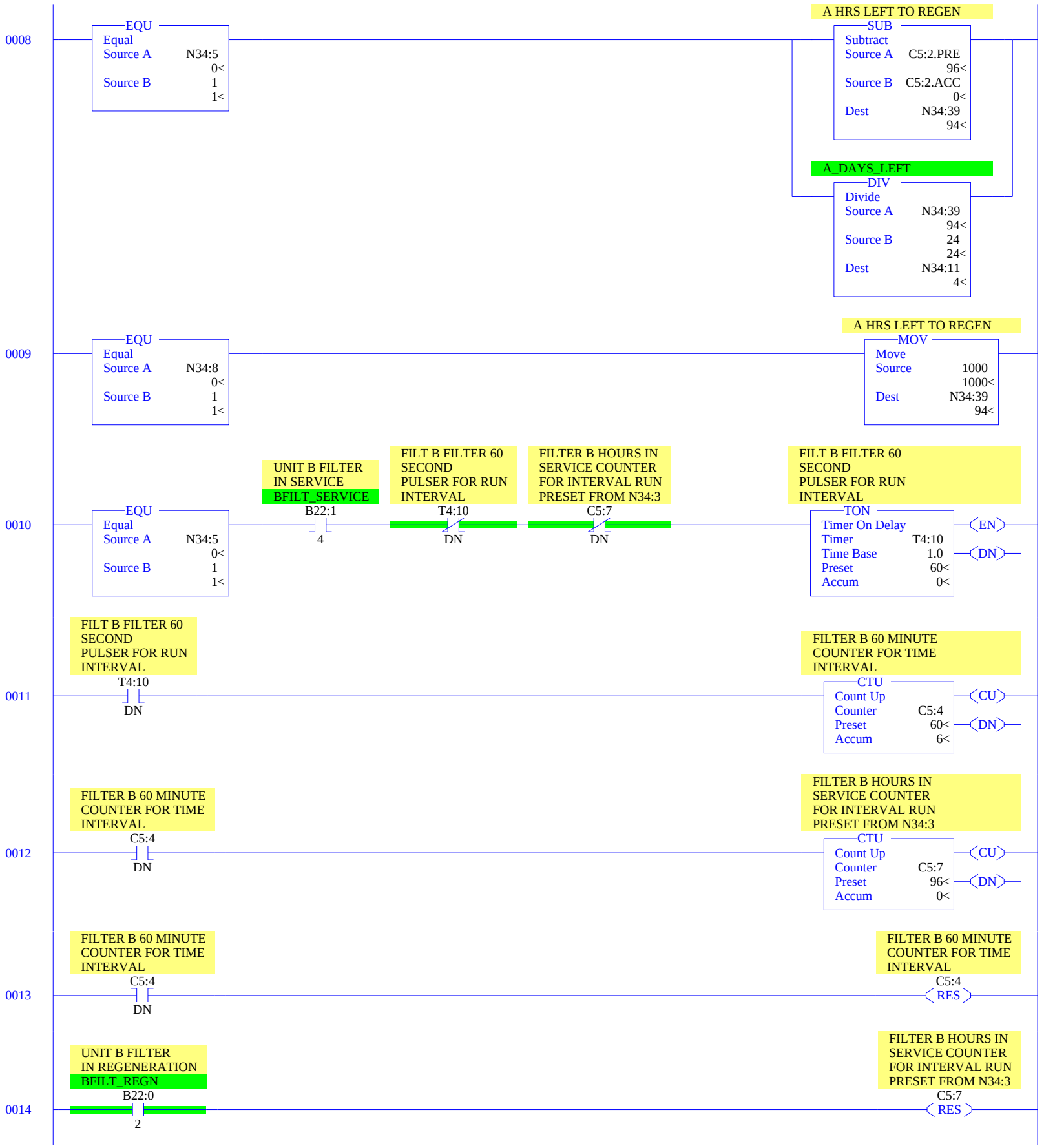



A_MINUS_B
0001
LES
Less Than (A<B)
Source A N34:73 60<
Source B 30 30<
FILTER B MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME
ADD
Add
Source A N34:13 0<
Source B 30 30<
Dest N34:0 0<
FILTER B MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME
LEQ
Less Than or Eql (A<=B)
Source A N34:0 0<
Source B 59 59<
FILTER B HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME OR DP
MOV
Move
Source N34:26 1<
Dest N34:1 2<
FILTER B MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME
GRT
Greater Than (A>B)
Source A N34:0 0<
Source B 59 59<
FILTER B MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME
SUB
Subtract
Source A N34:0 0<
Source B 60 60<
Dest N34:0 0<
FILTER B HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME OR DP
ADD
Add
Source A N34:26 1<
Source B 1 1<
Dest N34:1 2<
FILTER B HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME OR DP
EQU
Equal
Source A N34:1 2<
Source B 24 24<
FILTER B HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED TIME OR DP
MOV
Move
Source 0 0<
Dest N34:1 2<

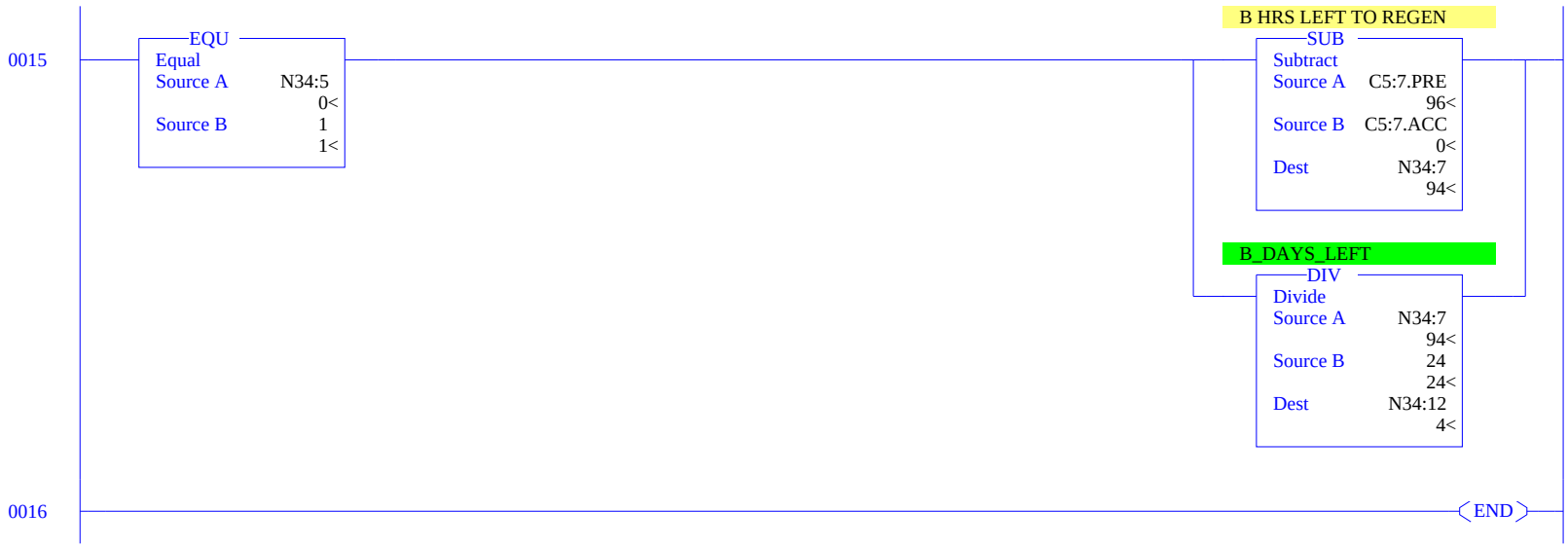
LAD 6 - TIME TRIG - TIME INTERVAL REGN INITIATION --- Total Rungs in File = 17



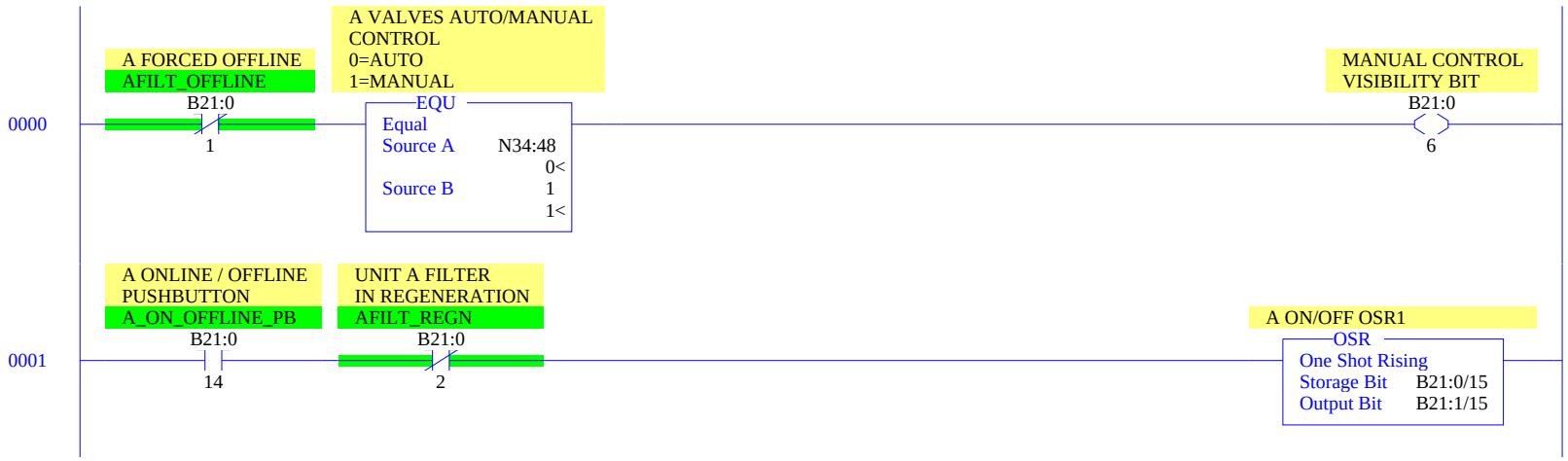
LAD 6 - TIME TRIG - TIME INTERVAL REGN INITIATION --- Total Rungs in File = 17

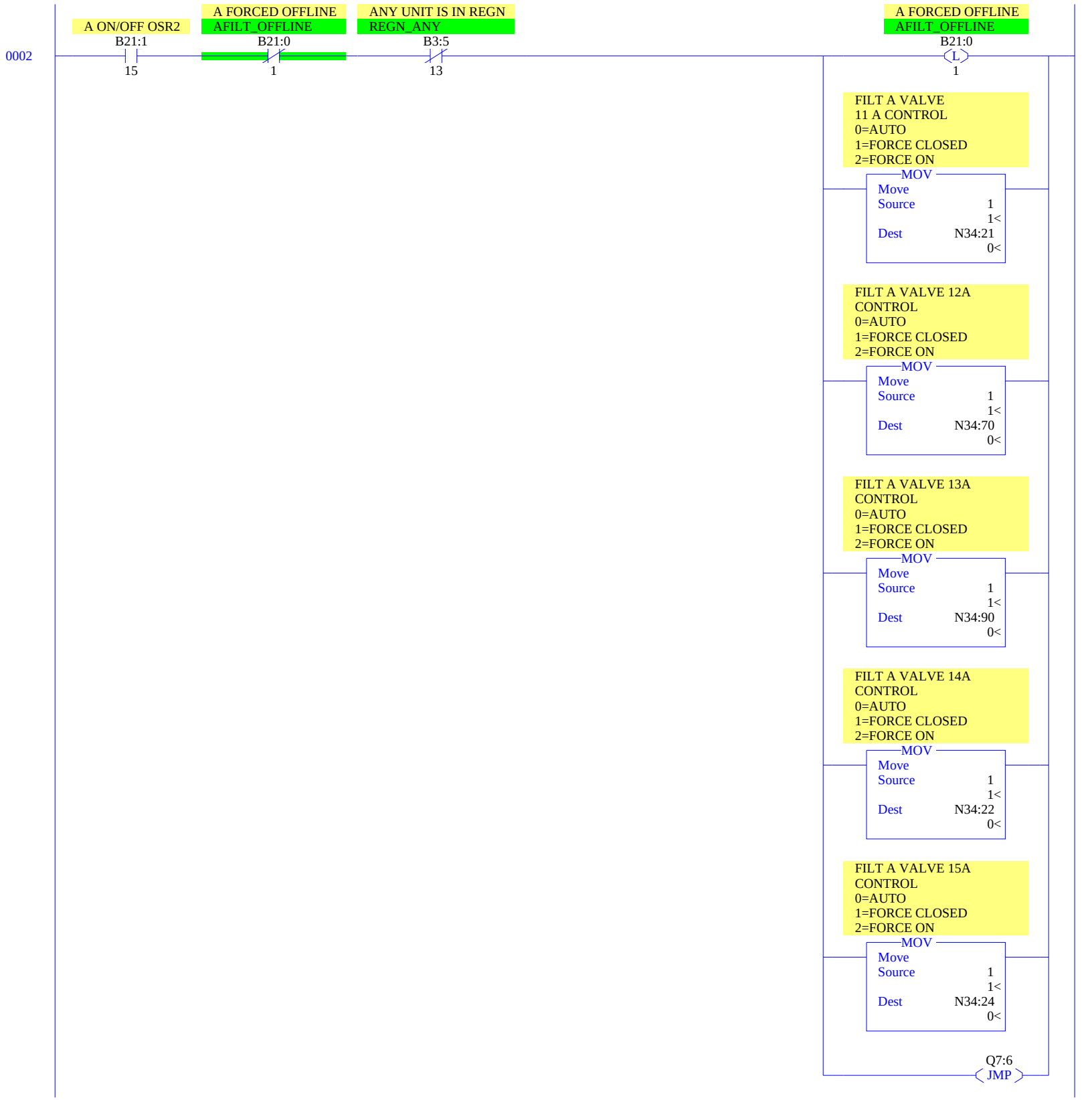


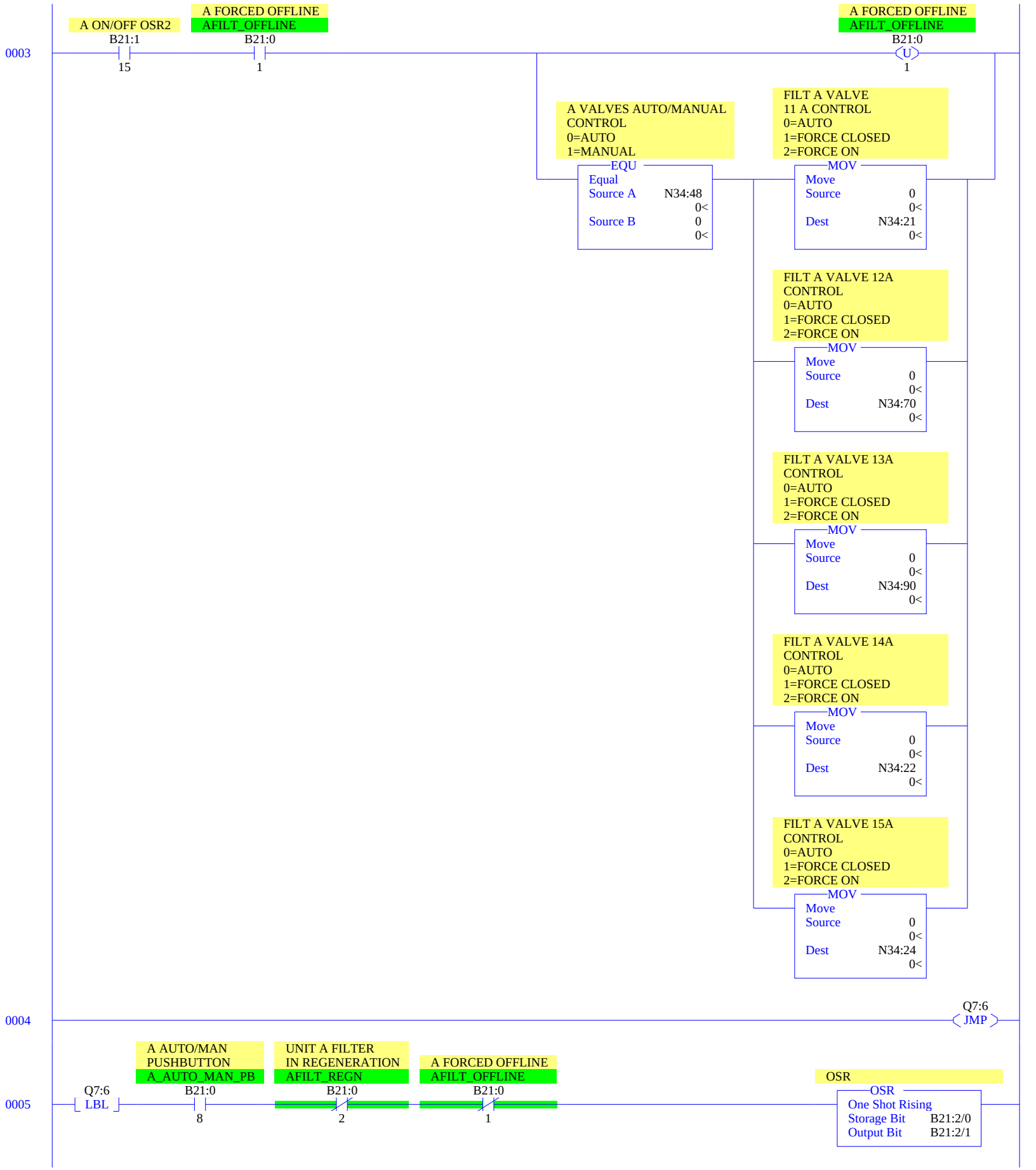
LAD 6 - TIME TRIG - TIME INTERVAL REGN INITIATION --- Total Rungs in File = 17

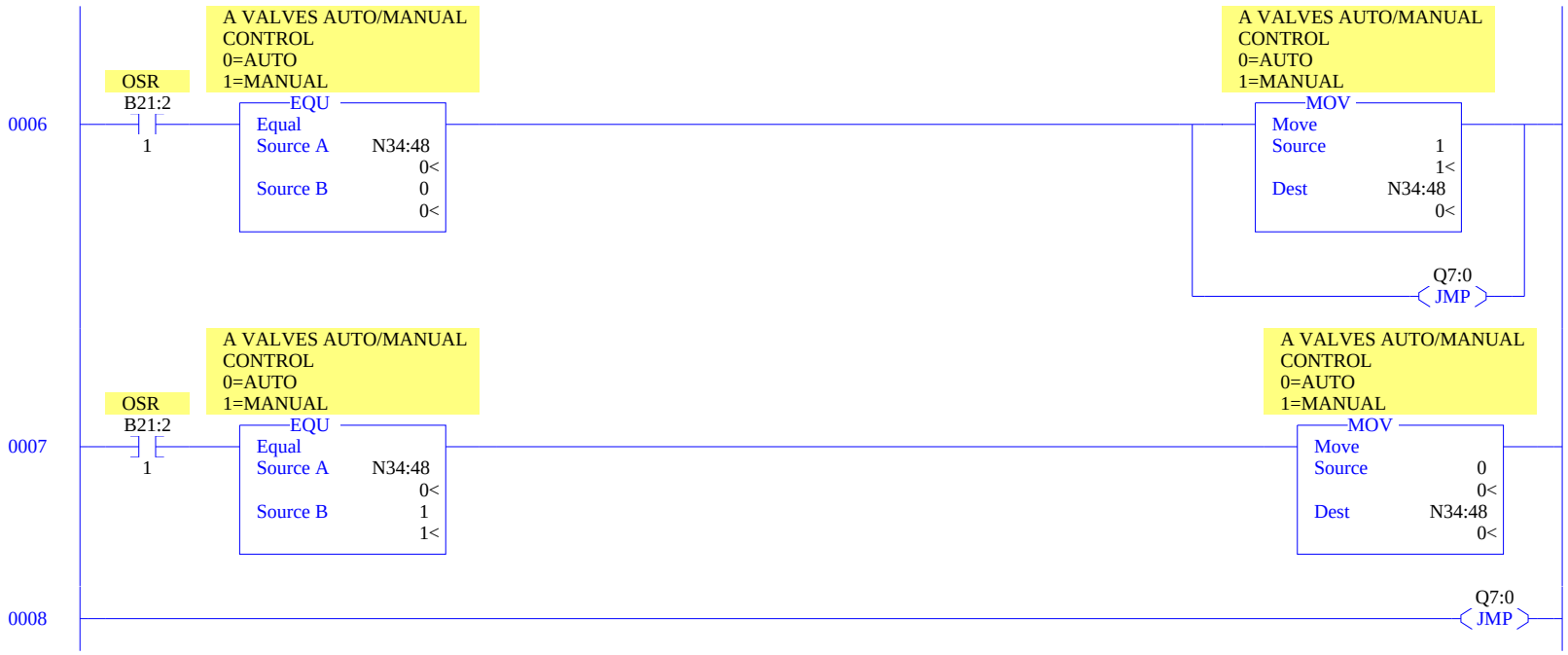


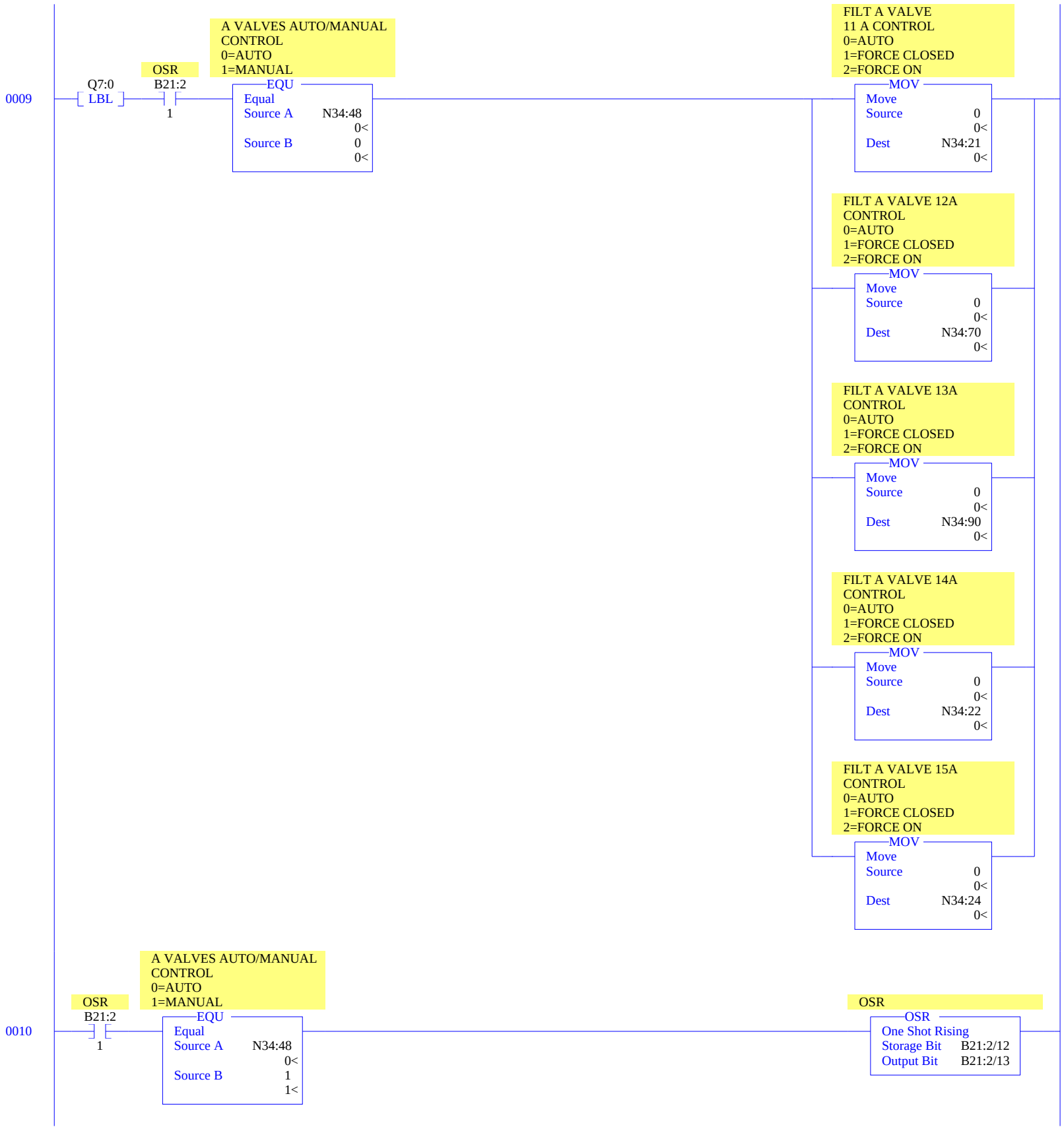
LAD 7 - A VESSEL --- Total Rungs in File = 38

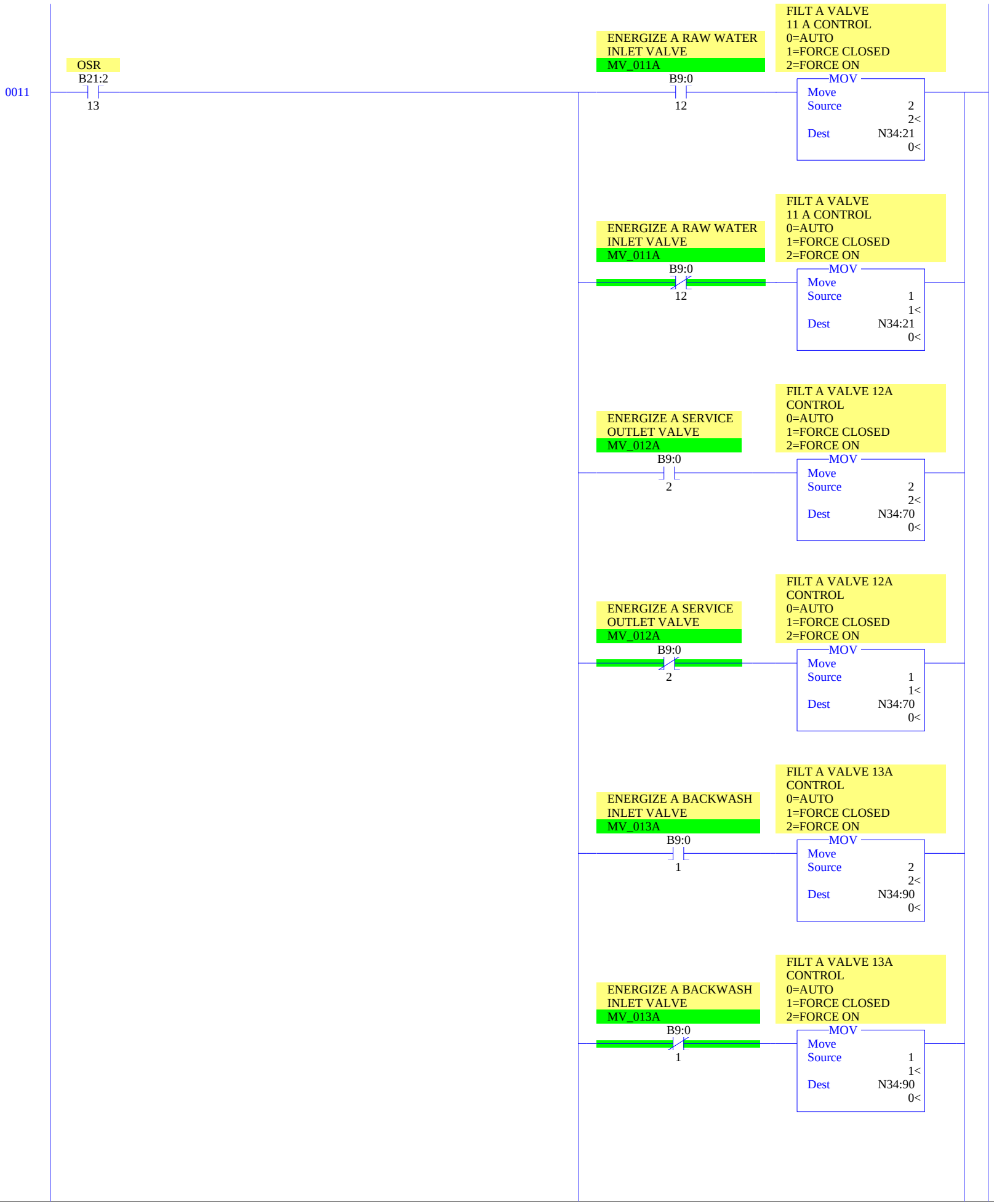


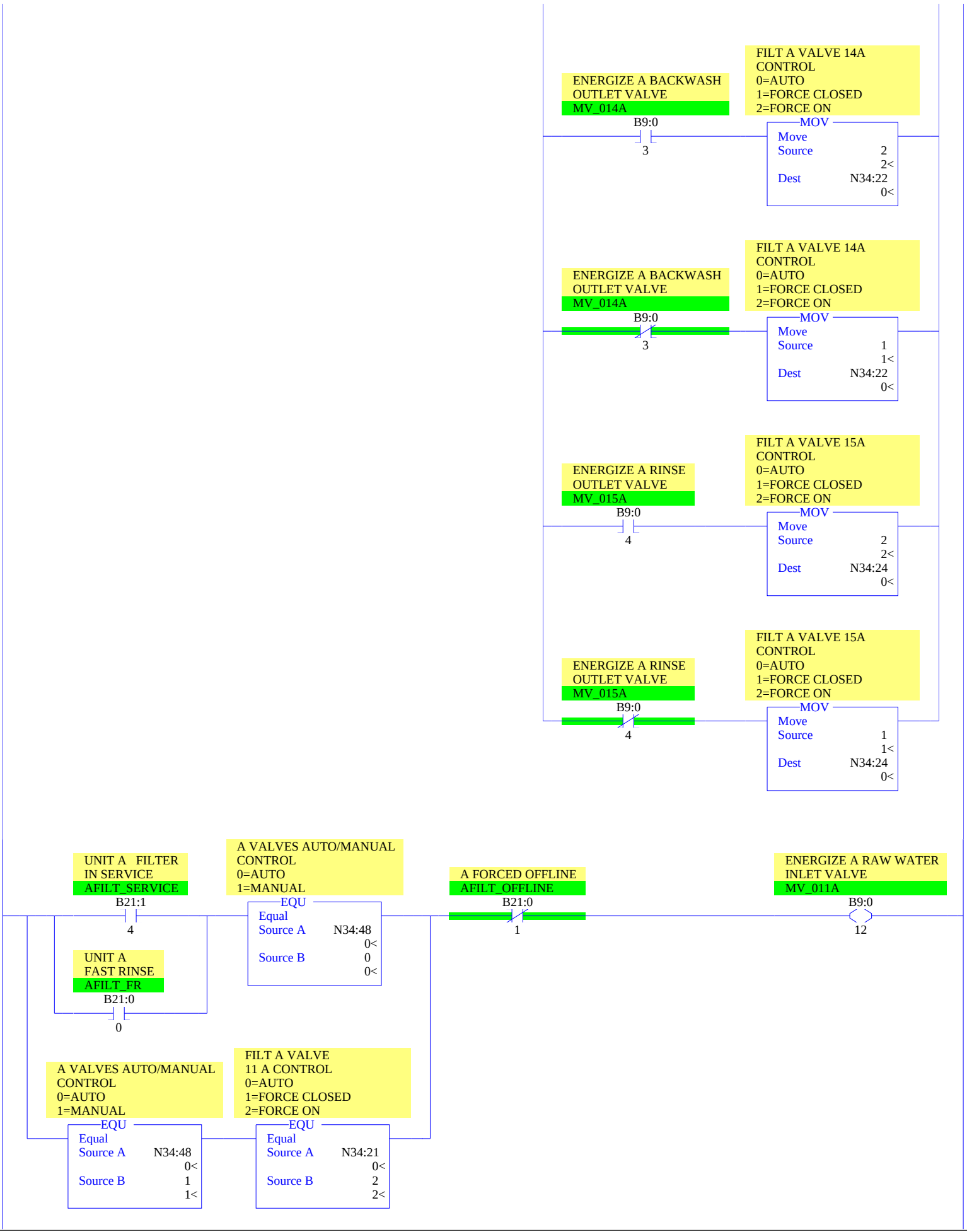

0002
A ON/OFF OSR2
B21:1
15
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ANY UNIT IS IN REGN
REGN_ANY
B3:5
13
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
L
1
FILT A VALVE
11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:21
0<
FILT A VALVE 12A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:70
0<
FILT A VALVE 13A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:90
0<
FILT A VALVE 14A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:22
0<
FILT A VALVE 15A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:24
0<
Q7:6
JMP

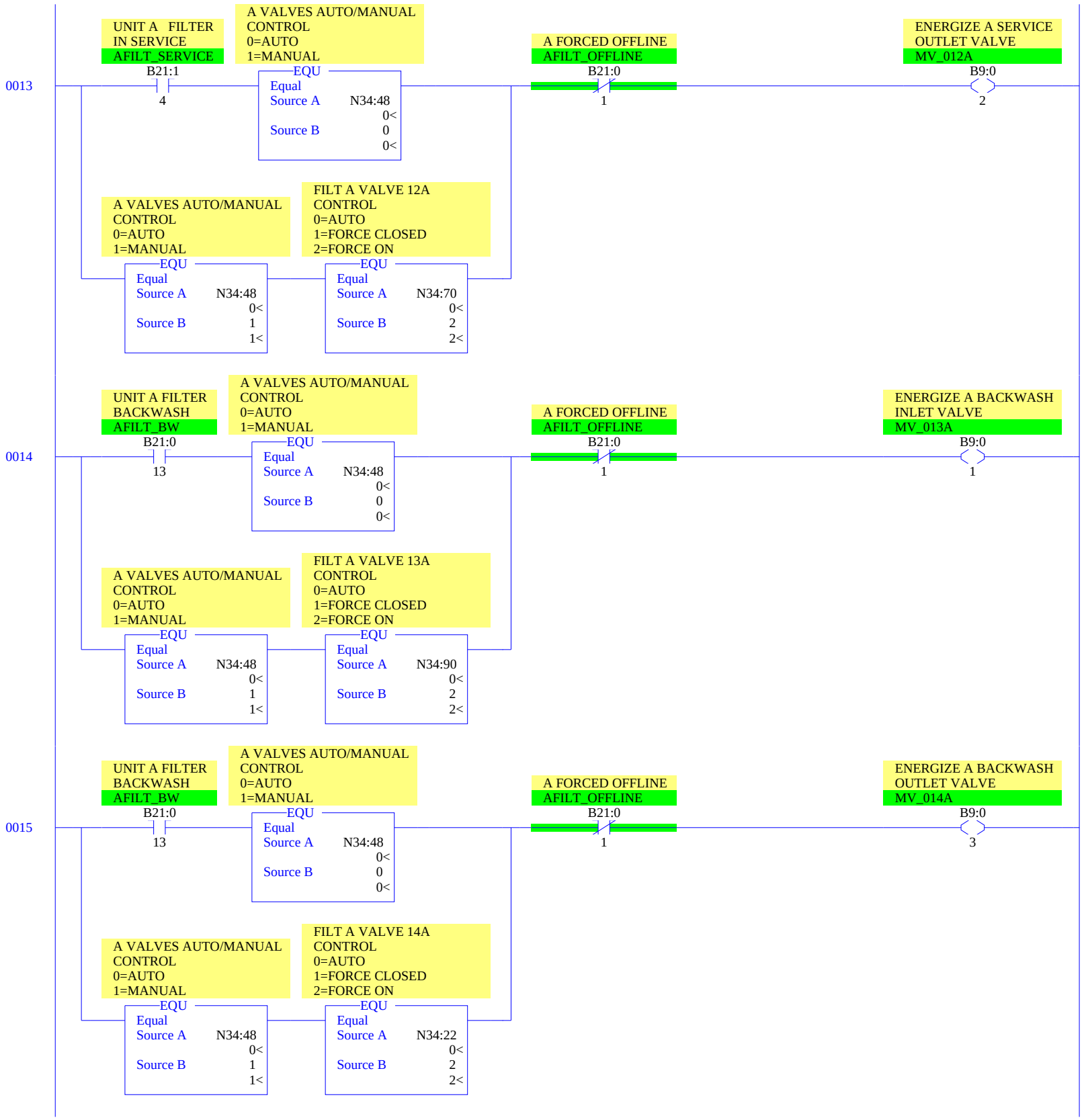

0003
A ON/OFF OSR2
B21:1
15
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
U
1
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48 0<
Source B 0 0<
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:21 0<
FILT A VALVE 12A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:70 0<
FILT A VALVE 13A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:90 0<
FILT A VALVE 14A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:22 0<
FILT A VALVE 15A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:24 0<
0004
Q7:6
JMP
0005
Q7:6
LBL
A AUTO/MAN PUSHBUTTON
A_AUTO_MAN_PB
B21:0
8
UNIT A FILTER IN REGENERATION
AFILT_REGN
B21:0
2
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
OSR
OSR
One Shot Rising
Storage Bit B21:2/0
Output Bit B21:2/1

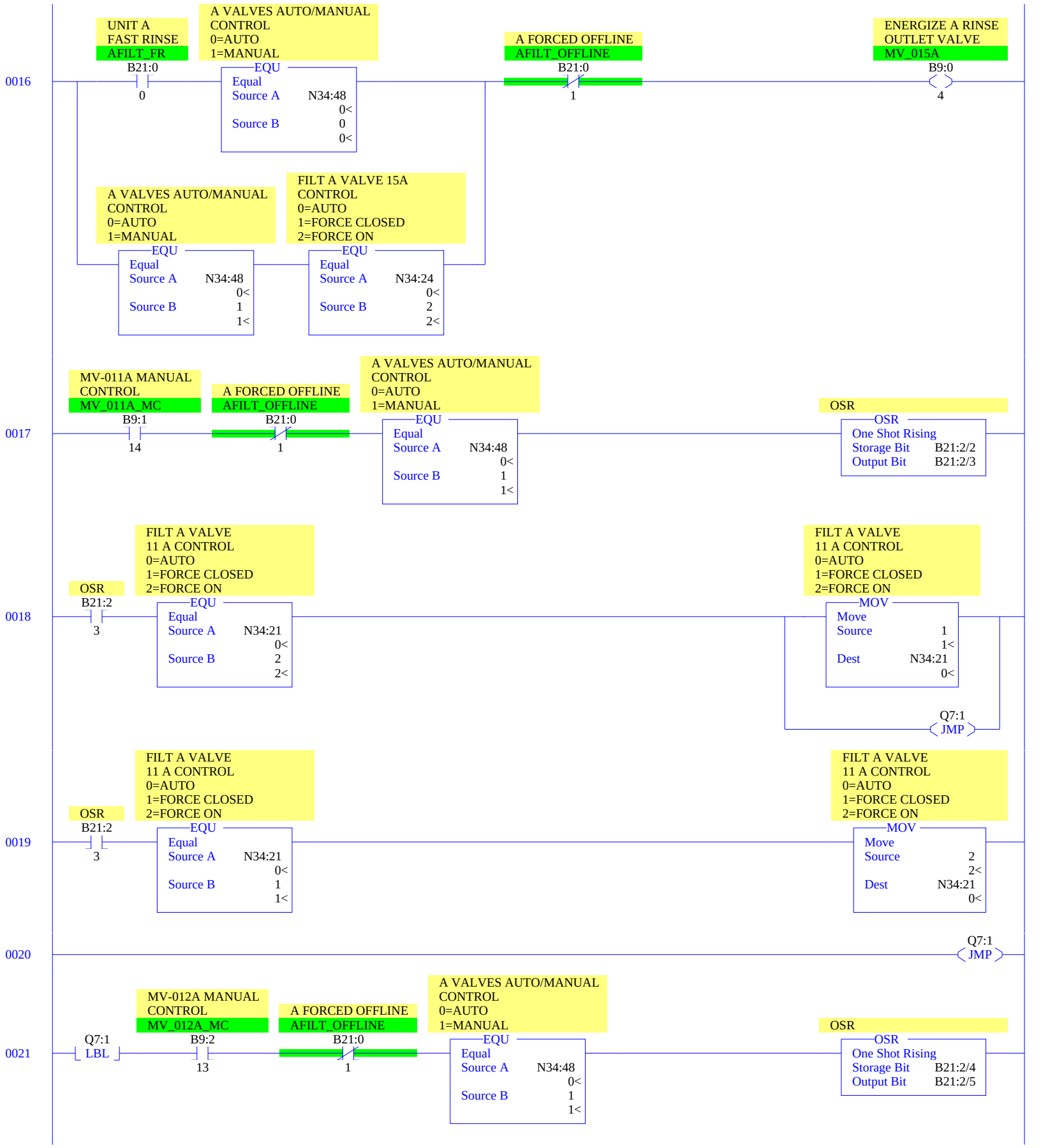

A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
OSR
B21:2
1
EQU
Equal
Source A N34:48 0<
Source B 0 0<
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
MOV
Move
Source 1 1<
Dest N34:48 0<
Q7:0
JMP
0006
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
OSR
B21:2
1
EQU
Equal
Source A N34:48 0<
Source B 1 1<
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
MOV
Move
Source 0 0<
Dest N34:48 0<
0007
Q7:0
JMP
0008

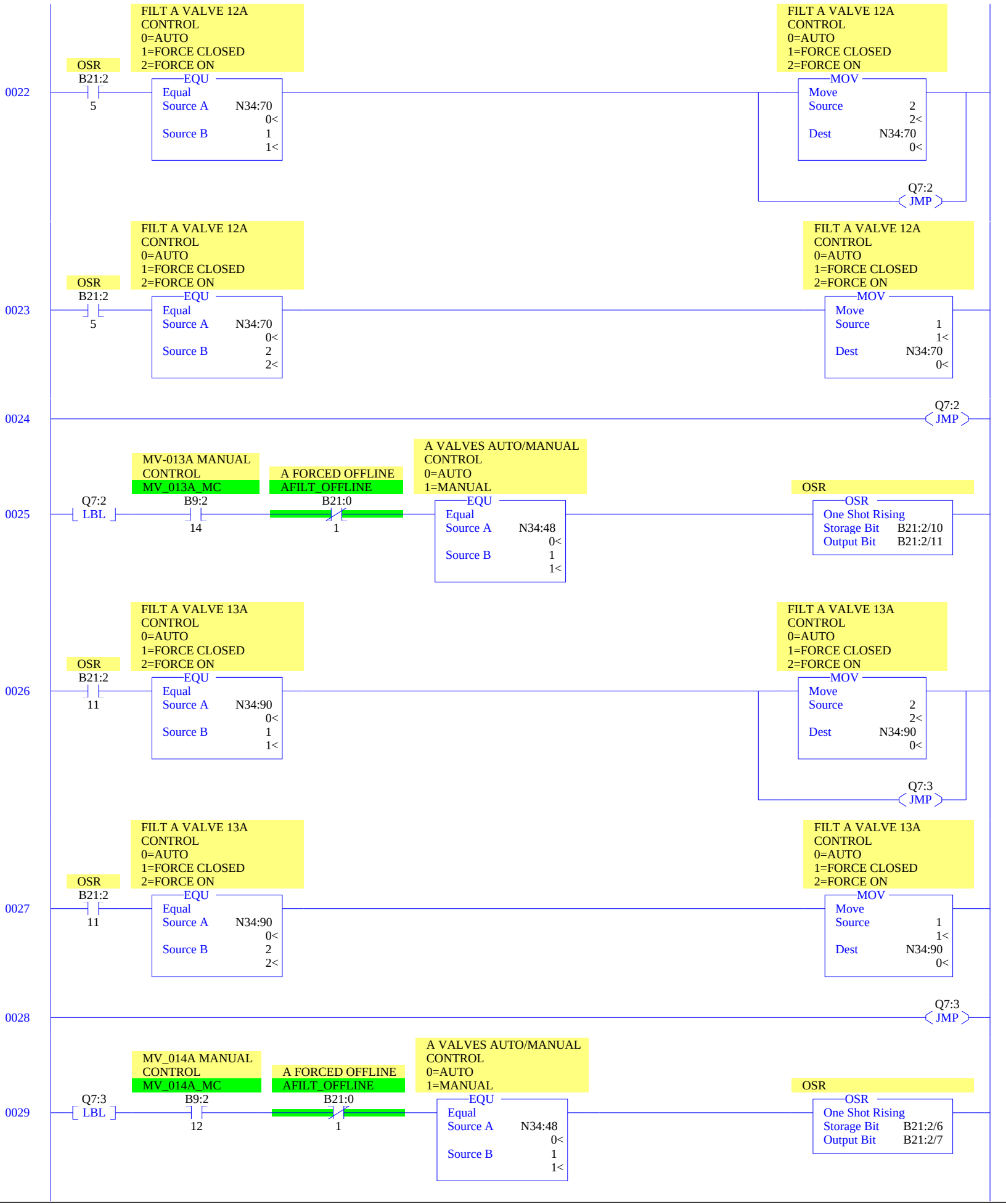

0009
Q7:0
LBL
OSR
B21:2
1
A VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 0
0<
FILT A VALVE
11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:21
0<
FILT A VALVE 12A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:70
0<
FILT A VALVE 13A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:90
0<
FILT A VALVE 14A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:22
0<
FILT A VALVE 15A
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:24
0<
0010
OSR
B21:2
1
A VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 1
1<
OSR
OSR
One Shot Rising
Storage Bit B21:2/12
Output Bit B21:2/13


0011
OSR
B21:2
13
ENERGIZE A RAW WATER INLET VALVE
MV_011A
B9:0
12
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:21
0<
ENERGIZE A RAW WATER INLET VALVE
MV_011A
B9:0
12
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:21
0<
ENERGIZE A SERVICE OUTLET VALVE
MV_012A
B9:0
2
FILT A VALVE 12A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:70
0<
ENERGIZE A SERVICE OUTLET VALVE
MV_012A
B9:0
2
FILT A VALVE 12A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:70
0<
ENERGIZE A BACKWASH INLET VALVE
MV_013A
B9:0
1
FILT A VALVE 13A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:90
0<
ENERGIZE A BACKWASH INLET VALVE
MV_013A
B9:0
1
FILT A VALVE 13A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:90
0<

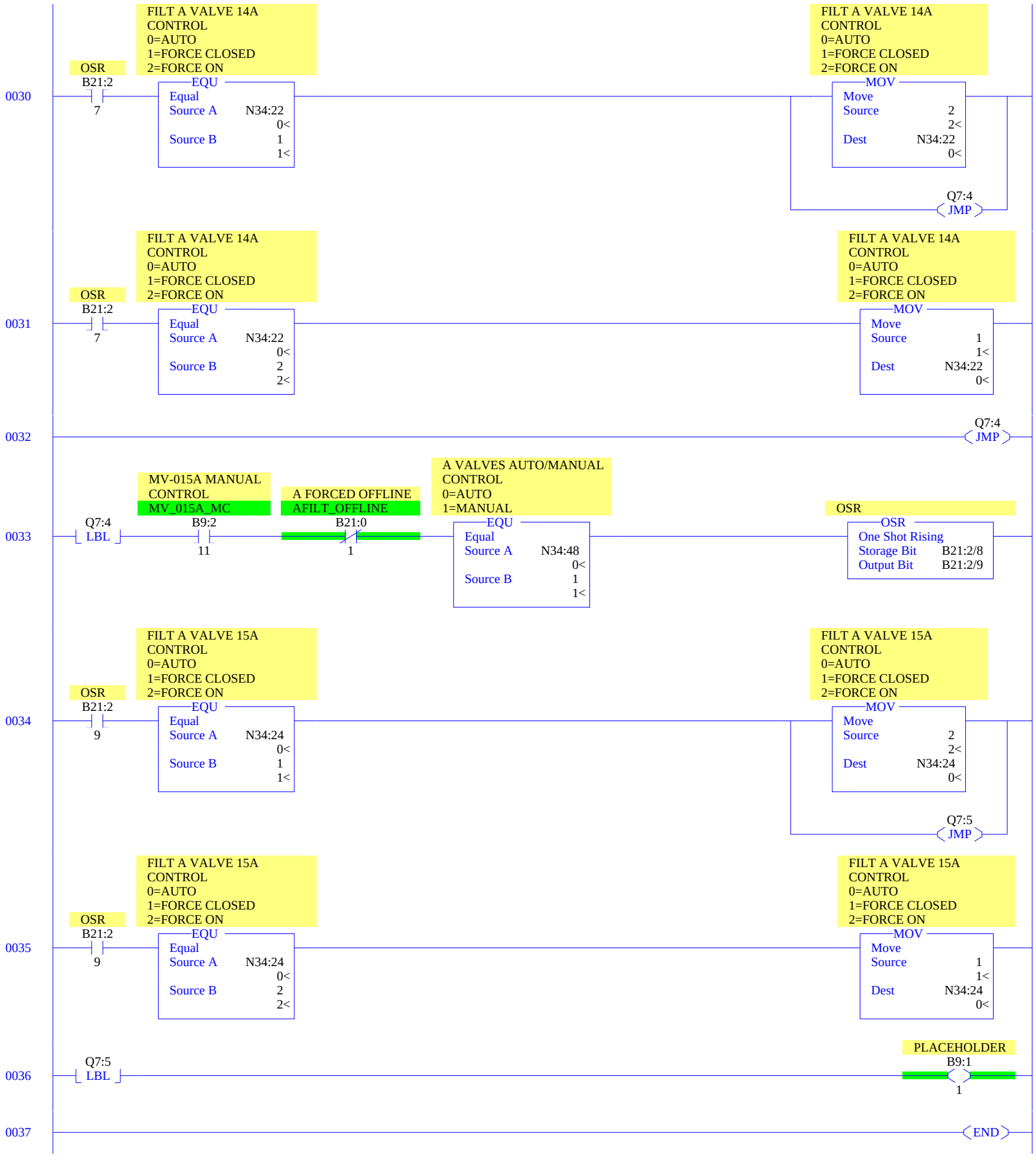

ENERGIZE A BACKWASH OUTLET VALVE
MV_014A
B9:0
3
FILT A VALVE 14A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:22
0<
ENERGIZE A BACKWASH OUTLET VALVE
MV_014A
B9:0
3
FILT A VALVE 14A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:22
0<
ENERGIZE A RINSE OUTLET VALVE
MV_015A
B9:0
4
FILT A VALVE 15A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:24
0<
ENERGIZE A RINSE OUTLET VALVE
MV_015A
B9:0
4
FILT A VALVE 15A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:24
0<
0012
UNIT A FILTER IN SERVICE
AFILT_SERVICE
B21:1
4
UNIT A FAST RINSE
AFILT_FR
B21:0
0
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 0
0<
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ENERGIZE A RAW WATER INLET VALVE
MV_011A
B9:0
12
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 1
1<
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:21
0<
Source B 2
2<


0013
UNIT A FILTER IN SERVICE
AFILT_SERVICE
B21:1
4
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 0
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ENERGIZE A SERVICE OUTLET VALVE
MV_012A
B9:0
2
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 1
FILT A VALVE 12A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:70
Source B 2
0014
UNIT A FILTER BACKWASH
AFILT_BW
B21:0
13
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 0
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ENERGIZE A BACKWASH INLET VALVE
MV_013A
B9:0
1
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 1
FILT A VALVE 13A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:90
Source B 2
0015
UNIT A FILTER BACKWASH
AFILT_BW
B21:0
13
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 0
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ENERGIZE A BACKWASH OUTLET VALVE
MV_014A
B9:0
3
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
Source B 1
FILT A VALVE 14A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:22
Source B 2


0016
UNIT A FAST RINSE
AFILT_FR
B21:0
0
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48 0<
Source B 0 0<
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
ENERGIZE A RINSE OUTLET VALVE
MV_015A
B9:0
4
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48 0<
Source B 1 1<
FILT A VALVE 15A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:24 0<
Source B 2 2<
0017
MV-011A MANUAL CONTROL
MV_011A_MC
B9:1
14
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B21:2/2
Output Bit B21:2/3
0018
OSR
B21:2
3
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:21 0<
Source B 2 2<
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:21 0<
Q7:1
JMP
0019
OSR
B21:2
3
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:21 0<
Source B 1 1<
FILT A VALVE 11 A CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:21 0<
0020
Q7:1
JMP
0021
Q7:1
LBL
MV-012A MANUAL CONTROL
MV_012A_MC
B9:2
13
A FORCED OFFLINE
AFILT_OFFLINE
B21:0
1
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B21:2/4
Output Bit B21:2/5


0022
FILT A VALVE 12A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B21:2
5
EQU
Equal
Source A N34:70 0<
Source B 1 1<
FILT A VALVE 12A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:70 0<
Q7:2
JMP
0023
FILT A VALVE 12A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B21:2
5
EQU
Equal
Source A N34:70 0<
Source B 2 2<
FILT A VALVE 12A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:70 0<
0024
Q7:2
JMP
0025
MV-013A MANUAL CONTROL
MV_013A_MC
A FORCED OFFLINE
AFILT_OFFLINE
A VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
Q7:2
LBL
B9:2
14
B21:0
1
EQU
Equal
Source A N34:48 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B21:2/10
Output Bit B21:2/11
0026
FILT A VALVE 13A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B21:2
11
EQU
Equal
Source A N34:90 0<
Source B 1 1<
FILT A VALVE 13A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:90 0<
Q7:3
JMP
0027
FILT A VALVE 13A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B21:2
11
EQU
Equal
Source A N34:90 0<
Source B 2 2<
FILT A VALVE 13A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:90 0<
0028
Q7:3
JMP
0029
MV_014A MANUAL CONTROL
MV_014A_MC
A FORCED OFFLINE
AFILT_OFFLINE
A VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
Q7:3
LBL
B9:2
12
B21:0
1
EQU
Equal
Source A N34:48 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B21:2/6
Output Bit B21:2/7

LAD 7 - A VESSEL --- Total Rungs in File = 38



LAD 8 - ANALOG IN --- Total Rungs in File = 6

0000

VESSEL A INLET PRESSURE SCALED VALUE

SCP
Scale w/Parameters

Input	I:2.0
	6266<
Input Min.	6240.0
	6240.0<
Input Max.	32760.0
	32760.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	145.0
	145.0<
Output	F8:119
	0.1312217<

0001

VESSEL A OUTLET PRESSURE SCALED VALUE

SCP
Scale w/Parameters

Input	I:2.1
	5<
Input Min.	6240.0
	6240.0<
Input Max.	32760.0
	32760.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	145.0
	145.0<
Output	F8:120
	-34.09031<

0002

VESSEL B INLET PRESSURE SCALED VALUE

SCP
Scale w/Parameters

Input	I:2.2
	7<
Input Min.	6240.0
	6240.0<
Input Max.	32760.0
	32760.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	145.0
	145.0<
Output	F8:5
	-34.07938<

0003

VESSEL B OUTLET PRESSURE SCALED VALUE

SCP
Scale w/Parameters

Input	I:2.3
	6<
Input Min.	6240.0
	6240.0<
Input Max.	32760.0
	32760.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	145.0
	145.0<
Output	F8:8
	-34.09031<

LAD 8 - ANALOG IN --- Total Rungs in File = 6

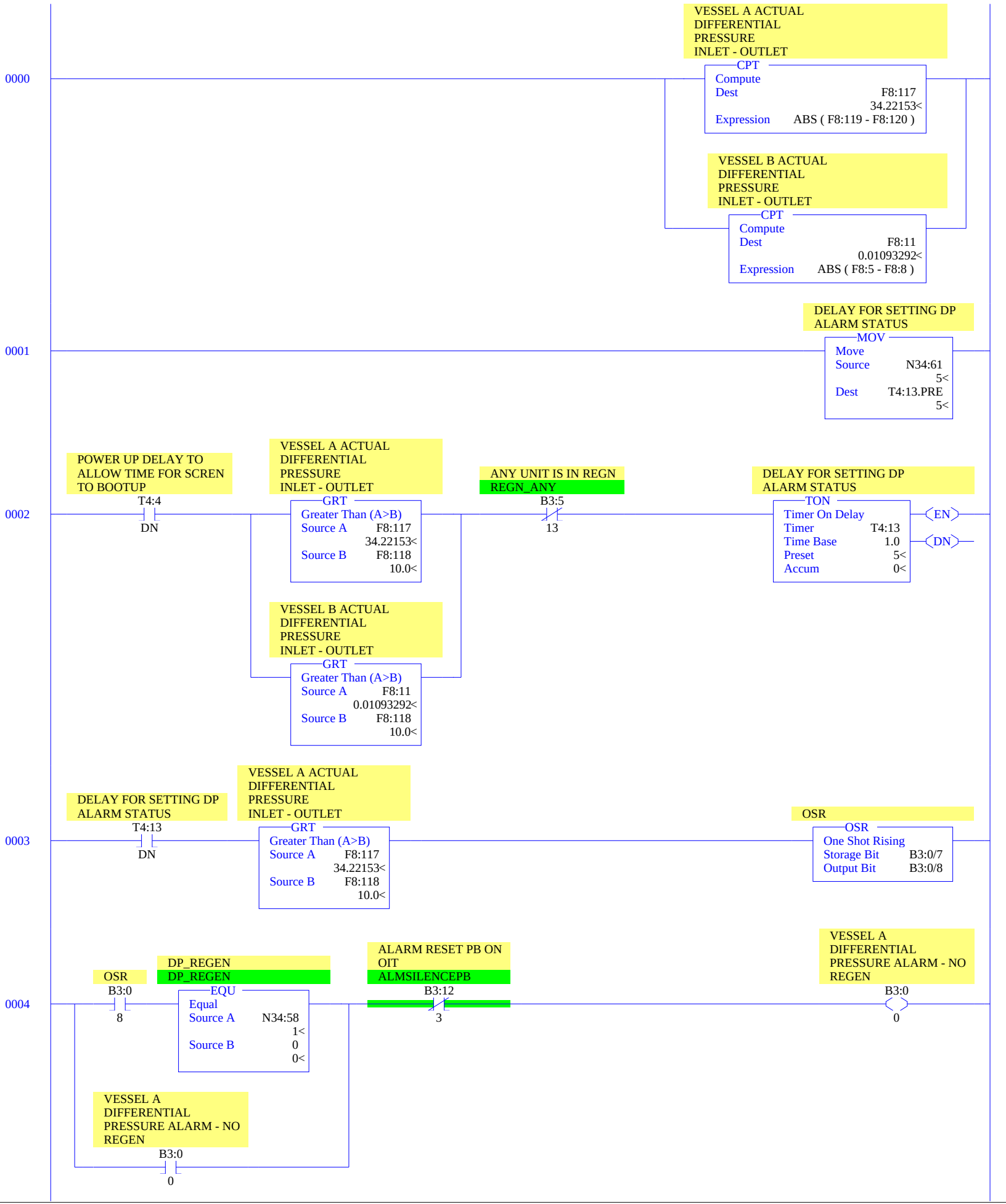
BACKWASH
RECYCLE/WASTE
TANK LEVEL DISPLAY

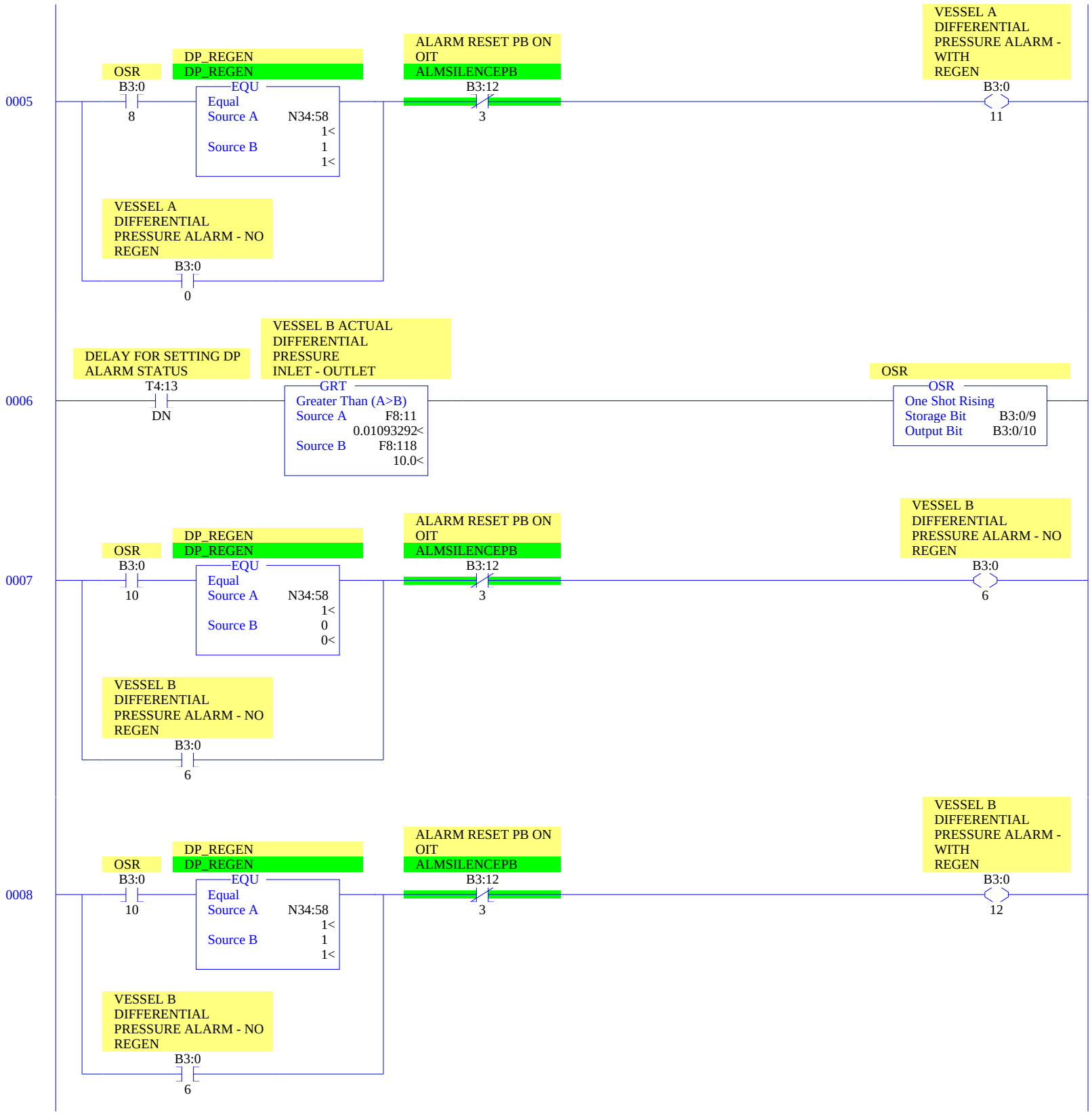
0004

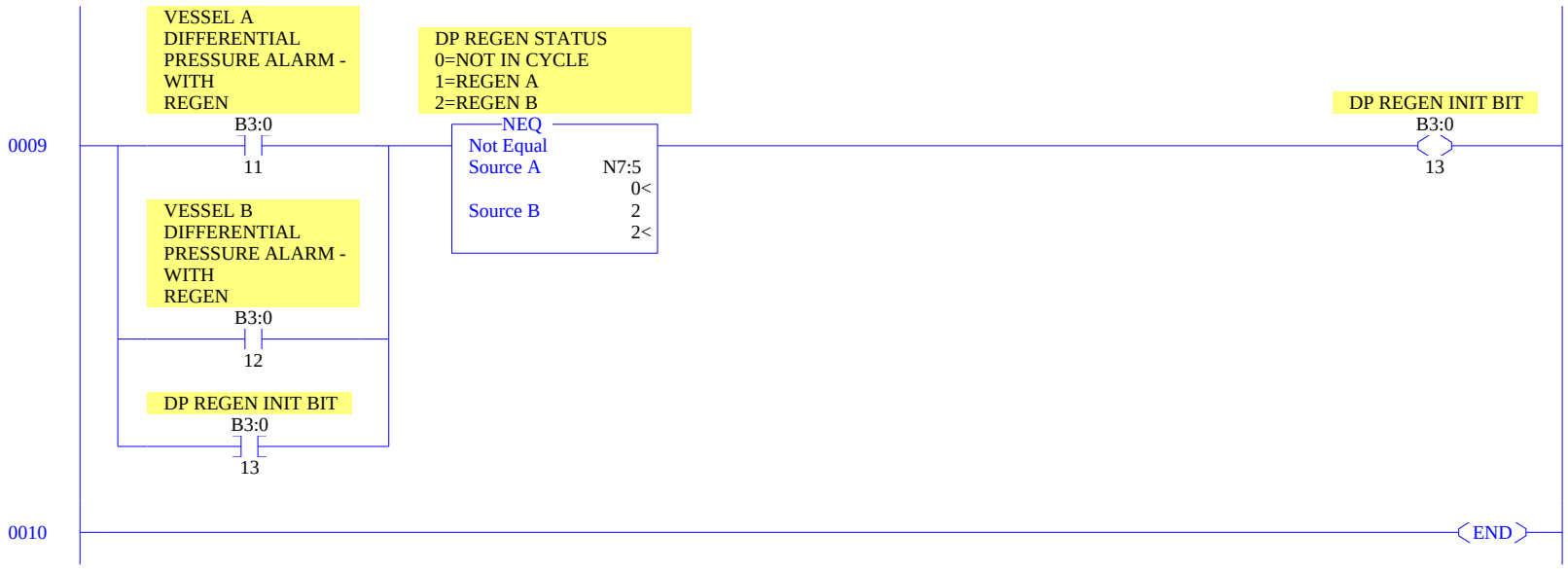
SCP
Scale w/Parameters
Input I:3.0
6<
Input Min. 6240.0
6240.0<
Input Max. 32760.0
32760.0<
Scaled Min. N34:63
0<
Scaled Max. N34:62
100<
Output F8:82
-23.50302<

0005

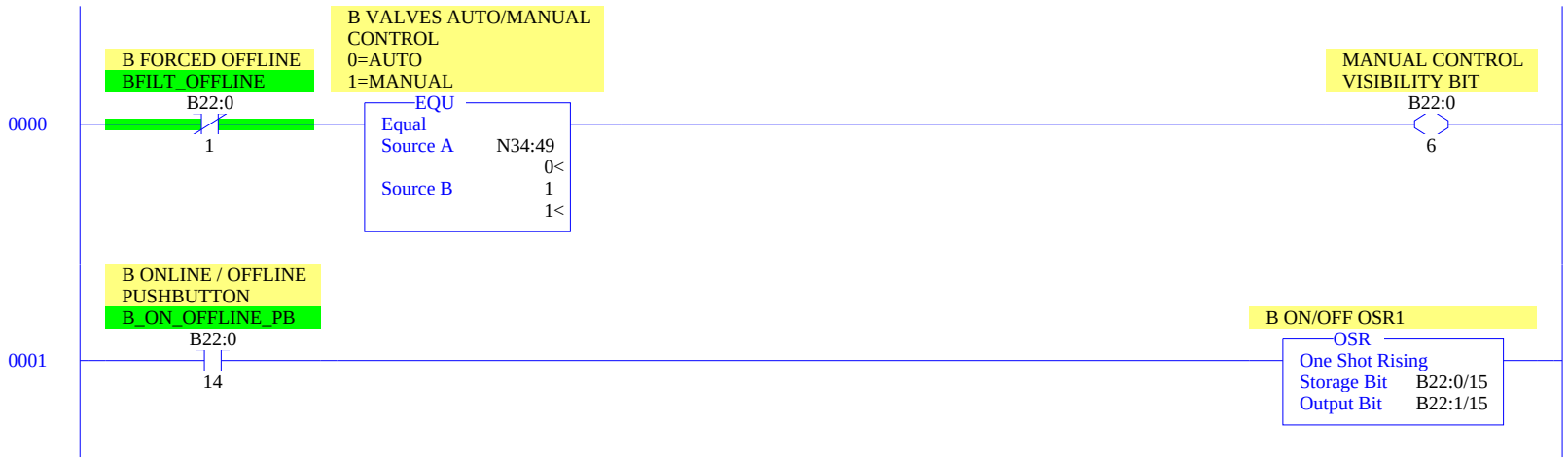
END

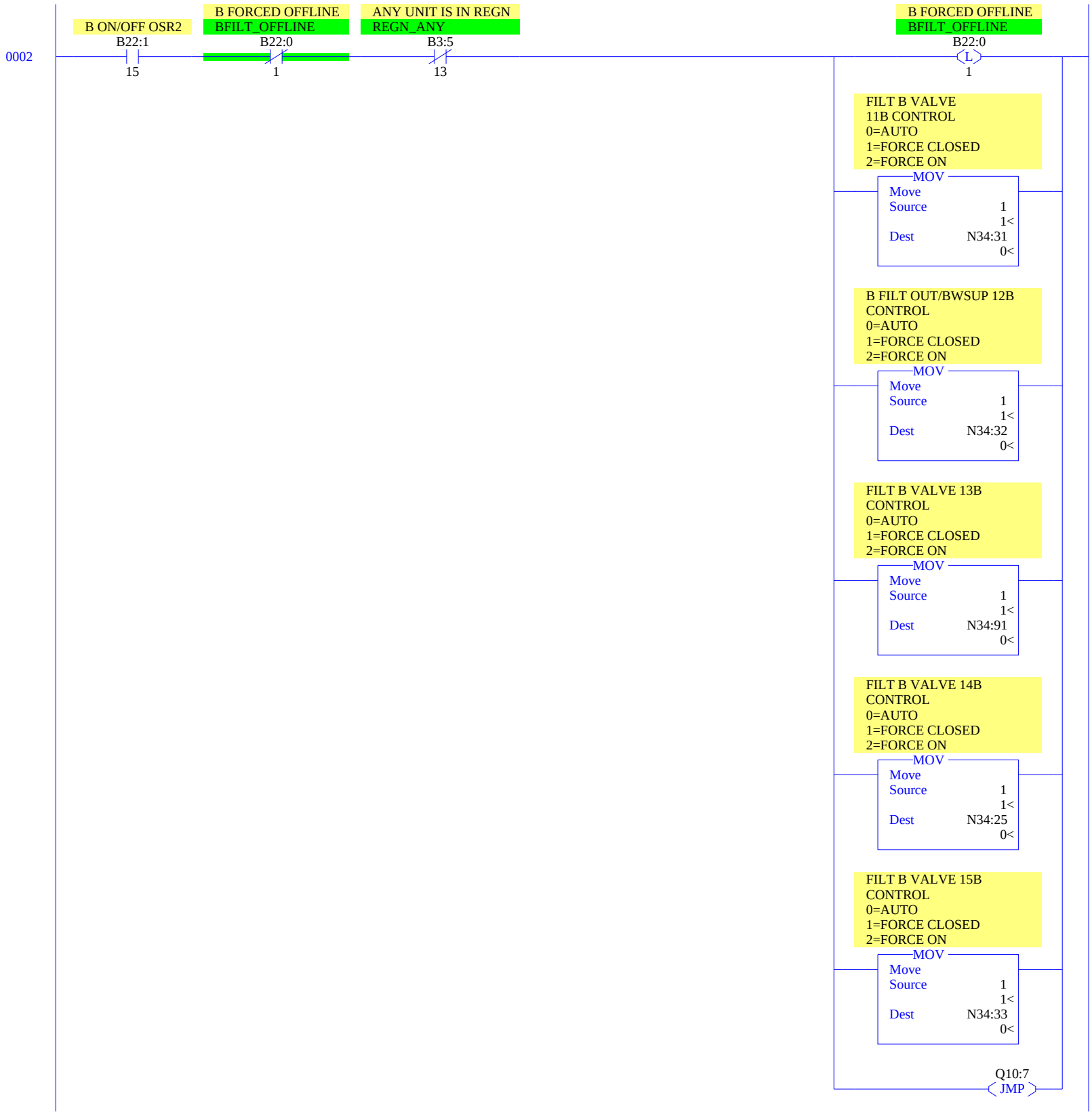

VESSEL A ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
0000
CPT
Compute
Dest F8:117
34.22153<
Expression ABS (F8:119 - F8:120)
VESSEL B ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
CPT
Compute
Dest F8:11
0.01093292<
Expression ABS (F8:5 - F8:8)
DELAY FOR SETTING DP ALARM STATUS
0001
MOV
Move
Source N34:61
5<
Dest T4:13.PRE
5<
POWER UP DELAY TO ALLOW TIME FOR SCREN TO BOOTUP
T4:4
DN
VESSEL A ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
0002
GRT
Greater Than (A>B)
Source A F8:117
34.22153<
Source B F8:118
10.0<
ANY UNIT IS IN REGN
REGN_ANY
B3:5
13
DELAY FOR SETTING DP ALARM STATUS
TON
Timer On Delay
Timer T4:13
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
VESSEL B ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
GRT
Greater Than (A>B)
Source A F8:11
0.01093292<
Source B F8:118
10.0<
DELAY FOR SETTING DP ALARM STATUS
T4:13
DN
VESSEL A ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
0003
GRT
Greater Than (A>B)
Source A F8:117
34.22153<
Source B F8:118
10.0<
OSR
OSR
One Shot Rising
Storage Bit B3:0/7
Output Bit B3:0/8
OSR
B3:0
8
DP_REGEN
DP_REGEN
EQU
Equal
Source A N34:58
1<
Source B 0
0<
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
0
0004
VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
0

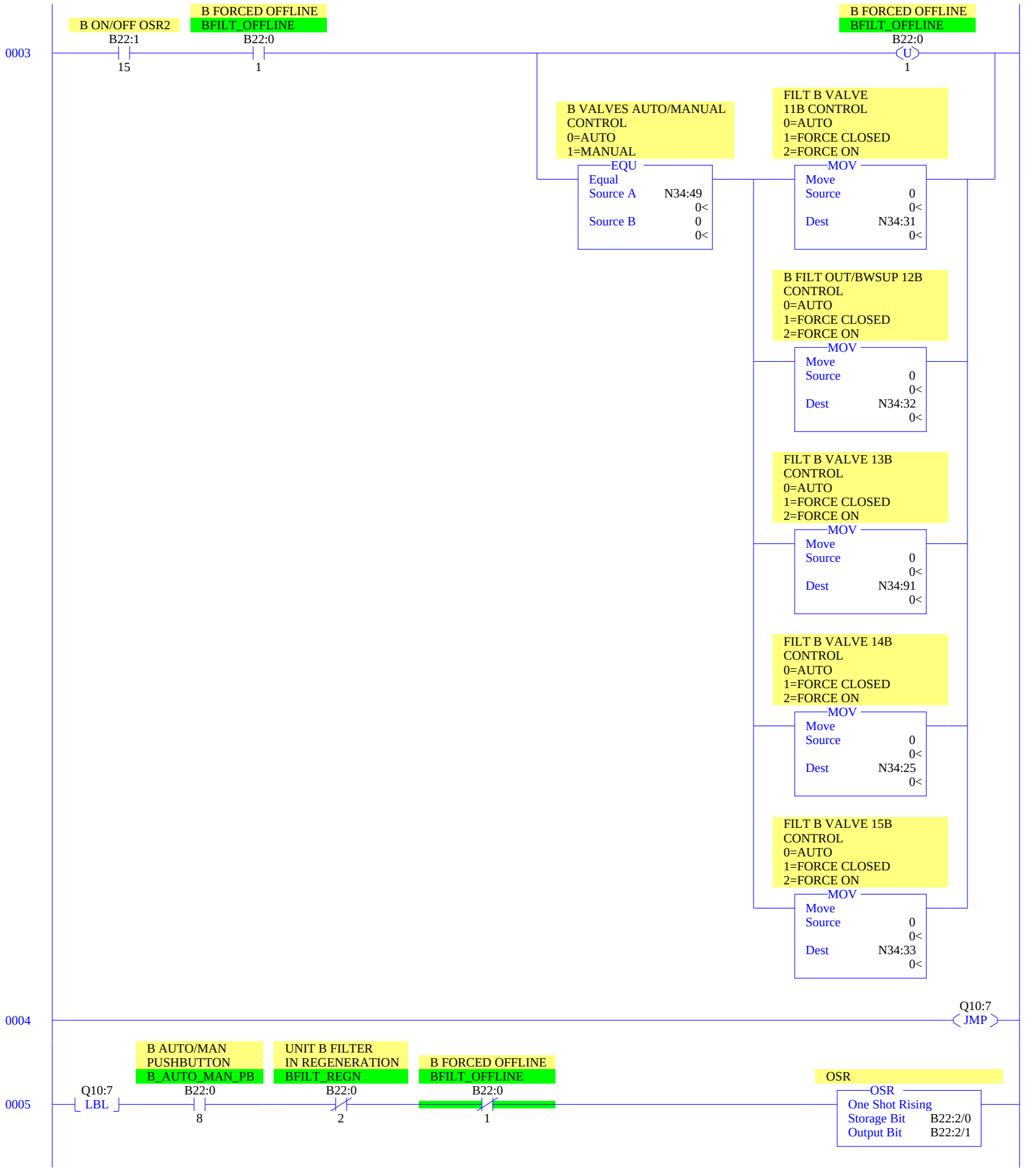

0005
OSR
B3:0
8
DP_REGEN
DP_REGEN
EQU
Equal
Source A N34:58
1<
Source B 1
1<
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
VESSEL A DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
11
VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
0
0006
DELAY FOR SETTING DP ALARM STATUS
T4:13
DN
VESSEL B ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
GRT
Greater Than (A>B)
Source A F8:11
0.01093292<
Source B F8:118
10.0<
OSR
OSR
One Shot Rising
Storage Bit B3:0/9
Output Bit B3:0/10
0007
OSR
B3:0
10
DP_REGEN
DP_REGEN
EQU
Equal
Source A N34:58
1<
Source B 0
0<
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
VESSEL B DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
6
VESSEL B DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
6
0008
OSR
B3:0
10
DP_REGEN
DP_REGEN
EQU
Equal
Source A N34:58
1<
Source B 1
1<
ALARM RESET PB ON OIT
ALMSILENCEPB
B3:12
3
VESSEL B DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
12
VESSEL B DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0
6

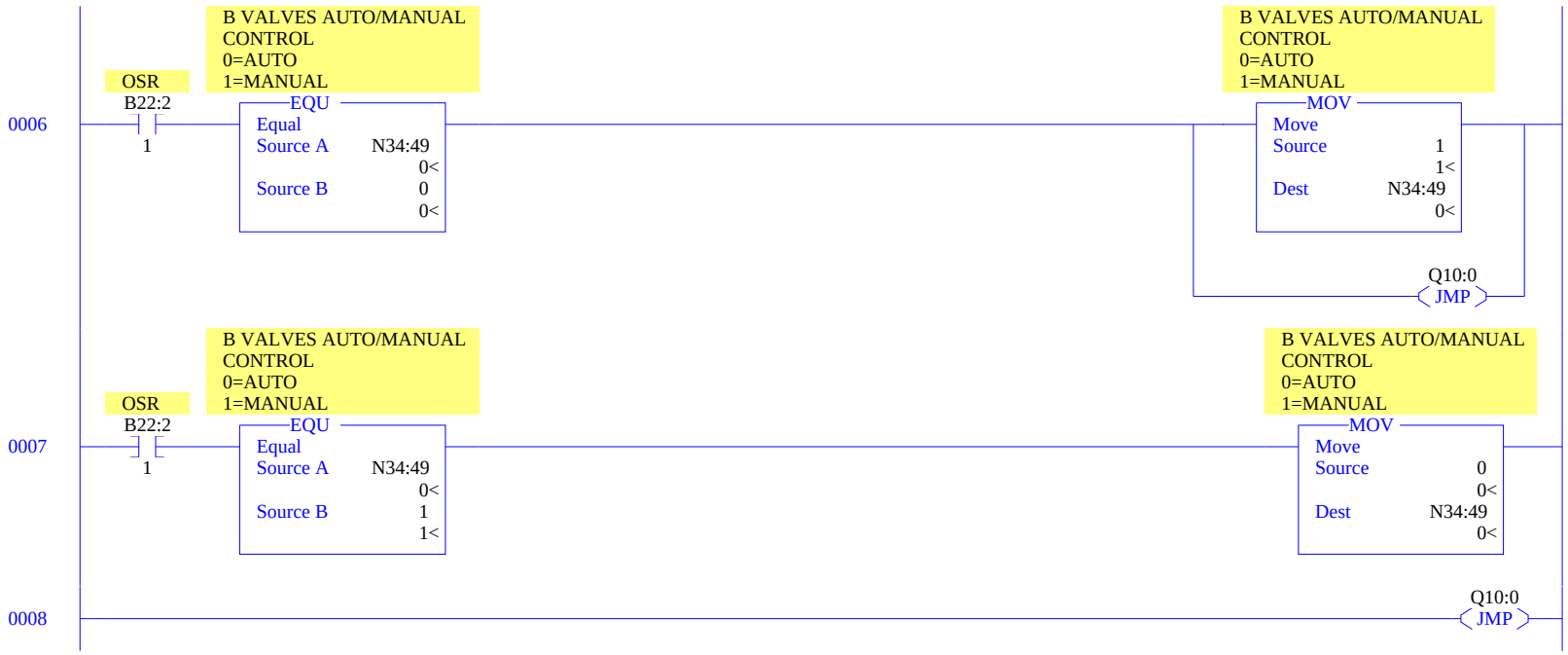

VESSEL A DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
11
DP REGEN STATUS
0=NOT IN CYCLE
1=REGEN A
2=REGEN B
NEQ
Not Equal
Source A
N7:5
0<
Source B
2
2<
DP REGEN INIT BIT
B3:0
13
0009
VESSEL B DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0
12
DP REGEN INIT BIT
B3:0
13
0010
END

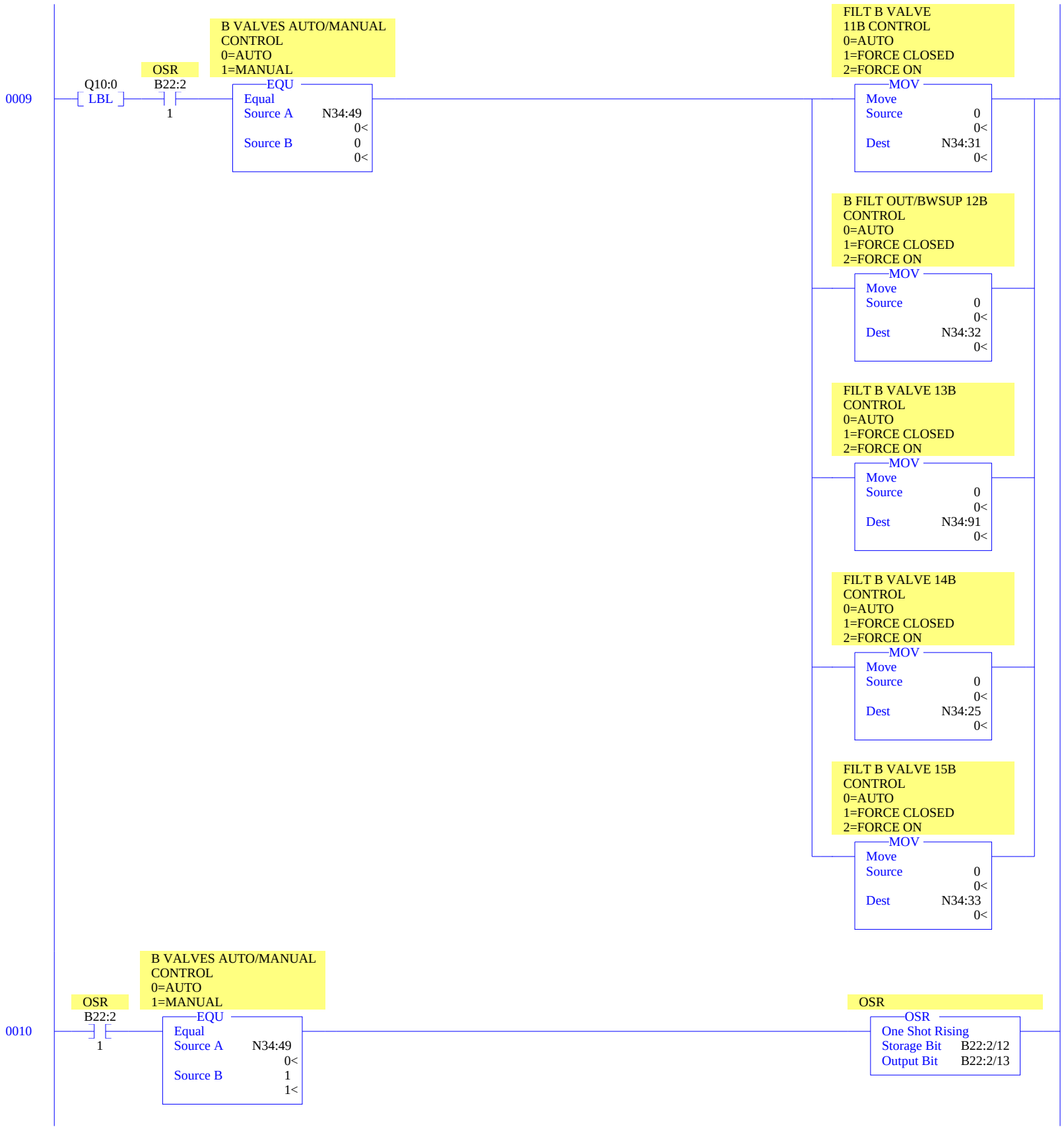
LAD 10 - B VESSEL --- Total Rungs in File = 38

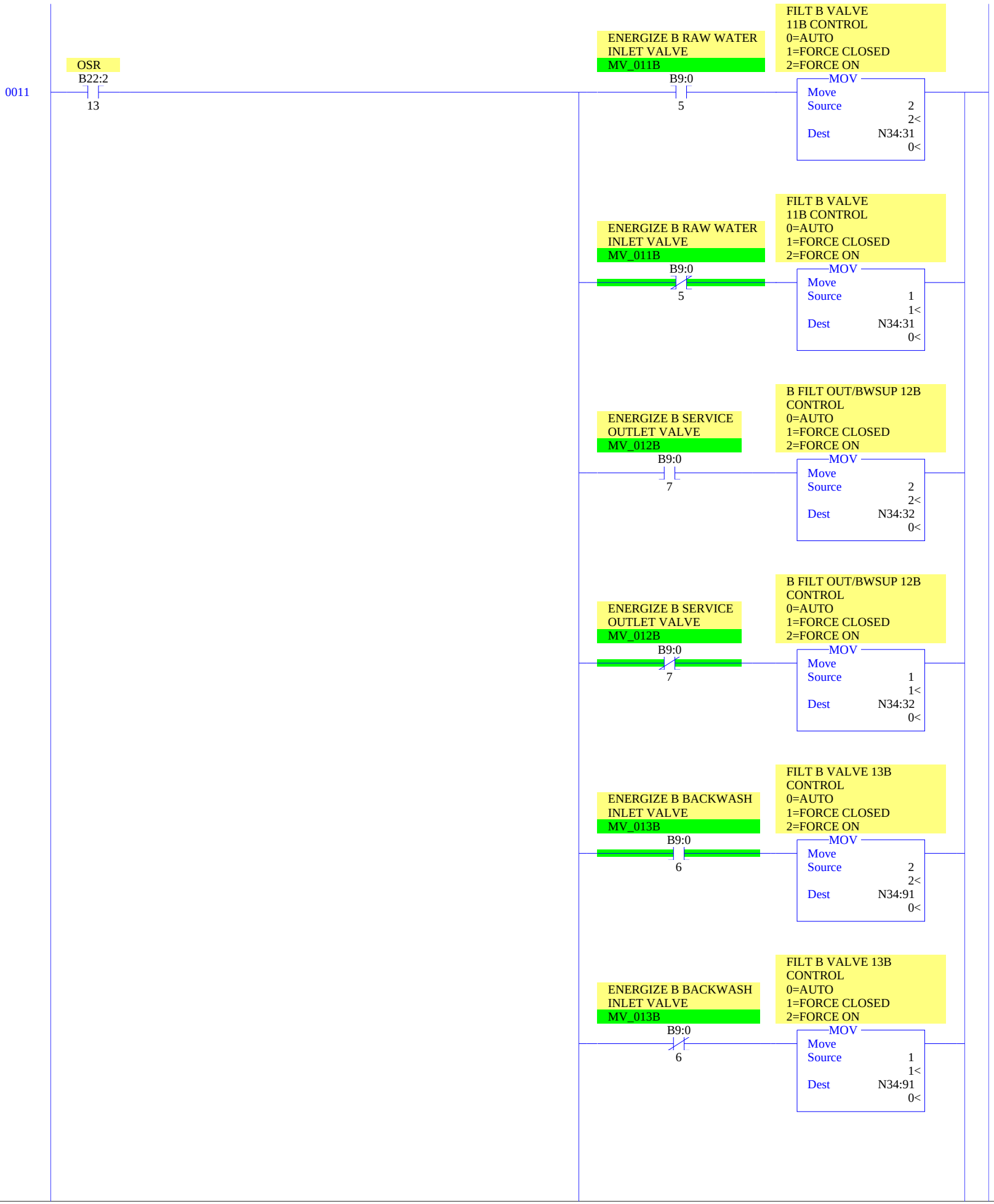


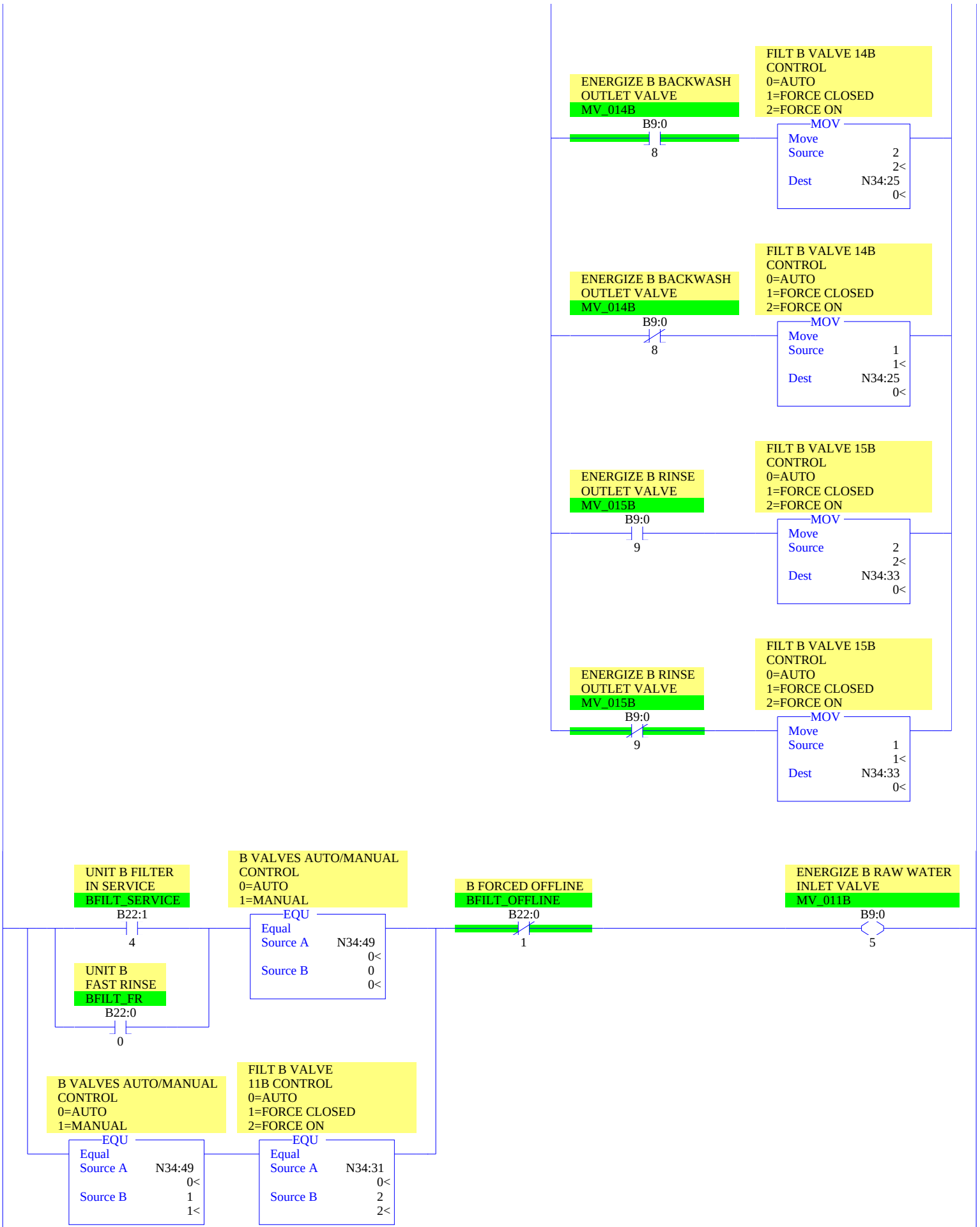

0002
B ON/OFF OSR2
B22:1
15
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ANY UNIT IS IN REGN
REGN_ANY
B3:5
13
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
L
1
FILT B VALVE
11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:31
0<
B FILT OUT/BWSUP 12B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:32
0<
FILT B VALVE 13B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:91
0<
FILT B VALVE 14B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:25
0<
FILT B VALVE 15B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:33
0<
Q10:7
JMP

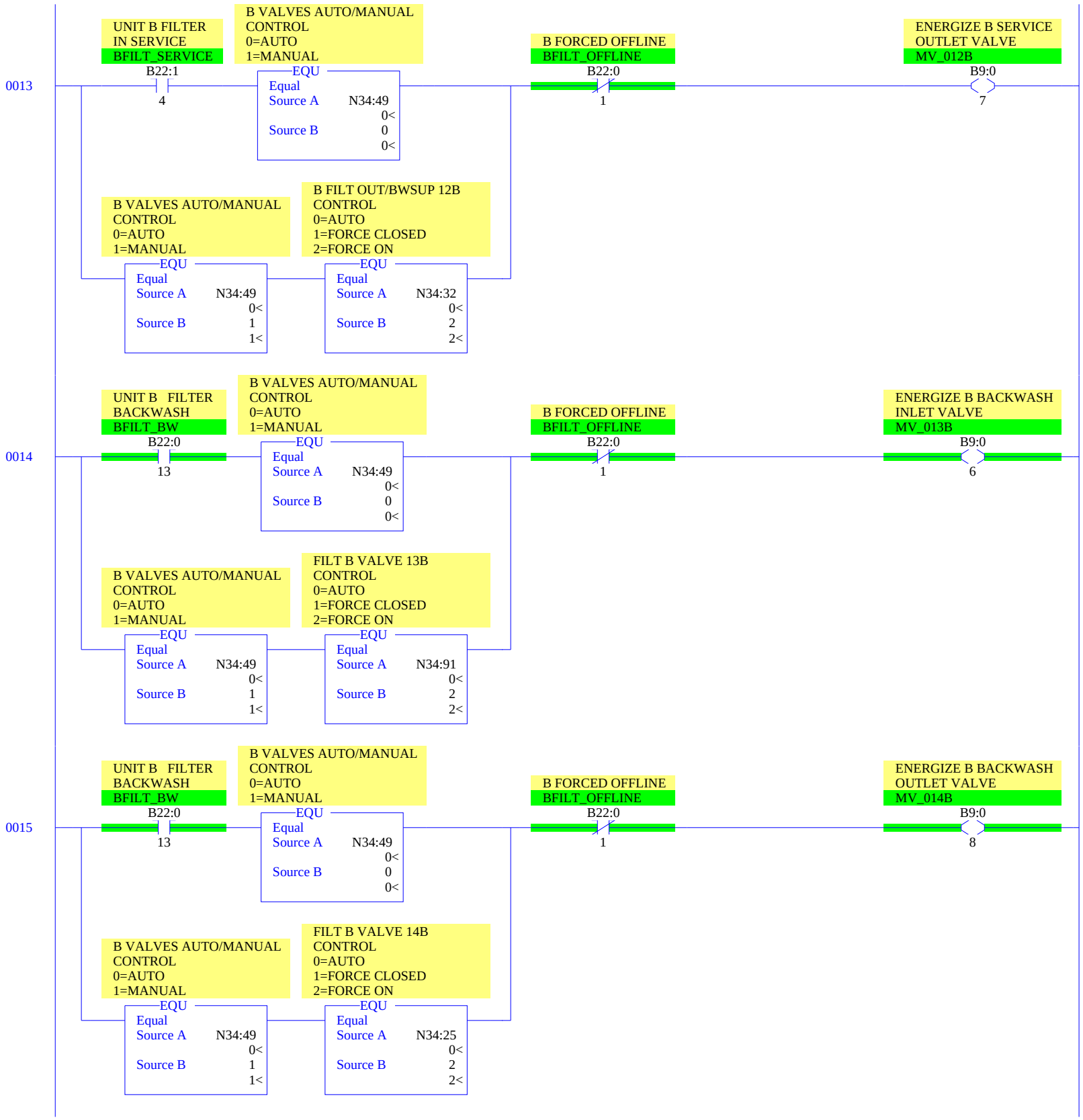

0003
B ON/OFF OSR2
B22:1
15
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
U
1
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49 0<
Source B 0 0<
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:31 0<
B FILT OUT/BWSUP 12B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:32 0<
FILT B VALVE 13B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:91 0<
FILT B VALVE 14B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:25 0<
FILT B VALVE 15B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0 0<
Dest N34:33 0<
0004
Q10:7
JMP
0005
Q10:7
LBL
B AUTO/MAN PUSHBUTTON
B_AUTO_MAN_PB
B22:0
8
UNIT B FILTER IN REGENERATION
BFILT_REGN
B22:0
2
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
OSR
OSR
One Shot Rising
Storage Bit B22:2/0
Output Bit B22:2/1

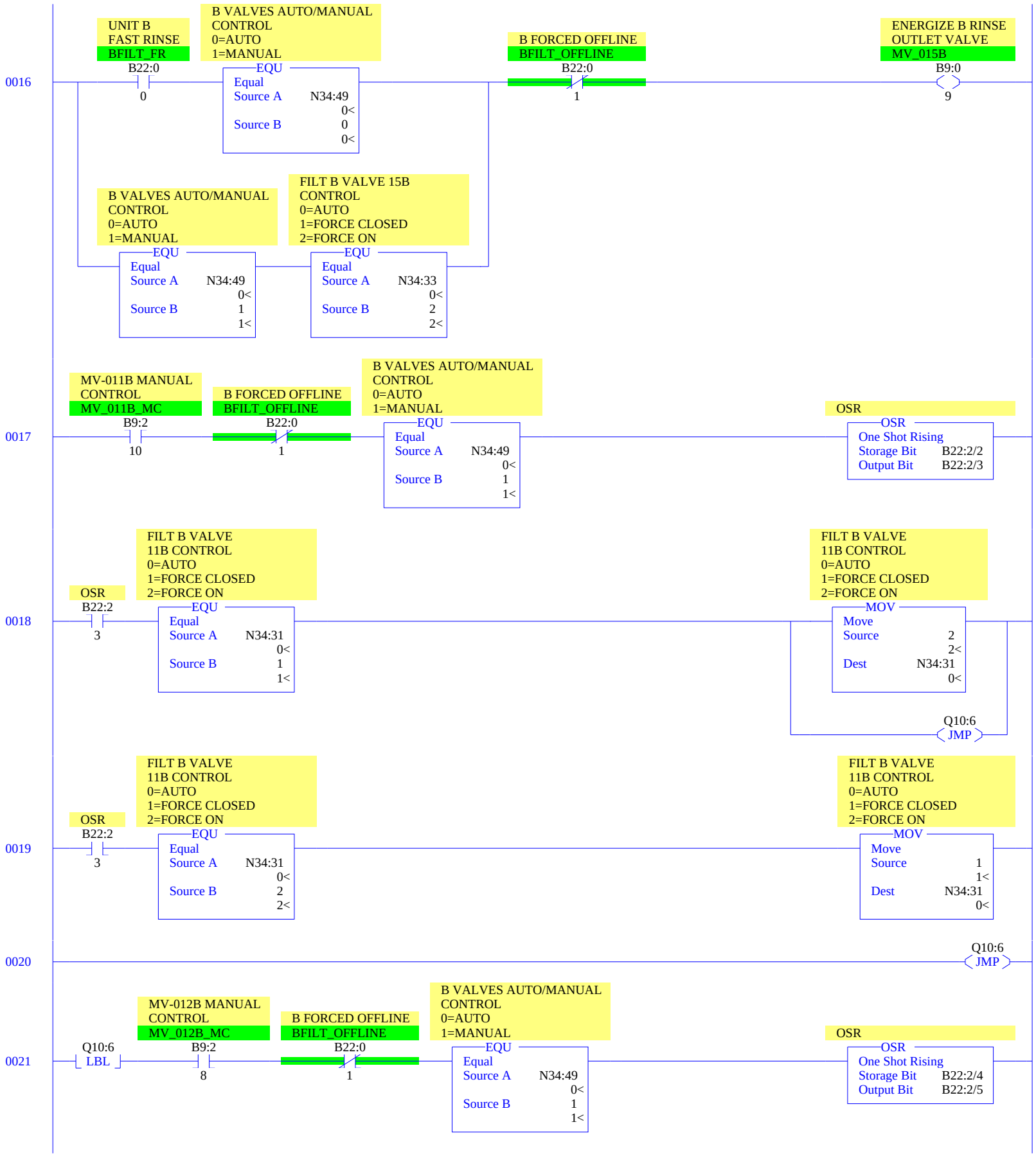

B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
OSR
B22:2
1
EQU
Equal
Source A N34:49 0<
Source B 0 0<
B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
MOV
Move
Source 1 1<
Dest N34:49 0<
Q10:0
JMP
0006
B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
OSR
B22:2
1
EQU
Equal
Source A N34:49 0<
Source B 1 1<
B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
MOV
Move
Source 0 0<
Dest N34:49 0<
0007
Q10:0
JMP
0008

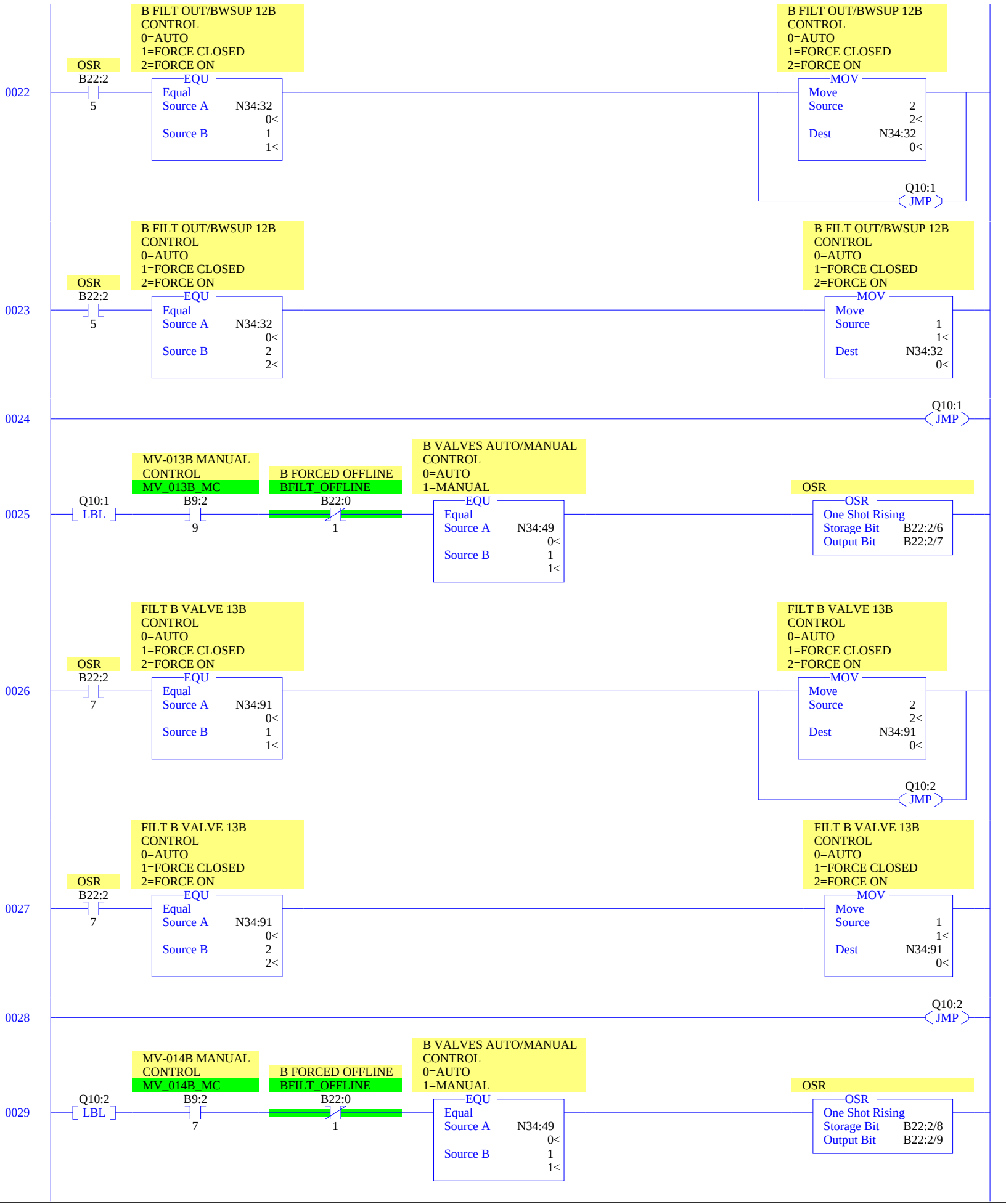

0009
Q10:0
LBL
OSR
B22:2
1
B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
0<
Source B 0
0<
FILT B VALVE
11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:31
0<
B FILT OUT/BWSUP 12B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:32
0<
FILT B VALVE 13B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:91
0<
FILT B VALVE 14B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:25
0<
FILT B VALVE 15B
CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 0
0<
Dest N34:33
0<
0010
OSR
B22:2
1
B VALVES AUTO/MANUAL
CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
0<
Source B 1
1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/12
Output Bit B22:2/13

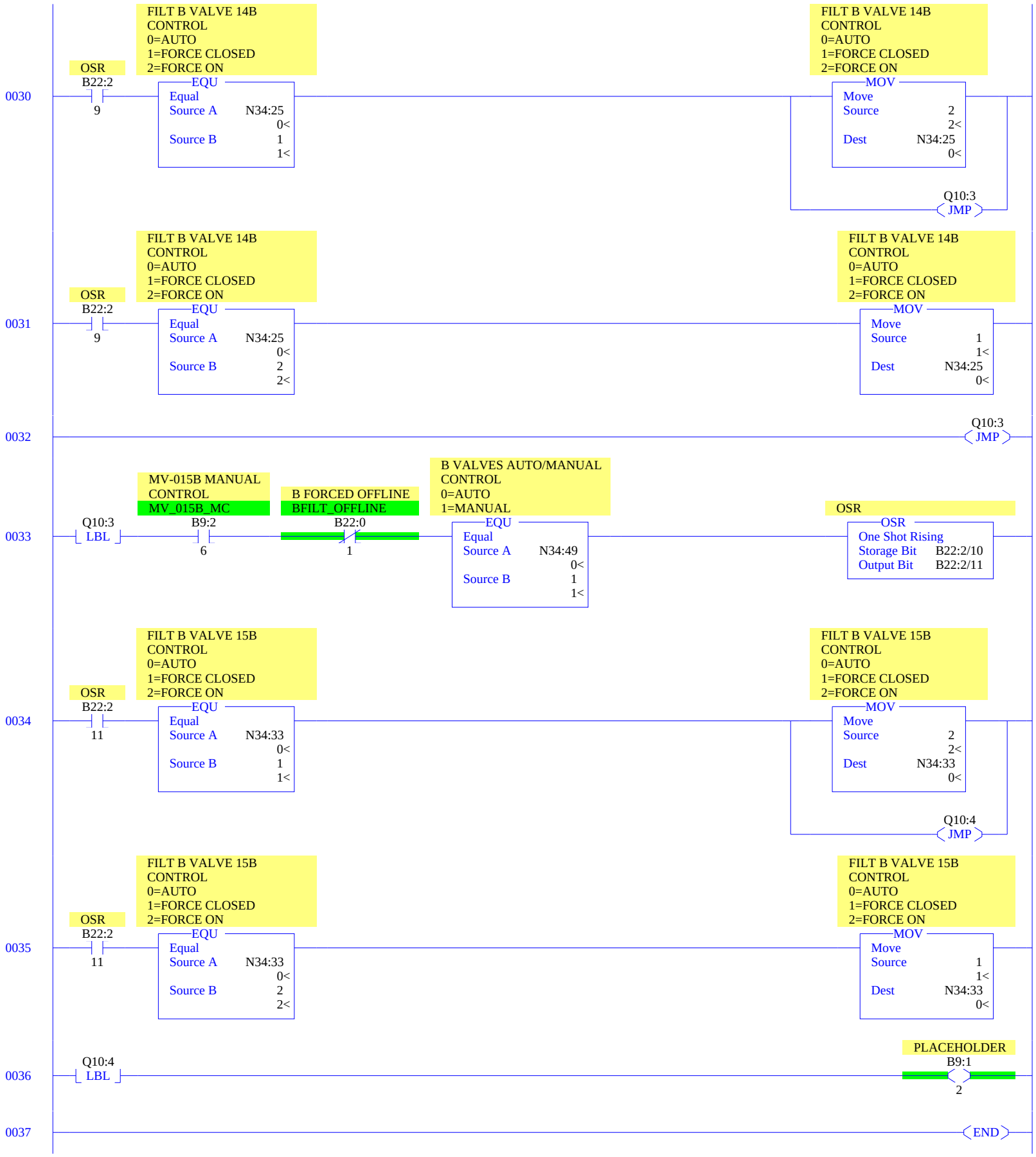

0011
OSR
B22:2
13
ENERGIZE B RAW WATER INLET VALVE
MV_011B
B9:0
5
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
Dest N34:31
ENERGIZE B RAW WATER INLET VALVE
MV_011B
B9:0
5
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
Dest N34:31
ENERGIZE B SERVICE OUTLET VALVE
MV_012B
B9:0
7
B FILT OUT/BWSUP 12B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
Dest N34:32
ENERGIZE B SERVICE OUTLET VALVE
MV_012B
B9:0
7
B FILT OUT/BWSUP 12B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
Dest N34:32
ENERGIZE B BACKWASH INLET VALVE
MV_013B
B9:0
6
FILT B VALVE 13B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
Dest N34:91
ENERGIZE B BACKWASH INLET VALVE
MV_013B
B9:0
6
FILT B VALVE 13B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
Dest N34:91


ENERGIZE B BACKWASH OUTLET VALVE
MV_014B
B9:0
8
FILT B VALVE 14B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:25
0<
ENERGIZE B BACKWASH OUTLET VALVE
MV_014B
B9:0
8
FILT B VALVE 14B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:25
0<
ENERGIZE B RINSE OUTLET VALVE
MV_015B
B9:0
9
FILT B VALVE 15B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2
2<
Dest N34:33
0<
ENERGIZE B RINSE OUTLET VALVE
MV_015B
B9:0
9
FILT B VALVE 15B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1
1<
Dest N34:33
0<
0012
UNIT B FILTER IN SERVICE
BFILT_SERVICE
B22:1
4
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
0<
Source B 0
0<
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ENERGIZE B RAW WATER INLET VALVE
MV_011B
B9:0
5
UNIT B FAST RINSE
BFILT_FR
B22:0
0
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
0<
Source B 1
1<
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:31
0<
Source B 2
2<

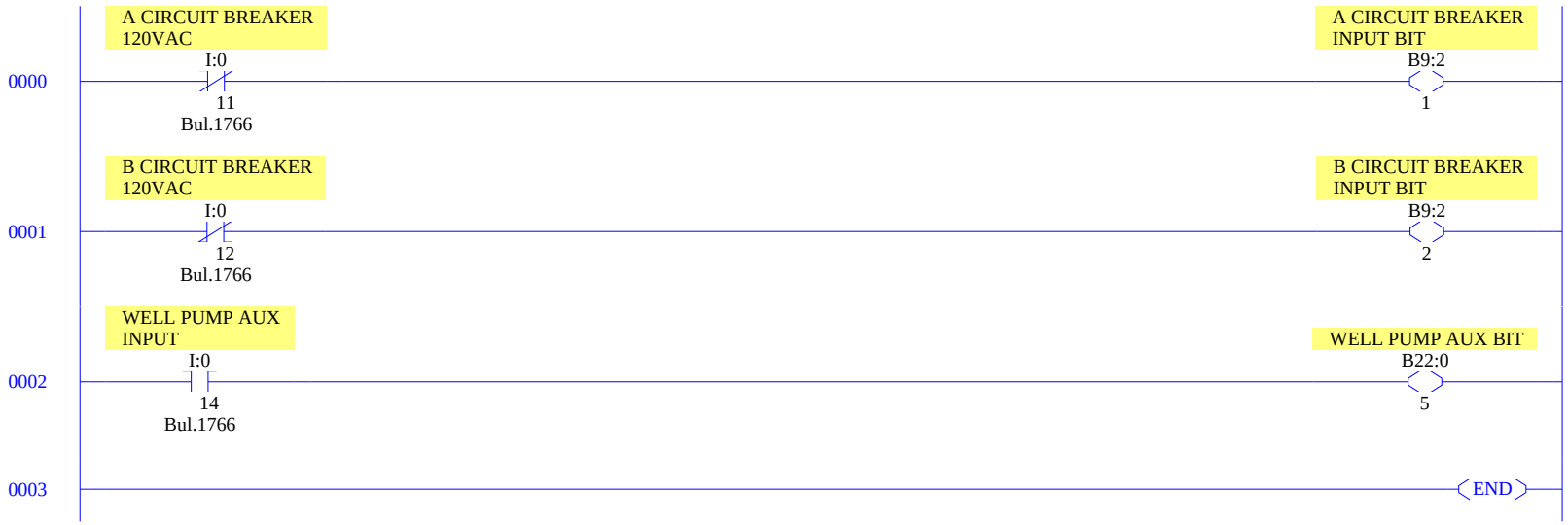

0013
UNIT B FILTER IN SERVICE
BFILT_SERVICE
B22:1
4
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 0
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ENERGIZE B SERVICE OUTLET VALVE
MV_012B
B9:0
7
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 1
B FILT OUT/BWSUP 12B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:32
Source B 2
0014
UNIT B FILTER BACKWASH
BFILT_BW
B22:0
13
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 0
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ENERGIZE B BACKWASH INLET VALVE
MV_013B
B9:0
6
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 1
FILT B VALVE 13B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:91
Source B 2
0015
UNIT B FILTER BACKWASH
BFILT_BW
B22:0
13
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 0
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ENERGIZE B BACKWASH OUTLET VALVE
MV_014B
B9:0
8
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49
Source B 1
FILT B VALVE 14B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:25
Source B 2


0016
UNIT B FAST RINSE
BFILT_FR
B22:0
0
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49 0<
Source B 0 0<
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
ENERGIZE B RINSE OUTLET VALVE
MV_015B
B9:0
9
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49 0<
Source B 1 1<
FILT B VALVE 15B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:33 0<
Source B 2 2<
0017
MV-011B MANUAL CONTROL
MV_011B_MC
B9:2
10
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/2
Output Bit B22:2/3
0018
OSR
B22:2
3
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:31 0<
Source B 1 1<
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:31 0<
Q10:6
JMP
0019
OSR
B22:2
3
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
EQU
Equal
Source A N34:31 0<
Source B 2 2<
FILT B VALVE 11B CONTROL
0=AUTO
1=FORCE CLOSED
2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:31 0<
0020
Q10:6
JMP
0021
Q10:6
LBL
MV-012B MANUAL CONTROL
MV_012B_MC
B9:2
8
B FORCED OFFLINE
BFILT_OFFLINE
B22:0
1
B VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:49 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/4
Output Bit B22:2/5

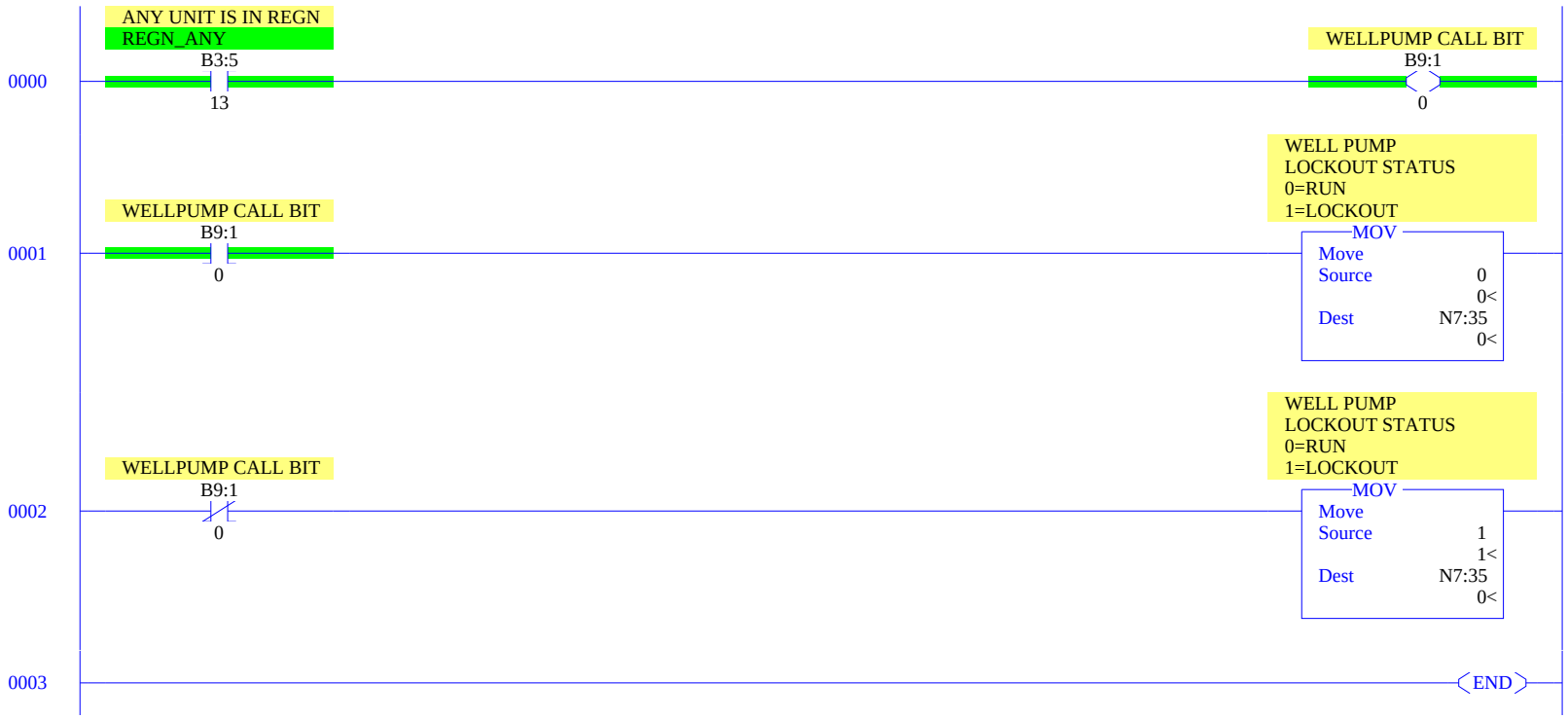

0022
B FILT OUT/BWSUP 12B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
5
EQU
Equal
Source A N34:32 0<
Source B 1 1<
B FILT OUT/BWSUP 12B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:32 0<
Q10:1
JMP
0023
B FILT OUT/BWSUP 12B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
5
EQU
Equal
Source A N34:32 0<
Source B 2 2<
B FILT OUT/BWSUP 12B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:32 0<
0024
Q10:1
JMP
0025
MV-013B MANUAL CONTROL
MV_013B_MC
B FORCED OFFLINE
BFILT_OFFLINE
B VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
Q10:1
LBL
B9:2
9
B22:0
1
EQU
Equal
Source A N34:49 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/6
Output Bit B22:2/7
0026
FILT B VALVE 13B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
7
EQU
Equal
Source A N34:91 0<
Source B 1 1<
FILT B VALVE 13B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:91 0<
Q10:2
JMP
0027
FILT B VALVE 13B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
7
EQU
Equal
Source A N34:91 0<
Source B 2 2<
FILT B VALVE 13B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:91 0<
0028
Q10:2
JMP
0029
MV-014B MANUAL CONTROL
MV_014B_MC
B FORCED OFFLINE
BFILT_OFFLINE
B VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
Q10:2
LBL
B9:2
7
B22:0
1
EQU
Equal
Source A N34:49 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/8
Output Bit B22:2/9


0030
FILT B VALVE 14B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
9
EQU
Equal
Source A N34:25 0<
Source B 1 1<
FILT B VALVE 14B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:25 0<
Q10:3
JMP
0031
FILT B VALVE 14B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
9
EQU
Equal
Source A N34:25 0<
Source B 2 2<
FILT B VALVE 14B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:25 0<
0032
Q10:3
JMP
0033
MV-015B MANUAL CONTROL
MV_015B_MC
B FORCED OFFLINE
BFILT_OFFLINE
B VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
Q10:3
LBL
B9:2
6
B22:0
1
EQU
Equal
Source A N34:49 0<
Source B 1 1<
OSR
OSR
One Shot Rising
Storage Bit B22:2/10
Output Bit B22:2/11
0034
FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
11
EQU
Equal
Source A N34:33 0<
Source B 1 1<
FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 2 2<
Dest N34:33 0<
Q10:4
JMP
0035
FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
OSR
B22:2
11
EQU
Equal
Source A N34:33 0<
Source B 2 2<
FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
MOV
Move
Source 1 1<
Dest N34:33 0<
0036
Q10:4
LBL
PLACEHOLDER
B9:1
2
0037
END

LAD 11 - DIGITAL IN --- Total Rungs in File = 4



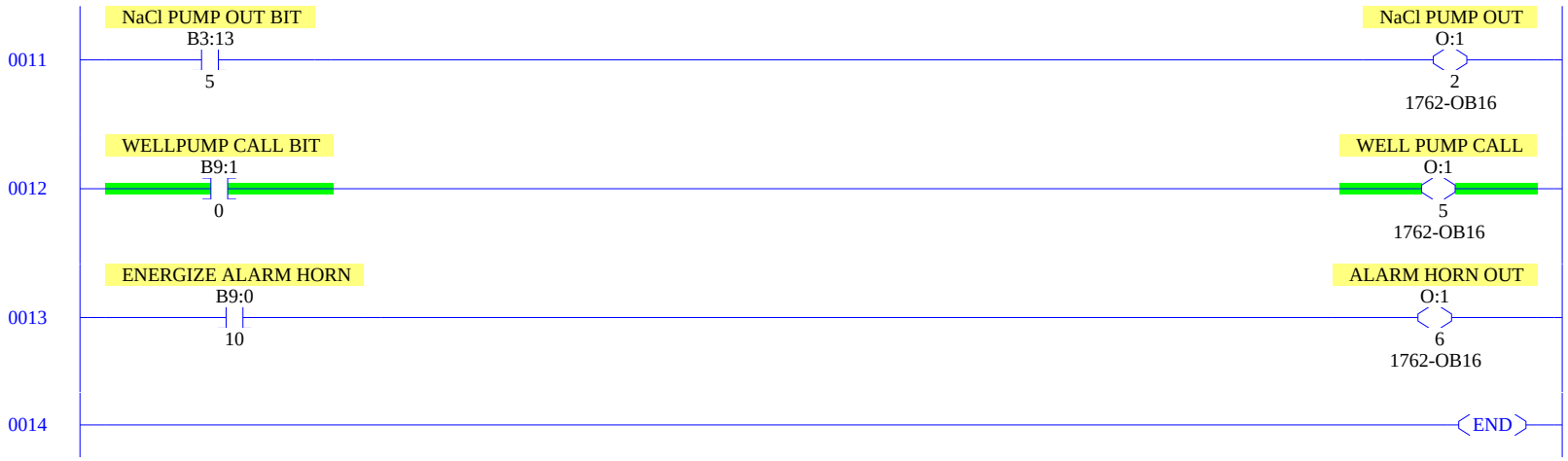
LAD 12 - WELL PUMP --- Total Rungs in File = 4



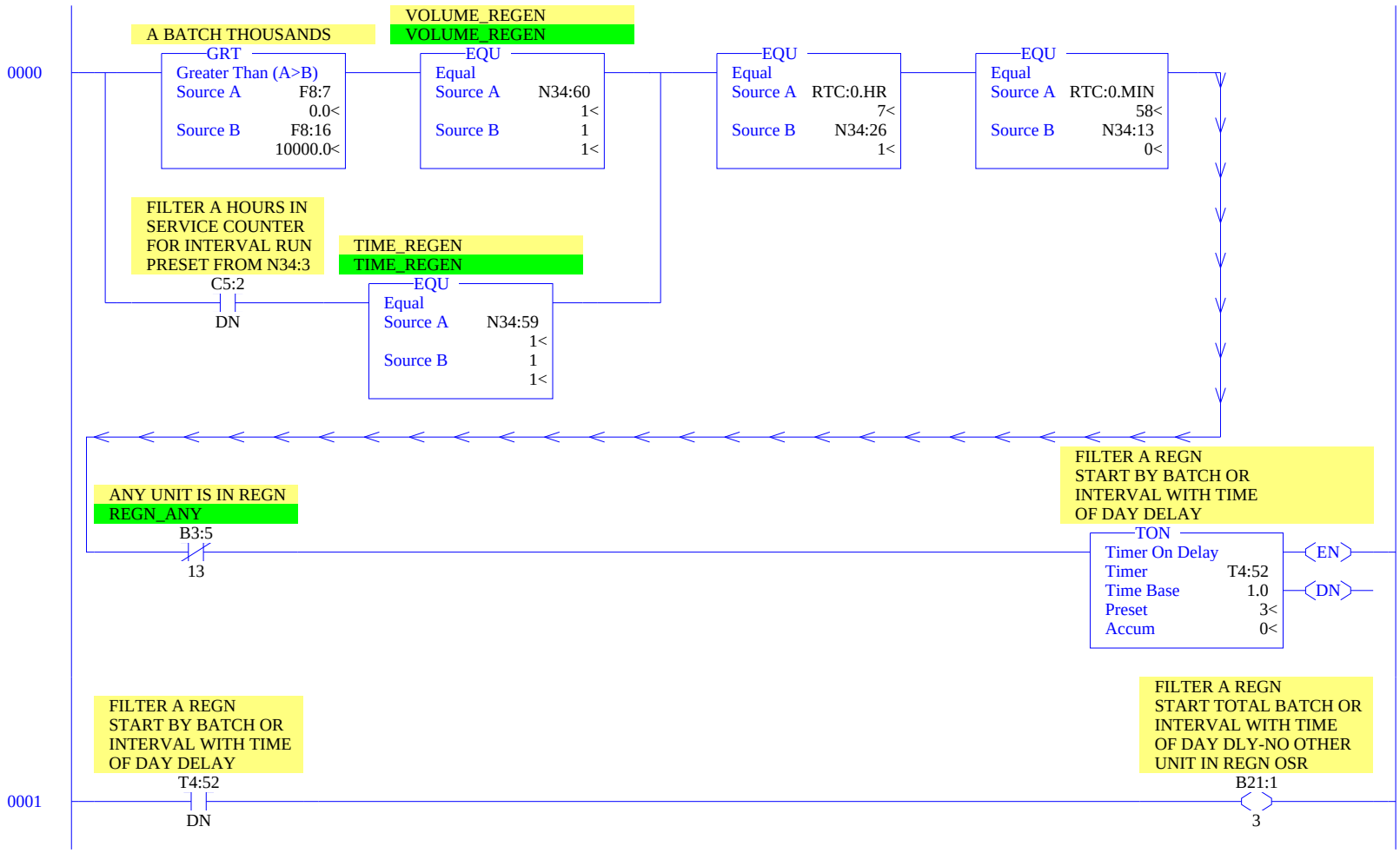
LAD 16 - DIGITL OUT --- Total Rungs in File = 15

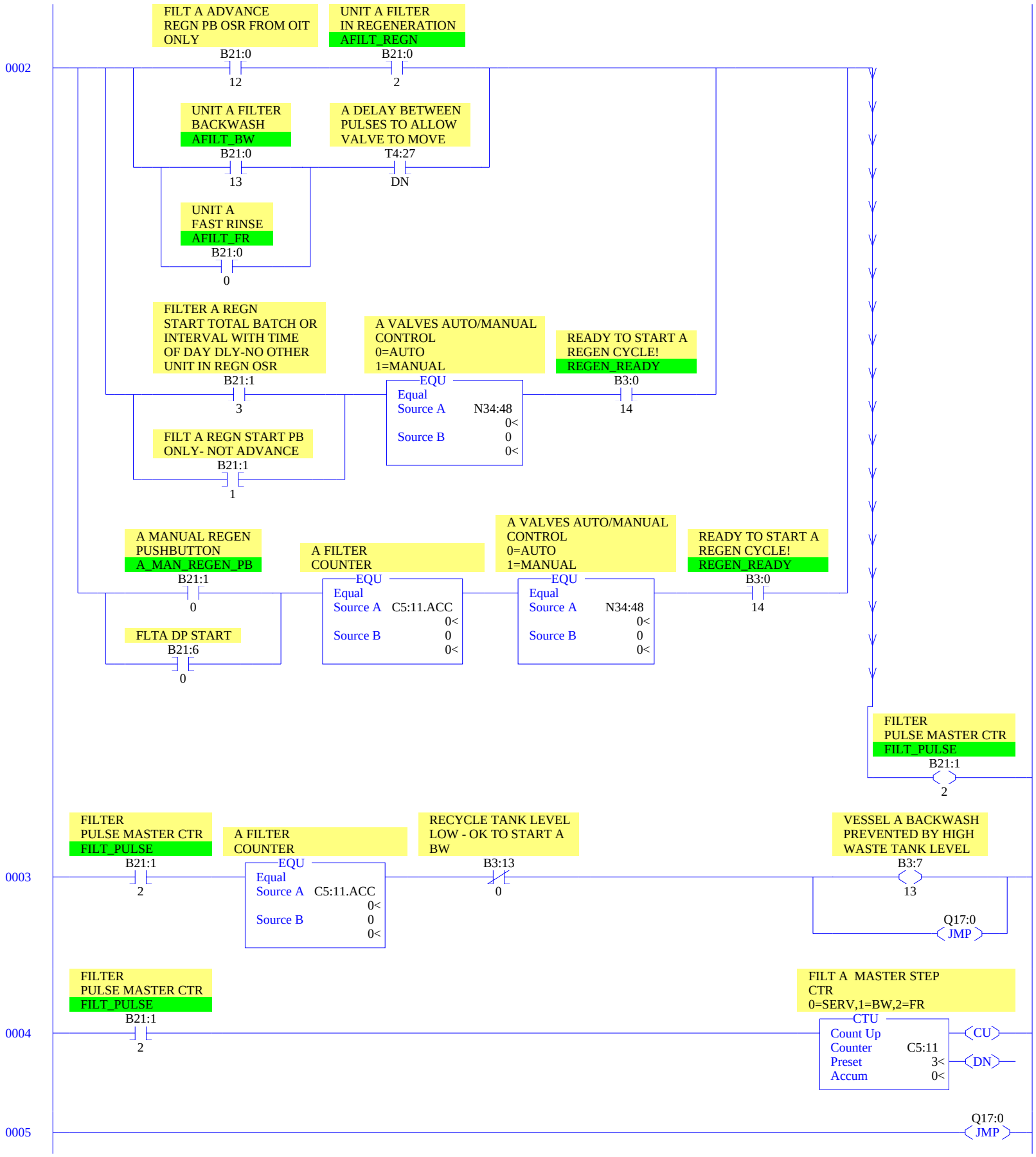


LAD 16 - DIGITL OUT --- Total Rungs in File = 15

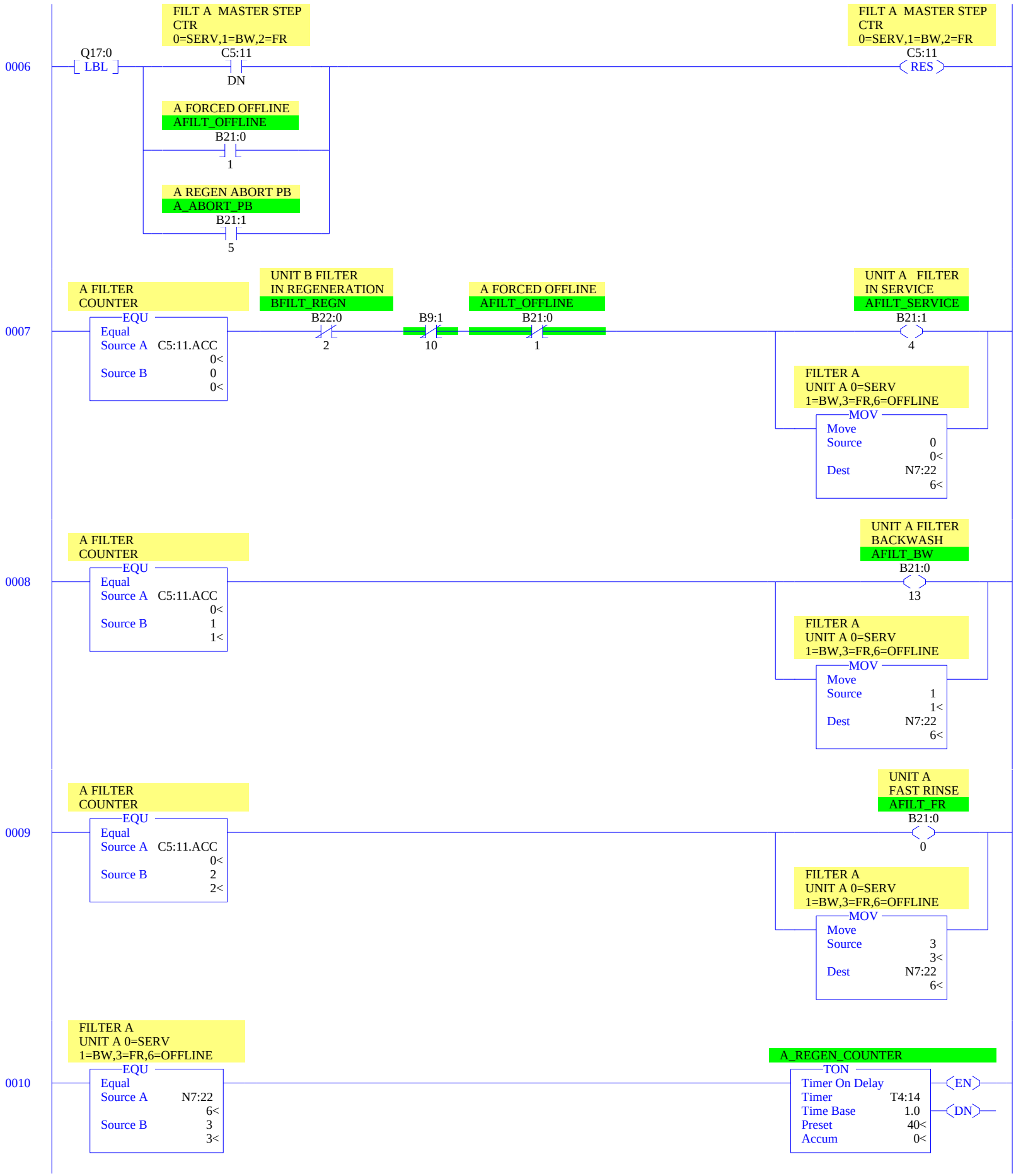


LAD 17 - REGEN A --- Total Rungs in File = 17

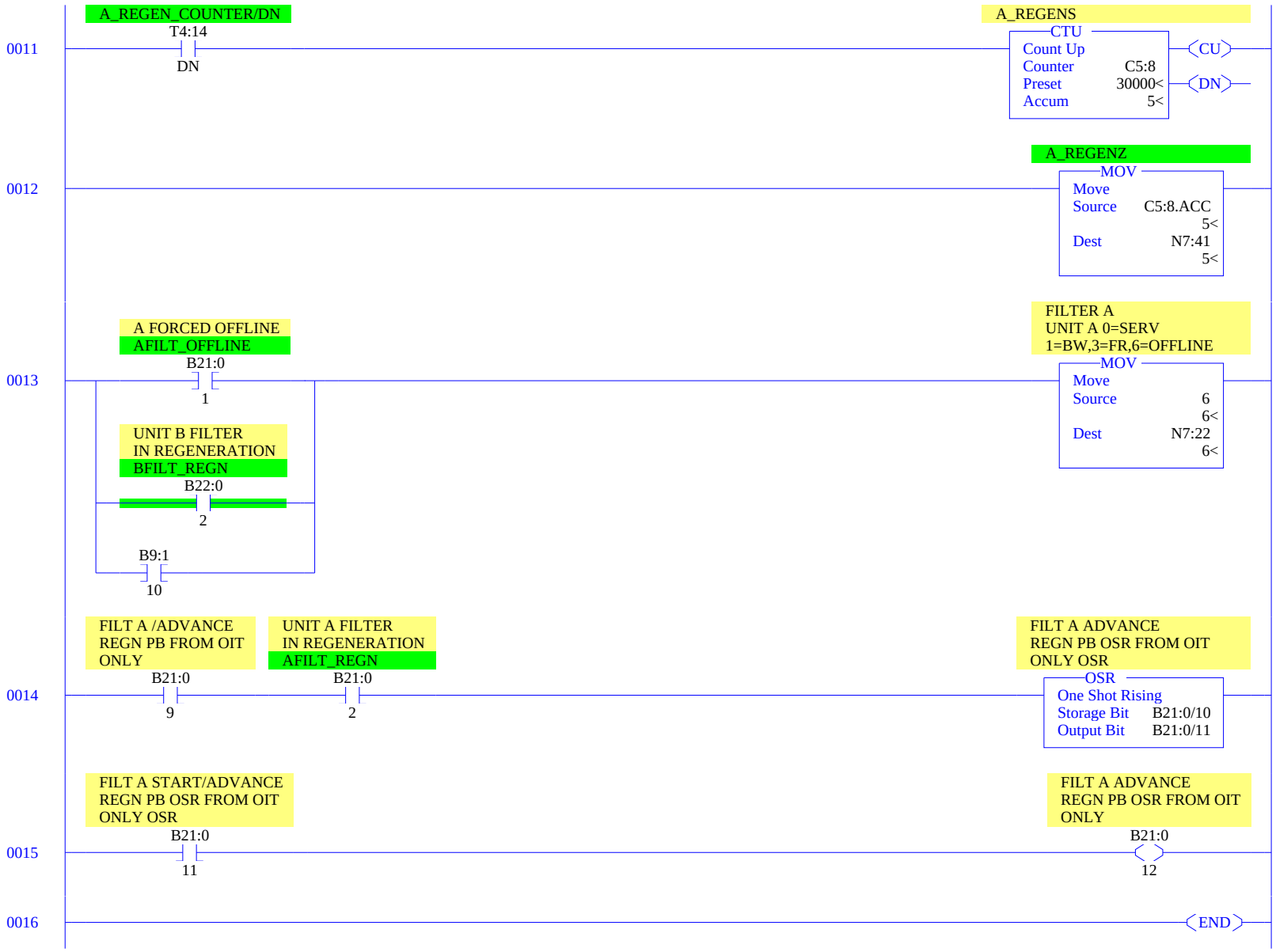



0002
FILT A ADVANCE REGN PB OSR FROM OIT ONLY
B21:0
12
UNIT A FILTER IN REGENERATION
AFILT_REGN
B21:0
2
UNIT A FILTER BACKWASH
AFILT_BW
B21:0
13
A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:27
DN
UNIT A FAST RINSE
AFILT_FR
B21:0
0
FILTER A REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DLY-NO OTHER UNIT IN REGN OSR
B21:1
3
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 0
0<
READY TO START A REGEN CYCLE!
REGEN_READY
B3:0
14
FILT A REGN START PB ONLY- NOT ADVANCE
B21:1
1
A MANUAL REGEN PUSHBUTTON
A_MAN_REGEN_PB
B21:1
0
A FILTER COUNTER
EQU
Equal
Source A C5:11.ACC
0<
Source B 0
0<
A VALVES AUTO/MANUAL CONTROL
0=AUTO
1=MANUAL
EQU
Equal
Source A N34:48
0<
Source B 0
0<
READY TO START A REGEN CYCLE!
REGEN_READY
B3:0
14
FLTA DP START
B21:6
0
FILTER PULSE MASTER CTR
FILT_PULSE
B21:1
2
0003
FILTER PULSE MASTER CTR
FILT_PULSE
B21:1
2
A FILTER COUNTER
EQU
Equal
Source A C5:11.ACC
0<
Source B 0
0<
RECYCLE TANK LEVEL LOW - OK TO START A BW
B3:13
0
VESSEL A BACKWASH PREVENTED BY HIGH WASTE TANK LEVEL
B3:7
13
Q17:0
JMP
0004
FILTER PULSE MASTER CTR
FILT_PULSE
B21:1
2
FILT A MASTER STEP CTR
0=SERV,1=BW,2=FR
CTU
Count Up
Counter C5:11
Preset 3<
Accum 0<
CU
DN
0005
Q17:0
JMP

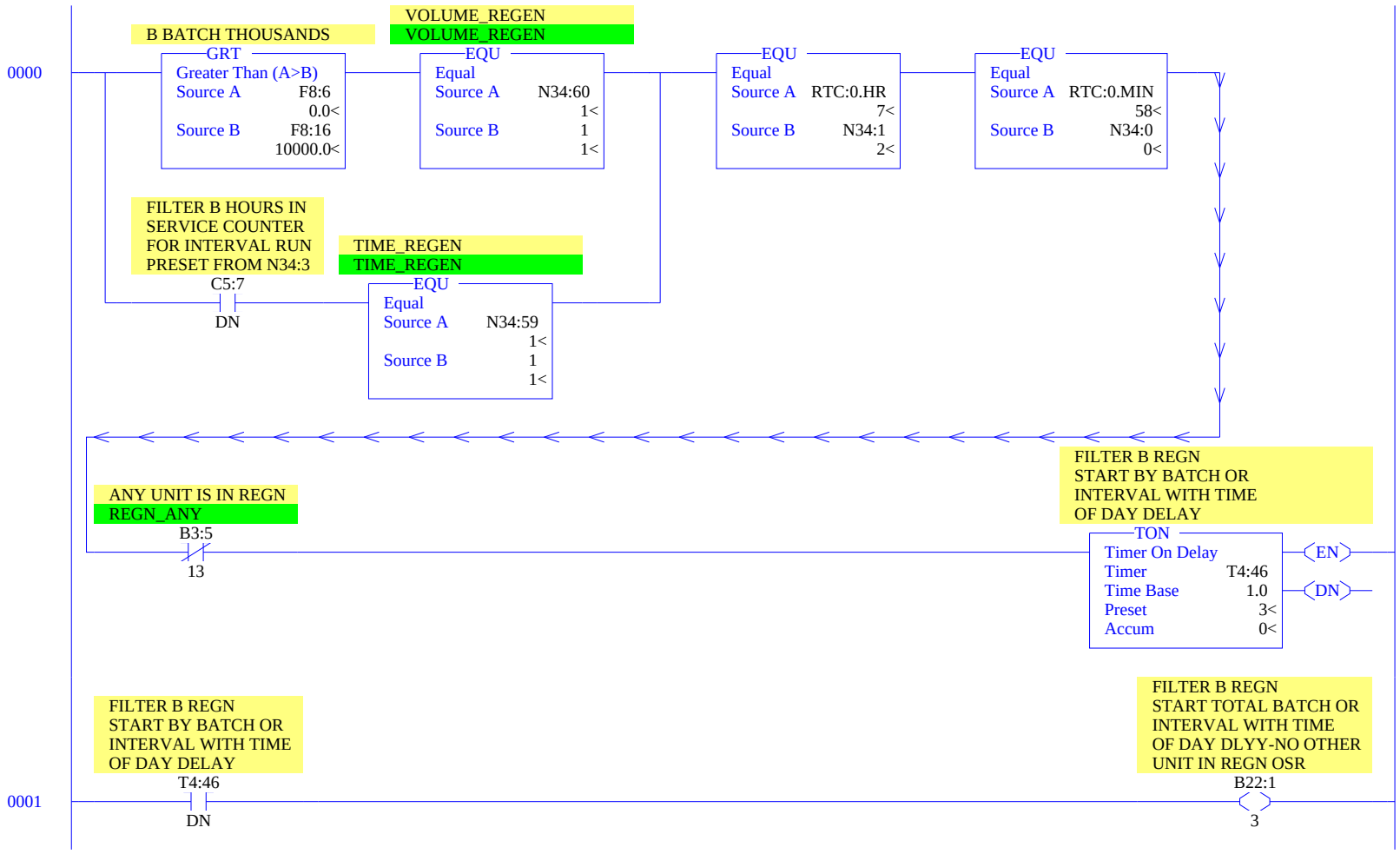
LAD 17 - REGEN A --- Total Rungs in File = 17



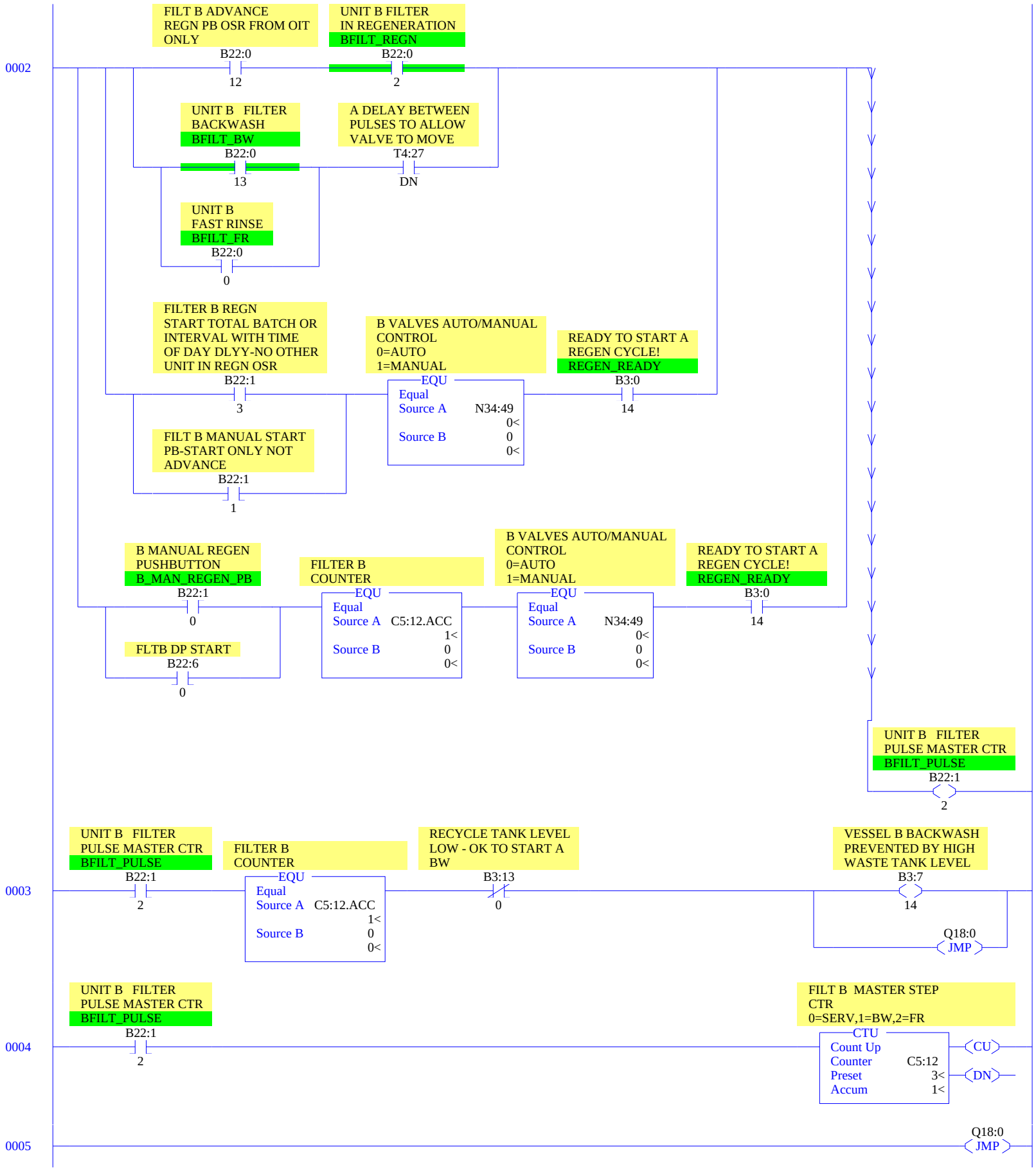
LAD 17 - REGEN A --- Total Rungs in File = 17



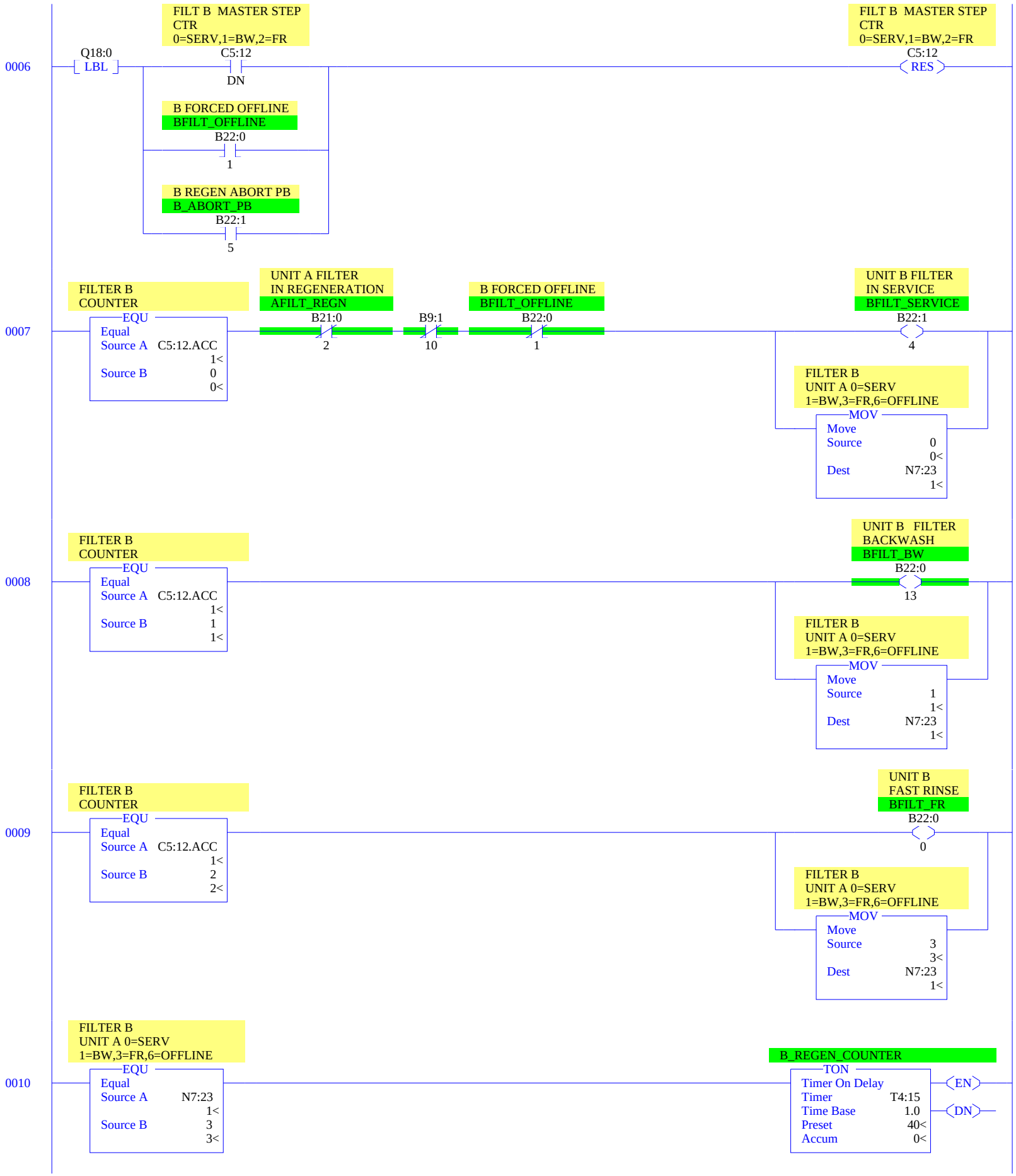
LAD 18 - REGEN B --- Total Rungs in File = 17



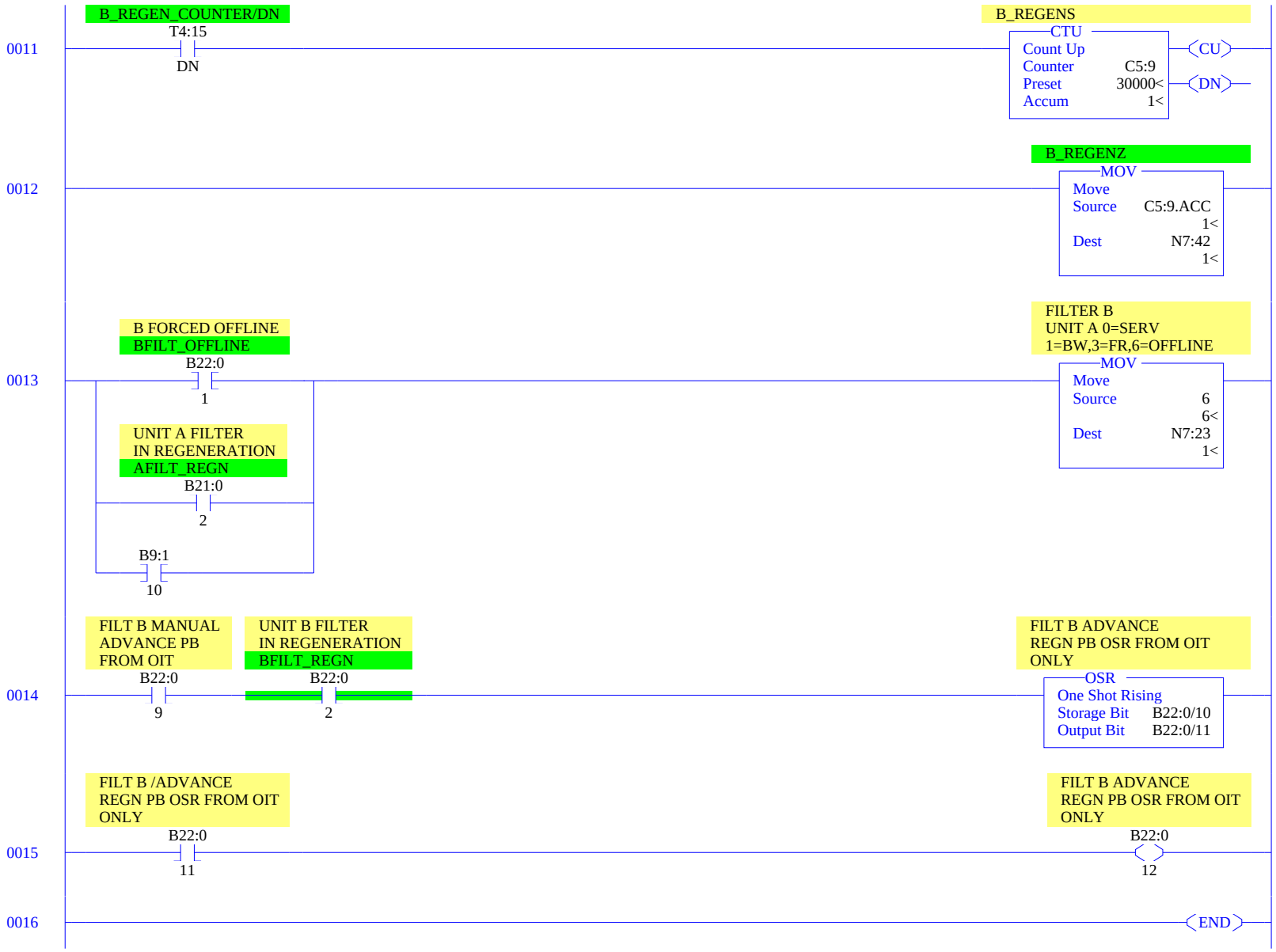
LAD 18 - REGEN B --- Total Rungs in File = 17



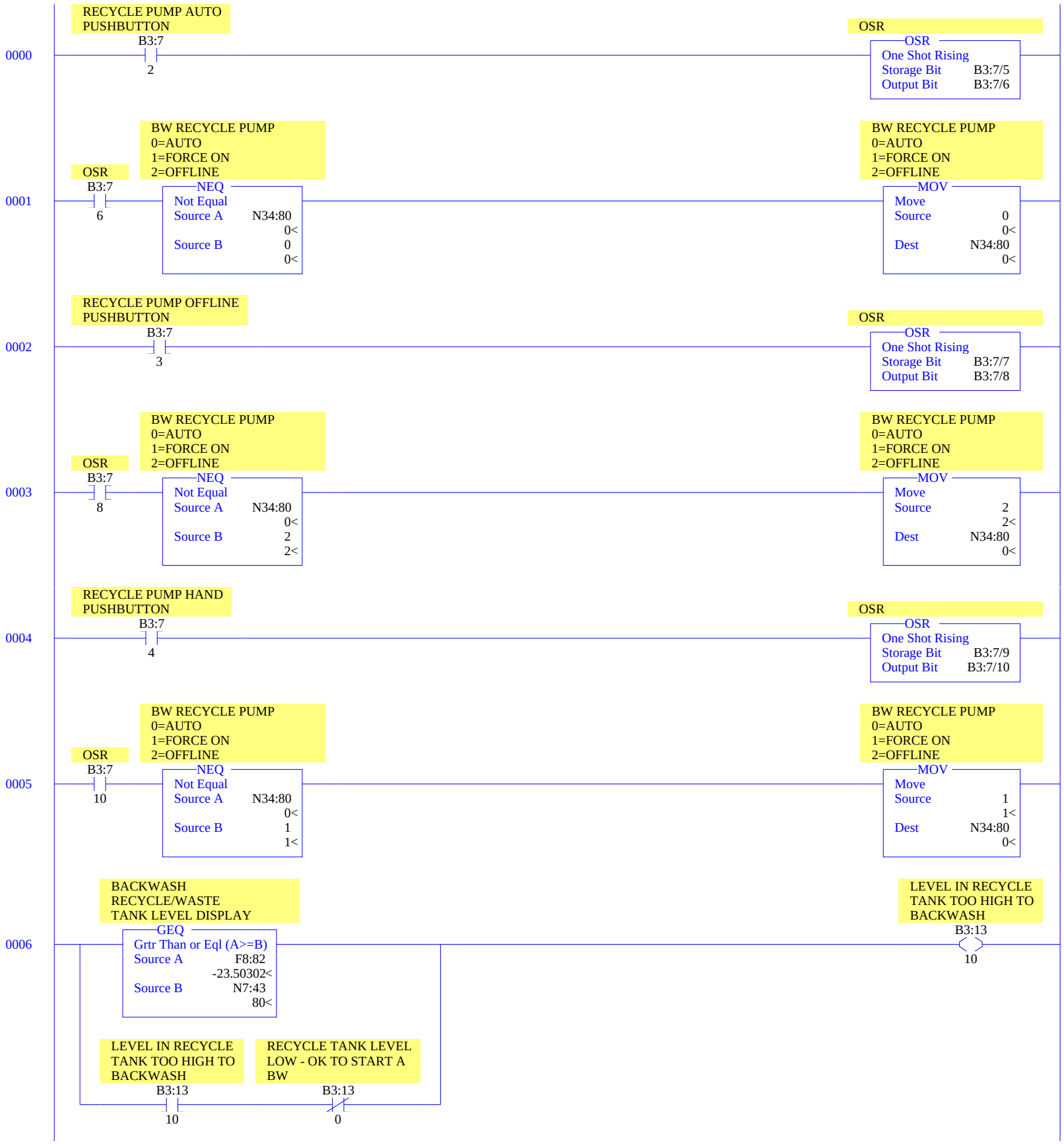
LAD 18 - REGEN B --- Total Rungs in File = 17

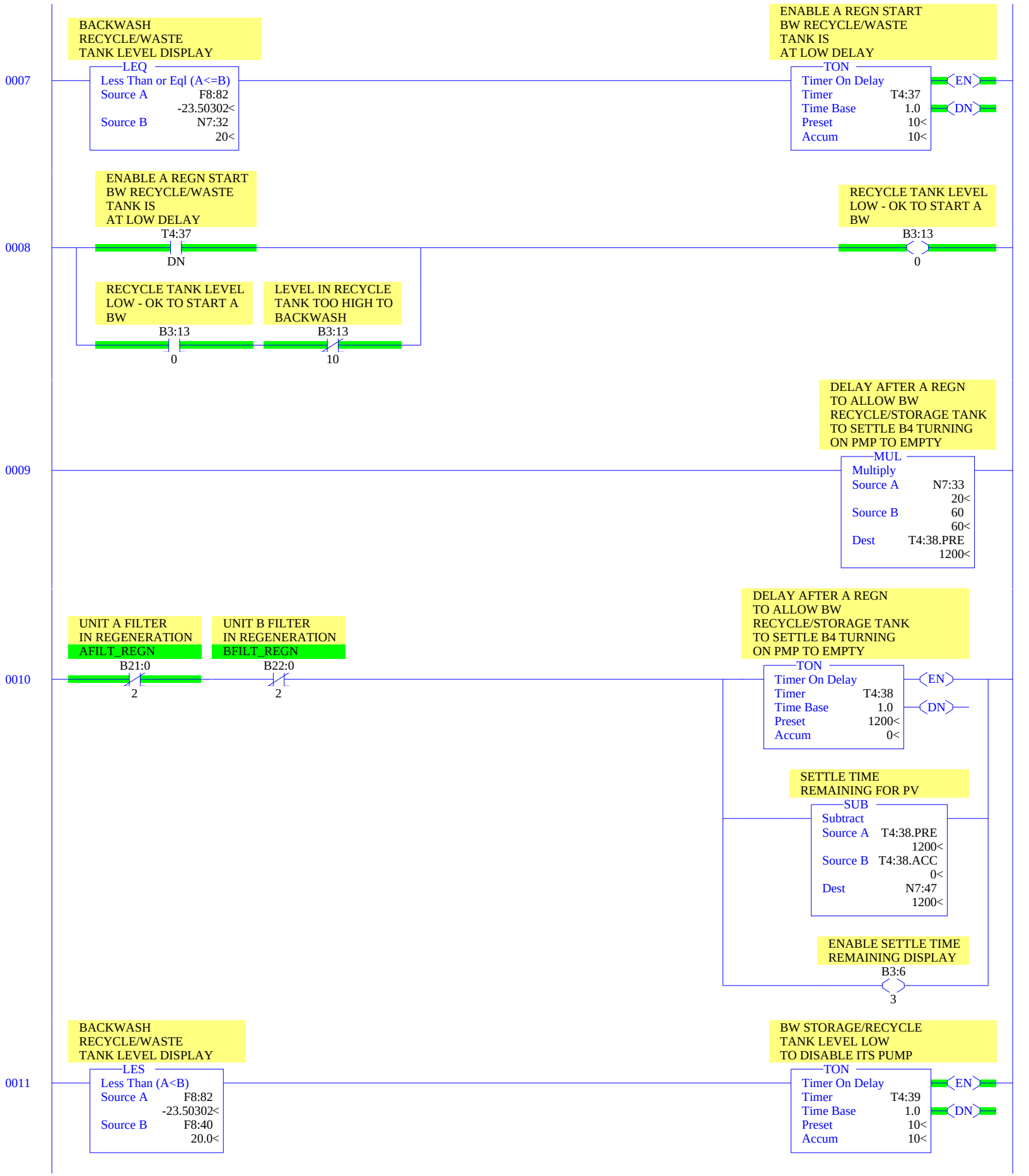


LAD 18 - REGEN B --- Total Rungs in File = 17

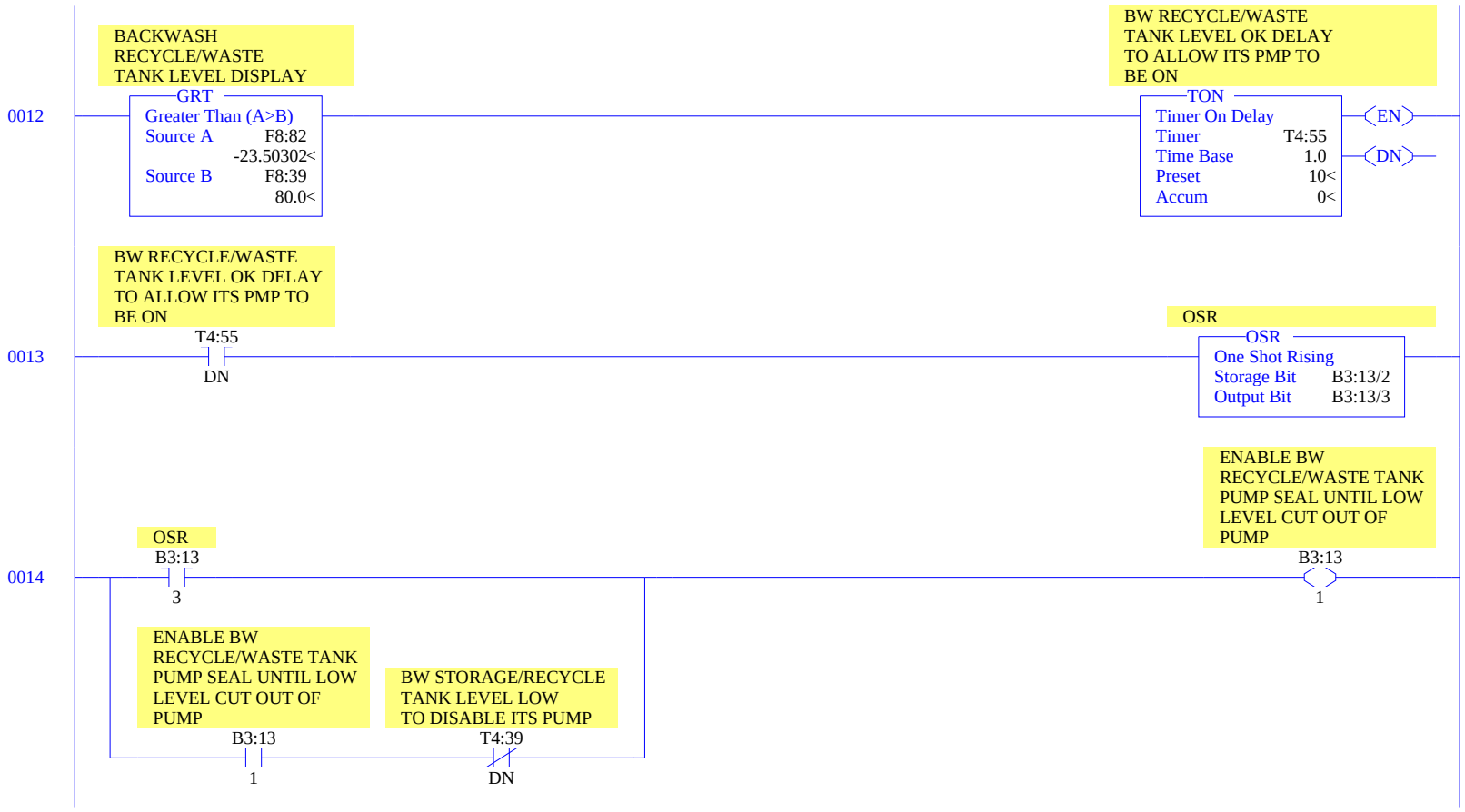


LAD 19 - RECYCLE --- Total Rungs in File = 17

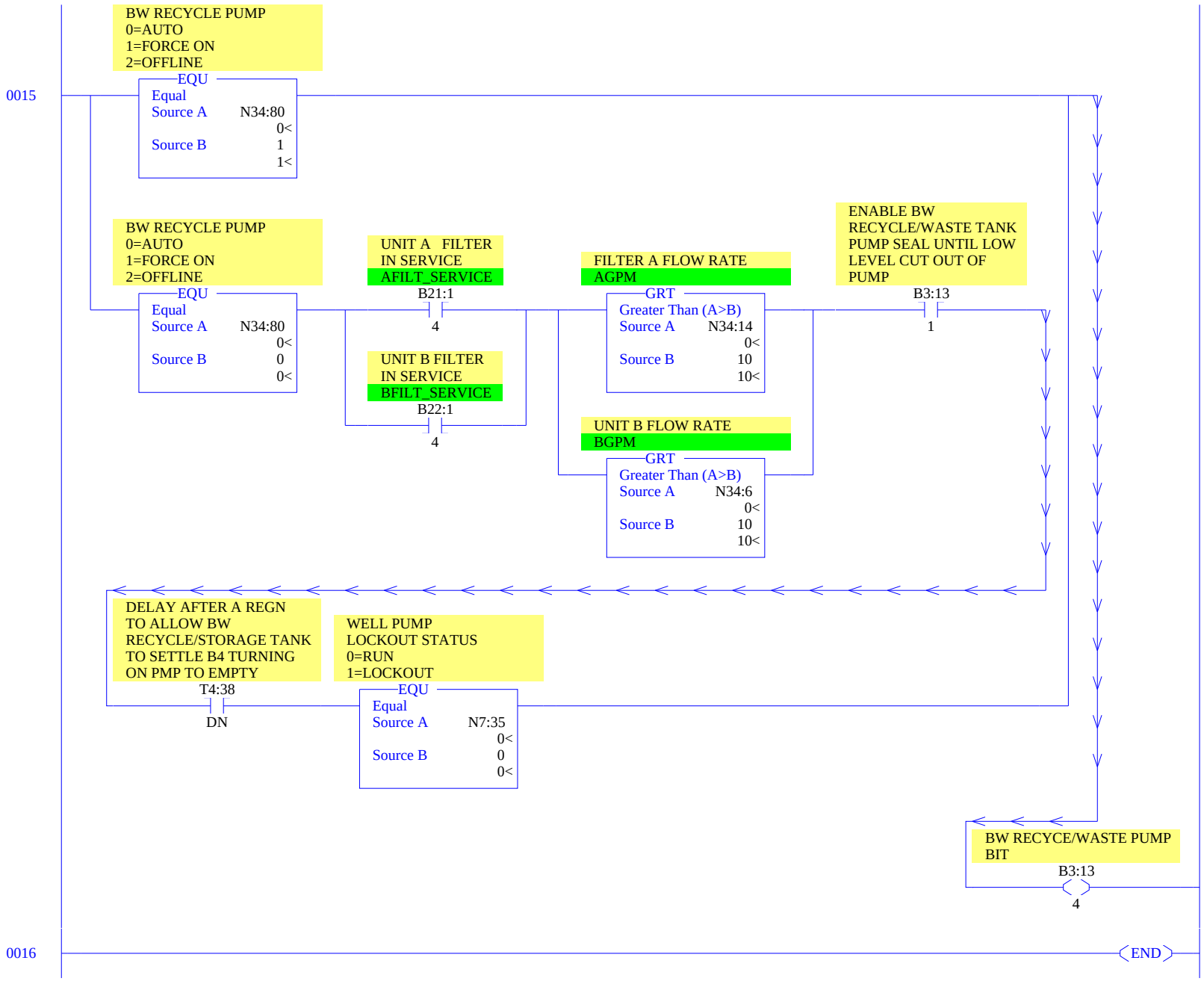



0007
BACKWASH RECYCLE/WASTE TANK LEVEL DISPLAY
LEQ
Less Than or Eql (A<=B)
Source A F8:82
-23.50302<
Source B N7:32
20<
ENABLE A REGN START BW RECYCLE/WASTE TANK IS AT LOW DELAY
TON
Timer On Delay
Timer T4:37
Time Base 1.0
Preset 10<
Accum 10<
EN
DN
0008
ENABLE A REGN START BW RECYCLE/WASTE TANK IS AT LOW DELAY
T4:37
DN
RECYCLE TANK LEVEL LOW - OK TO START A BW
B3:13
0
RECYCLE TANK LEVEL LOW - OK TO START A BW
B3:13
0
LEVEL IN RECYCLE TANK TOO HIGH TO BACKWASH
B3:13
10
0009
DELAY AFTER A REGN TO ALLOW BW RECYCLE/STORAGE TANK TO SETTLE B4 TURNING ON PMP TO EMPTY
MUL
Multiply
Source A N7:33
20<
Source B 60
60<
Dest T4:38.PRE
1200<
0010
UNIT A FILTER IN REGENERATION AFILT_REGN
B21:0
2
UNIT B FILTER IN REGENERATION BFILT_REGN
B22:0
2
DELAY AFTER A REGN TO ALLOW BW RECYCLE/STORAGE TANK TO SETTLE B4 TURNING ON PMP TO EMPTY
TON
Timer On Delay
Timer T4:38
Time Base 1.0
Preset 1200<
Accum 0<
EN
DN
SETTLE TIME REMAINING FOR PV
SUB
Subtract
Source A T4:38.PRE
1200<
Source B T4:38.ACC
0<
Dest N7:47
1200<
ENABLE SETTLE TIME REMAINING DISPLAY
B3:6
3
0011
BACKWASH RECYCLE/WASTE TANK LEVEL DISPLAY
LES
Less Than (A<B)
Source A F8:82
-23.50302<
Source B F8:40
20.0<
BW STORAGE/RECYCLE TANK LEVEL LOW TO DISABLE ITS PUMP
TON
Timer On Delay
Timer T4:39
Time Base 1.0
Preset 10<
Accum 10<
EN
DN

LAD 19 - RECYCLE --- Total Rungs in File = 17



LAD 19 - RECYCLE --- Total Rungs in File = 17



Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
O:0.0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
O:0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
O:0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
O:1.0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1762-OB16 - 16-Output (TRANS-SRC) 10/50 VDC

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
I:0.0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B
I:0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
I:0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
I:0.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
I:0.7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1766 MicroLogix 1400 Series B-Analog
I:2.0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	1	0	1762-IF4 - Analog 4 Chan. Input
I:2.1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1762-IF4 - Analog 4 Chan. Input
I:2.2	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1762-IF4 - Analog 4 Chan. Input
I:2.3	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1762-IF4 - Analog 4 Chan. Input
I:2.4	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1762-IF4 - Analog 4 Chan. Input
I:2.5	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:2.6	1	0	1	0	0	0	0	1	1	1	1	1	0	0	1	1	1762-IF4 - Analog 4 Chan. Input
I:3.0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1762-IF4 - Analog 4 Chan. Input
I:3.1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1762-IF4 - Analog 4 Chan. Input
I:3.2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:3.3	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1762-IF4 - Analog 4 Chan. Input
I:3.4	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1762-IF4 - Analog 4 Chan. Input
I:3.5	1	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:3.6	1	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	1762-IF4 - Analog 4 Chan. Input

Data File S2 (hex) -- STATUS

Main

Processor Mode S:1/0 - S:1/4 = RUN (keyswitch = RUN)
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 1111-0100-0111-0011

Proc

OS Catalog Number S:57 = 1400 User Program Type S:63 = 9108h
OS Series S:58 = B Compiler Revision Number S:64 =
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = B
Processor FRN S:62 =

Scan Times

Maximum (x10 ms) S:22 = 46
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 32
Scan Toggle Bit S:33/9 = 1

Math

Math Overflow Selected S:2/14 = 0 Math Register (lo word) S:13 = 1
Overflow Trap S:5/0 = 0 Math Register (high word) S:14-S:13 = 0
Carry S:0/0 = 0 Math Register (32 Bit) S:14-S:13 = 1
Overflow S:0/1 = 0
Zero Bit S:0/2 = 0
Sign Bit S:0/3 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = RUN (keyswitch = RUN)
Node Address S:15 (low byte) = 0 Outgoing Msg Cmd Pending S:33/2 = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0 Fault Routine S:29 = 0
Startup Protection Fault S:1/9 = 0 Major Error S:6 = 0h
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0 Error Description:
Control Register Error S:5/2 = 0
Major Error Executing User Fault Rtn. S:5/3 = 0
Battery Low S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
ASCII String Manipulation error S:5/15 = 0

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- COMMON

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B3:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
B3:2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:3	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	
B3:4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:5	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:6	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	
B3:7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
B3:10	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	
B3:11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
B3:12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
B3:13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol) Description
T4:0	0	0	0	1.0 sec	60	0	FILT A FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:1	0	0	0	1.0 sec	15	0	DELAY BW FEED PUMP ON TILL THE BW INLET VALVE OPENS FIRST
T4:2	1	1	0	1.0 sec	60	50	FILT C FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:3	0	0	0	1.0 sec	3	0	
T4:4	1	1	0	1.0 sec	135	32	POWER UP DELAY TO ALLOW TIME FOR SCREN TO BOOTUP
T4:5	0	0	0	1.0 sec	10	0	UNIT A FLOW ALARM DELAY
T4:6	0	0	0	1.0 sec	10	0	
T4:7	0	0	0	1.0 sec	10	0	
T4:8	0	0	0	1.0 sec	10	0	UNIT B FLOW RATE ALARM DELAY
T4:9	0	0	0	1.0 sec	40	0	
T4:10	0	0	0	1.0 sec	60	0	FILT B FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:11	0	0	0	1.0 sec	10	0	RESET FLOW TOTAL DELAY FOR OP INTENT
T4:12	0	0	0	1.0 sec	300	0	INTERVESSEL DP REGEN TIMER 5 MIN
T4:13	0	0	0	1.0 sec	5	0	DELAY FOR SETTING DP ALARM STATUS
T4:14	0	0	0	1.0 sec	40	0	(A_REGEN_COUNTER)
T4:15	0	0	0	1.0 sec	40	0	(B_REGEN_COUNTER)
T4:16	0	0	0	1.0 sec	5	0	(SHOW_REGENS)
T4:17	0	0	0	1.0 sec	20	0	
T4:18	0	0	0	1.0 sec	20	0	
T4:19	0	0	0	1.0 sec	10	0	LEVEL IN BW SUPPLY TANK FALLS BELOW PERMISSIVE BW START LEVEL-
T4:20	0	0	0	1.0 sec	5	0	BW RECYCL TANK HI LEVEL DELAY FOR ALM
T4:21	0	0	0	.001 sec	50	0	PULSE OUTPUT LENGTH FOR NAOH PUMP PRESET FROM N7:53
T4:22	0	0	0	1.0 sec	15	0	TREATED WATER/BW SUPPLY STORAGE TANK HI LVL DELAY FOR ALM
T4:23	0	0	0	.001 sec	50	0	PULSE OUTPUT LENGTH FOR POLYMER PUMP PRESET FROM N7:54
T4:24	0	0	0	1.0 sec	2	0	B DRIVE MULTIPORT VALVE TIMER
T4:25	0	0	0	1.0 sec	60	0	VALVE A HOMING DELAY FRO ALM
T4:26	0	0	0	1.0 sec	20	0	
T4:27	0	0	0	1.0 sec	5	0	A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:28	0	0	0	1.0 sec	1	0	
T4:29	0	0	0	1.0 sec	1	0	VALVE C DELAY IN BETWEEN PB ADVANCES TO ALLOW VALVE TO TURN
T4:30	0	0	0	1.0 sec	1	0	VALVE D DELAY IN BETWEEN PB ADVANCES TO ALLOW VALVE TO TURN
T4:31	0	0	0	1.0 sec	20	0	LEVEL OK TO RUN BW FEED PUMP DELAY
T4:32	0	0	0	1.0 sec	60	0	VALVE B HOMING DELAY FRO ALM
T4:33	0	0	0	1.0 sec	10	0	HI LEVEL STOP WELL PUMP DELAY
T4:34	0	0	0	1.0 sec	10	0	TREATED WATER STORAGE TANK/BW SUPPLY WATER AT WELL PUMP RE-STA
T4:35	0	0	0	1.0 sec	60	0	VALVE C HOMING DELAY FRO ALM
T4:36	0	0	0	1.0 sec	60	0	NOT IN DP START MODE FILTER SYSTEM DP DELAY ONE MINUTE TIMER
T4:37	1	0	1	1.0 sec	10	10	ENABLE A REGN START BW RECYCLE/WASTE TANK IS AT LOW DELAY
T4:38	0	0	0	1.0 sec	1200	0	DELAY AFTER A REGN TO ALLOW BW RECYCLE/STORAGE TANK TO SETTLE
T4:39	1	0	1	1.0 sec	10	10	BW STORAGE/RECYCLE TANK LEVEL LOW TO DISABLE ITS PUMP
T4:40	0	0	0	1.0 sec	2	0	
T4:41	0	0	0	1.0 sec	2	0	
T4:42	1	1	0	.01 sec	100	34	ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:43	0	0	0	1.0 sec	3600	0	FILT B DP DELAY TO AVOID SIMUTANEOUS REGN STARTS DUE TO DP IF
T4:44	0	0	0	1.0 sec	4	0	
T4:45	0	0	0	1.0 sec	5	0	
T4:46	0	0	0	1.0 sec	3	0	FILTER B REGN START BY BATCH OR INTERVAL WITH TIME OF DAY DELA
T4:47	0	0	0	1.0 sec	7	0	
T4:48	0	0	0	1.0 sec	9	0	
T4:49	0	0	0	1.0 sec	3	0	FILT B DELAY AFTER LOCKOUT TO AVOID DP START OF BOTH FILTER AT
T4:50	0	0	0	1.0 sec	4	0	
T4:51	0	0	0	1.0 sec	5	0	
T4:52	0	0	0	1.0 sec	3	0	FILTER A REGN START BY BATCH OR INTERVAL WITH TIME OF DAY DELA
T4:53	0	0	0	1.0 sec	60	0	
T4:54	1	1	0	1.0 sec	60	32	ONE MINUTE PULSER FOR MASTER CYCLE COUNTER
T4:55	0	0	0	1.0 sec	10	0	BW RECYCLE/WASTE TANK LEVEL OK DELAY TO ALLOW ITS PMP TO BE ON

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol) Description
C5:0	0	0	0	0	0	0	60	7	FILTER A 60 MINUTE COUNTER FOR TIME INTERVAL
C5:1	0	0	0	0	0	0	10	0	COMMON FILTER CYCLE TIME COUNTER
C5:2	0	0	0	0	0	0	96	0	FILTER A HOURS IN SERVICE COUNTER FOR INTERVAL RUN PRESET FROM
C5:3	0	0	0	0	0	0	1000	0	FLOW SENSOR A PULS CTR
C5:4	0	0	0	0	0	0	60	6	FILTER B 60 MINUTE COUNTER FOR TIME INTERVAL
C5:5	0	0	0	0	0	0	1000	0	FLOW SENSOR B PULS CTR
C5:6	0	0	0	0	0	0	60	8	FILTER C 60 MINUTE COUNTER FOR TIME INTERVAL
C5:7	0	0	0	0	0	0	96	0	FILTER B HOURS IN SERVICE COUNTER FOR INTERVAL RUN PRESET FROM
C5:8	0	0	0	0	0	0	30000	5	A_REGENS
C5:9	0	0	0	0	0	0	30000	1	B_REGENS
C5:10	0	0	0	0	0	0	96	2	
C5:11	0	0	0	0	0	0	3	0	FILT A MASTER STEP CTR 0=SERV,1=BW,2=FR
C5:12	0	0	0	0	0	0	3	1	FILT B MASTER STEP CTR 0=SERV,1=BW,2=FR

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol)	Description
R6:0	0	0	0	0	0	0	0	0	0	0		

Data File N7 (dec)

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	568	19	18	5	10	0	2	1	0	0
N7:10	0	95	12	30000	0	2	28	9	9	30000
N7:20	0	0	6	1	0	10	20	999	0	0
N7:30	0	100	20	20	10	0	1	2	4993	3200
N7:40	0	5	1	80	20	12	0	1200	0	0
N7:50	0	500								

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0	0	0	0	0
F8:5	-34.07938	0	0	-34.09031	10
F8:10	0	0.01093292	0	78.7	10000
F8:15	10000	10000	0.4955527	889.3606	172.7955
F8:20	0.03	33.33334	0	0	0
F8:25	78.7	78.7	0	730.2866	0
F8:30	0	0	0	0	0
F8:35	0	0	0	0	80
F8:40	20	0	0	0	0
F8:45	0	0	0	63.35838	0
F8:50	2873.137	0	0	0	1000000
F8:55	777314.6	1000000	0	10.5	0
F8:60	10	9.5	8	0	2
F8:65	0	0	0	0	0
F8:70	100	0.7099996	0	0	8.657895
F8:75	0.6	0	0	2	0
F8:80	9.46875	1	-23.50302	1000014	1000014
F8:85	0	0	0	0	10
F8:90	0	0	0	0	0
F8:95	14	14	0	0	0
F8:100	0	0	0	0	0
F8:105	12.6	2.3	0	0	0
F8:110	0	0	0	0	0
F8:115	0	0	34.22153	10	0.1312217
F8:120	-34.09031	4.1			

Data File B9 (bin) -- GENCTRL

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B9:0	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	
B9:1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	
B9:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File C10

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol) Description
C10:0	0	0	0	0	0	0	3	0	
C10:1	0	0	0	0	0	0	30000	1	
C10:2	0	0	0	0	0	0	0	0	
C10:3	0	0	0	0	0	0	0	0	
C10:4	0	0	0	0	0	0	0	0	

Data File B21 (bin) -- A BITS

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B21:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B21:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B21:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B21:3	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	1	
B21:4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	
B21:5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B21:6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File B22 (bin) -- B BITS

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B22:0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	
B22:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B22:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B22:3	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	
B22:4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	
B22:5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B22:6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File N34 (dec)

Offset	0	1	2	3	4	5	6	7	8	9
N34:0	0	2	0	4	0	0	0	94	0	0
N34:10	3	4	4	0	0	1	0	1	3	35
N34:20	2	0	0	0	0	0	1	0	0	0
N34:30	7	0	0	0	0	30	58	3	0	94
N34:40	125	0	125	0	60	0	2	2	0	0
N34:50	120	94	102	4	125	0	1000	1000	1	1
N34:60	1	5	100	0	1	15	25	1	0	2
N34:70	0	60	120	60	180	0	0	0	0	0
N34:80	0	0	0	2	0	0	0	0	0	0
N34:90	0	0								

Address (Symbol) = Value [Description]

T4:53.ACC = 0 []
C5:8.ACC = 5 []
B21:0/6 = 0 [MANUAL CONTROLVISIBILITY BIT]
B22:0/6 = 0 [MANUAL CONTROLVISIBILITY BIT]
B23:0/6 = Address does not exist. [C WENT INTO REGN DUETO HI DPMEMORY SEAL]
N34:5 = 0 []
N34:6 (BGPM) = 0 [UNIT B FLOW RATE]

Address/Symbol Database

Address	Symbol	Scope	Description
	FILT_B_END_REGNPB	Global	
	1F	Global	
B3:0/0			VESSEL A DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0/1	CB1_ALM_BIT	Global	ALARM BIT FOR CB1 TRIP
B3:0/2	CB2	Global	ALARM BIT FOR CB2 TRIP
B3:0/3			DP-BASED REGEN CYCLE PUSHBUTTON
B3:0/4			VOLUME-BASED REGEN CYCLE PUSHBUTTON
B3:0/5			TIME-BASED REGEN CYCLE PUSHBUTTON
B3:0/6			VESSEL B DIFFERENTIAL PRESSURE ALARM - NO REGEN
B3:0/7			OSR
B3:0/8			OSR
B3:0/9			OSR
B3:0/10			OSR
B3:0/11			VESSEL A DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0/12			VESSEL B DIFFERENTIAL PRESSURE ALARM - WITH REGEN
B3:0/13			DP REGEN INIT BIT
B3:0/14	REGEN_READY	Global	READY TO START A REGEN CYCLE!
B3:0/15			RESET SV-800-09 DUE TO LOW CO2 PSI
B3:1/0			WELL PUMP ENABLED DISPLAY ON PV
B3:1/1			
B3:1/2			
B3:1/3			
B3:1/4			
B3:1/5	SHOW_REGEN_TTL	Global	
B3:1/6			
B3:1/7			
B3:1/8			
B3:1/9			
B3:1/11			
B3:1/12			
B3:1/14			
B3:1/15	SHOW_BIT	Global	
B3:2/8			
B3:2/9			
B3:2/11			
B3:2/12			SEND NEW TIME AND DATE BUTTON ON PV
B3:2/13			FLOW SENSOR A PULS OSR
B3:2/14			FLOW SENSOR A PULS OSR
B3:3/0			FLOW SENSOR B PULSE OSR
B3:3/1			FLOW SENSOR B PULSE OSR
B3:3/2			
B3:3/3			
B3:3/4			
B3:3/5			
B3:3/14			
B3:3/15	WP_LO	Global	
B3:4/5			
B3:4/6			
B3:4/7			
B3:4/8			
B3:4/9			
B3:4/10			DP CHOSEN AND DELAY DONE OSR
B3:4/11			DP CHOSEN AND DELAY DONE OSR
B3:4/12			
B3:4/13			
B3:4/15			ANY UNIT IN FR PLUS NUMBER OF REGN COUNTERS FOR TRACKING SEQUENTIAL
B3:5/13	REGN_ANY	Global	ANY UNIT IS IN REGN
B3:5/14			
B3:5/15			
B3:6/0			
B3:6/1			
B3:6/2			
B3:6/3			ENABLE SETTLE TIME REMAINING DISPLAY
B3:6/4			
B3:6/5			
B3:6/6			
B3:6/7			
B3:6/8			
B3:6/9			
B3:6/10			
B3:6/11			
B3:6/12			

Address/Symbol Database

Address	Symbol	Scope	Description
B3:6/13			
B3:6/14			A PULS CTR OSR
B3:6/15			RESET TOTALS DELAY FROM ALM SILENCE FOR 10 SECS
B3:7/0			B PULS CTR OSR
B3:7/1			B PULS CTR OSR
B3:7/2			RECYCLE PUMP AUTO PUSHBUTTON
B3:7/3			RECYCLE PUMP OFFLINE PUSHBUTTON
B3:7/4			RECYCLE PUMP HAND PUSHBUTTON
B3:7/5			OSR
B3:7/6			OSR
B3:7/7			OSR
B3:7/8			OSR
B3:7/9			OSR
B3:7/10			OSR
B3:7/11			BW PREVENT ALT B
B3:7/12			BW PREVENT ALT A
B3:7/13			VESSEL A BACKWASH PREVENTED BY HIGH WASTE TANK LEVEL
B3:7/14			VESSEL B BACKWASH PREVENTED BY HIGH WASTE TANK LEVEL
B3:7/15			VESSEL A DELAYED BW ALARM
B3:8/0			OSR
B3:8/1			OSR
B3:8/2			OSR
B3:8/3			OSR
B3:8/4			OSR
B3:8/5			OSR
B3:8/6			OSR
B3:8/7			OSR
B3:8/8			VESSEL B DELAYED BW ALARM
B3:8/9			
B3:9/0			
B3:9/1			
B3:9/2			
B3:9/3			MODULATING IS BEING FORCED
B3:9/10			
B3:9/11			
B3:9/12			
B3:9/13			
B3:10/0			
B3:10/1			C FILT PULS OSR
B3:10/2			C FILT PULS OSR
B3:10/3			FILT A MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/4			FILT A MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/5			
B3:10/6			FILT B MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/7			FILT B MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/8			FILT B MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC DISPLAY
B3:10/9			FILT C MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/10			FILT C MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC
B3:10/11			FILT C MULTIPORT VALVE OUTPUT FOR ADV PB LOCKOUT LOGIC DISPLAY
B3:10/12			FV101A VLV INDICATOR ON PV
B3:10/13			FV102A VLV INDICATOR ON PV
B3:10/14			FV101B VLV INDICATOR ON PV
B3:10/15			FV102B VLV INDICATOR ON PV
B3:11/0			FV101C VLV INDICATOR ON PV
B3:11/1			FV102C VLV INDICATOR ON PV
B3:11/2			DP START MODE CHOSEN AND DP DELAY IS DONE OSR FOR ALM
B3:11/3			DP START MODE CHOSEN AND DP DELAY IS DONE OSR FOR ALM
B3:11/4			HI DP AND DP START MODE CHOSEN
B3:11/5			
B3:11/6			VLV A HOME ALM OSR
B3:11/7			VLV A HOME ALM OSR
B3:11/8			VLV A HOME ALM SEAL
B3:11/9			VLV B HOME ALM OSR
B3:11/10			VLV B HOME ALM OSR
B3:11/11			VLV B HOME ALM SEAL
B3:11/12			
B3:11/13			
B3:11/14			
B3:12/0			REGN START MODE IS NOT DP BUT DP IS HI ALM OSR
B3:12/1			REGN START MODE IS NOT DP BUT DP IS HI ALM OSR
B3:12/2			REGN START MODE IS NOT DP BUT DP IS HI ALM SEAL
B3:12/3	ALMSILENCEPB	Global	ALARM RESET PB ON OIT

Address/Symbol Database

Address	Symbol	Scope	Description
B3:12/4			
B3:12/5			
B3:12/6			
B3:12/7			
B3:12/8			D FILT HI DP ALM SEAL
B3:12/9			D FILT HI DP RESET BUTTON
B3:12/10			E FILTER DP HI ALM OSR
B3:12/11			E FILTER DP HI ALM OSR
B3:12/12			E FILT HI DP
B3:12/13			REGN START MODE IS NOT DP BUT DP IS HI ALM MSG
B3:12/14			HI DP / NO DP START HORN
B3:13/0			RECYCLE TANK LEVEL LOW - OK TO START A BW
B3:13/1			ENABLE BW RECYCLE/WASTE TANK PUMP SEAL UNTIL LOW LEVEL CUT OUT OF F
B3:13/2			OSR
B3:13/3			OSR
B3:13/4			BW RECYCE/WASTE PUMP BIT
B3:13/5			NaCl PUMP OUT BIT
B3:13/6			OSR
B3:13/7			OSR
B3:13/8			RECYCLE TANK HH LEVEL ALARM BIT
B3:13/9			
B3:13/10			LEVEL IN RECYCLE TANK TOO HIGH TO BACKWASH
B3:14/0			
B3:14/1			
B3:19/0			
B3:20/0			
B9:0/0			
B9:0/1	MV_013A	Global	ENERGIZE A BACKWASH INLET VALVE
B9:0/2	MV_012A	Global	ENERGIZE A SERVICE OUTLET VALVE
B9:0/3	MV_014A	Global	ENERGIZE A BACKWASH OUTLET VALVE
B9:0/4	MV_015A	Global	ENERGIZE A RINSE OUTLET VALVE
B9:0/5	MV_011B	Global	ENERGIZE B RAW WATER INLET VALVE
B9:0/6	MV_013B	Global	ENERGIZE B BACKWASH INLET VALVE
B9:0/7	MV_012B	Global	ENERGIZE B SERVICE OUTLET VALVE
B9:0/8	MV_014B	Global	ENERGIZE B BACKWASH OUTLET VALVE
B9:0/9	MV_015B	Global	ENERGIZE B RINSE OUTLET VALVE
B9:0/10			ENERGIZE ALARM HORN
B9:0/11			
B9:0/12	MV_011A	Global	ENERGIZE A RAW WATER INLET VALVE
B9:0/13			
B9:0/14			
B9:0/15			
B9:1/0			WELLPUMP CALL BIT
B9:1/1			PLACEHOLDER
B9:1/2			PLACEHOLDER
B9:1/3			
B9:1/4			
B9:1/5			
B9:1/6			
B9:1/7			
B9:1/8			
B9:1/9			
B9:1/10			
B9:1/11			
B9:1/12			
B9:1/13			
B9:1/14	MV_011A_MC	Global	MV-011A MANUAL CONTROL
B9:1/15			
B9:2/0			
B9:2/1			A CIRCUIT BREAKER INPUT BIT
B9:2/2			B CIRCUIT BREAKER INPUT BIT
B9:2/3			
B9:2/4			MANUAL CONTROL VISIBILITY BIT
B9:2/5			
B9:2/6	MV_015B_MC	Global	MV-015B MANUAL CONTROL
B9:2/7	MV_014B_MC	Global	MV-014B MANUAL CONTROL
B9:2/8	MV_012B_MC	Global	MV-012B MANUAL CONTROL
B9:2/9	MV_013B_MC	Global	MV-013B MANUAL CONTROL
B9:2/10	MV_011B_MC	Global	MV-011B MANUAL CONTROL
B9:2/11	MV_015A_MC	Global	MV-015A MANUAL CONTROL
B9:2/12	MV_014A_MC	Global	MV_014A MANUAL CONTROL
B9:2/13	MV_012A_MC	Global	MV-012A MANUAL CONTROL

Address/Symbol Database

Address	Symbol	Scope	Description
B9:2/14	MV_013A_MC	Global	MV-013A MANUAL CONTROL
B9:2/15			
B10:0			
B12:3/11			B FORCE OPEN VALVE FV12B WHEN IN MANUAL
B12:3/12			B FORCE OPEN VALVE FV13B WHEN IN MANUAL
B12:3/13			B FORCE ON VALVE FV11B WHEN IN MANUAL
B12:3/14			B FORCE OPEN VALVE FV14B WHEN IN MANUAL
B12:3/15			B FORCE OPEN VALVE FV15B WHEN IN MANUAL
B13:0/3			FILTER MEMORY OF SEQUENTIAL START REGN
B13:0/11			FILTER SYSTEM START REGN DUE TO DP W/TIME OF DAY DELAY OSR
B13:2/1			FILT B START REGN DUE TO DP W/TIME OF DAY DELAY OSR
B13:2/5	ALMBANNERCLOSEPB	Global	ALARM BANNER CLOSE BUTTTON
B13:2/7			FILT B DP DELAY COUNTER DONE ALRM OSR
B13:2/8			FILT B DP DELAY COUNTER DONE ALRM OSR
B13:2/9			FILT B HIGH DP ALARM BANNER
B13:2/10			FILT B HIGH DP SEAL FOR ALM
B13:2/13			RESET FILT D SERVICE/REGN GALLON TOT ON A NEW SERVICE RUN
B13:2/14			RESET FILT D SERVICE/REGN GALLON TOT ON A NEW SERVICE RUN
B13:3/5			B DAYS LEFT DISPLAY FOR OIT IF INTERVAL START IS ENABLED
B13:3/11			A FORCE ON RAW WATER INLET VALVE FV101A WHEN IN MANUAL
B13:3/12			A FORCE OFF VALVE FV12A WHEN IN MANUAL
B13:3/13			A FORCE OFF VALVE FV3A WHEN IN MANUAL
B13:3/14			A FORCE OFF VALVE FV4A WHEN IN MANUAL
B13:3/15			A FORCE OFF VALVE FV5A WHEN IN MANUAL
B13:4/0			E BYPASS VALVE BV116 FORCE ON IN MANUAL MODE
B13:4/1			COMMON BW INLET VALVE FORCE OFF VALVE FV115 WHEN IN MANUAL
B13:4/2			B FORCE OFF VALVE FV12B WHEN IN MANUAL
B13:4/3			B FORCE OFF VALVE FV3B WHEN IN MANUAL
B13:4/4			B FORCE OFF VALVE FV4B WHEN IN MANUAL
B13:4/5			B FORCE OFF VALVE FV5B WHEN IN MANUAL
B13:4/6			RESET ALL FLOW TOTAL PB ON OIT
B13:4/15			C FORCE OFF VALVE FV11C WHEN IN MANUAL
B13:5/9			FILT D ADVANCE REGN PB OSR FROM OIT ONLY OSR
B13:5/10			FILT D ADVANCE REGN PB OSR FROM OIT ONLY OSR
B13:6/5			FILT D END REGN PB
B13:6/10			C FORCE OFF VALVE FV12C WHEN IN MANUAL
B13:6/11			D FORCE OFF VALVE FV11D WHEN IN MANUAL
B13:6/12			D FORCE OFF VALVE FV12D WHEN IN MANUAL
B13:15/5			UNIT A FILTER SETTLE
B13:15/15			FILT SEQUENTIAL REGN MEMORY START B AFTER A IS DONE
B13:16/5			UNIT B FILTER SETTLE
B21:0/0	AFILT_FR	Global	UNIT A FAST RINSE
B21:0/1	AFILT_OFFLINE	Global	A FORCED OFFLINE
B21:0/2	AFILT_REGN	Global	UNIT A FILTER IN REGENERATION
B21:0/3			CB1 ALARM OSR STORAGE
B21:0/4			CB1 ALARM OSR OUTPUT
B21:0/5			
B21:0/6			MANUAL CONTROL VISIBILITY BIT
B21:0/7			FILTER A DP WITH TIME OF DAY DELAY IMMED OSR START
B21:0/8	A_AUTO_MAN_PB	Global	A AUTO/MAN PUSHBUTTON
B21:0/9			FILT A /ADVANCE REGN PB FROM OIT ONLY
B21:0/10			FILT A ADVANCE REGN PB OSR FROM OIT ONLY OSR
B21:0/11			FILT A START/ADVANCE REGN PB OSR FROM OIT ONLY OSR
B21:0/12			FILT A ADVANCE REGN PB OSR FROM OIT ONLY
B21:0/13	AFILT_BW	Global	UNIT A FILTER BACKWASH
B21:0/14	A_ON_OFFLINE_PB	Global	A ONLINE / OFFLINE PUSHBUTTON
B21:0/15			A ON/OFF OSR1
B21:1/0	A_MAN_REGEN_PB	Global	A MANUAL REGEN PUSHBUTTON
B21:1/1			FILT A REGN START PB ONLY- NOT ADVANCE
B21:1/2	FILT_PULSE	Global	FILTER PULSE MASTER CTR
B21:1/3			FILTER A REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DLY-NO
B21:1/4	AFILT_SERVICE	Global	UNIT A FILTER IN SERVICE
B21:1/5	A_ABORT_PB	Global	A REGEN ABORT PB
B21:1/6			OSR
B21:1/7			UNIT A HIGH FLOW OSR
B21:1/8			UNIT A HIGH FLOW OSR
B21:1/9			UNIT A HIGH FLOW ALARM MSG
B21:1/10			
B21:1/11			UNIT A LO FLOW OSR
B21:1/12			UNIT A LO FLOW OSR
B21:1/13			UNIT A LO FLOW ALARM MSG
B21:1/14			

Address/Symbol Database

Address	Symbol	Scope	Description
B21:1/15			A ON/OFF OSR2
B21:2/0			OSR
B21:2/1			OSR
B21:2/2			OSR
B21:2/3			OSR
B21:2/4			OSR
B21:2/5			OSR
B21:2/6			OSR
B21:2/7			OSR
B21:2/8			OSR
B21:2/9			OSR
B21:2/10			OSR
B21:2/11			OSR
B21:2/12			OSR
B21:2/13			OSR
B21:2/14			OSR
B21:2/15			OSR
B21:3/0			FLOW SENSOR A PULS OSR
B21:3/1			FLOW SENSOR A PULS OSR
B21:3/2			
B21:3/3			
B21:3/4			
B21:3/5			FILT A RAW WATER INLET VALVE GRAPHIC DISPLAY
B21:3/6			FILT A SERVICE OUTLET GRAPHIC DISPLAY
B21:3/7			FILT A BW OUT GRAPHIC DISPLAY
B21:3/8			FILT A RINSE OUT GRAPHIC DISPLAY
B21:3/9			A SERVICE THOUSANDS > 1 MILLION
B21:3/10			UNIT A ONGOING SERVICE FLOW CTR IS GREATER TAHN 1 MILLION OSR
B21:3/11			OSR FLOW MATH
B21:3/12			OSR FLOW MATH2
B21:3/13			
B21:3/14			
B21:3/15			
B21:4/0			
B21:4/1			FILT A BW IN GRAPHIC DISPLAY
B21:4/2			
B21:4/3			
B21:4/4			
B21:4/5			
B21:4/6			
B21:4/7			
B21:4/8			
B21:4/9			
B21:4/10			
B21:4/11			
B21:4/12			
B21:4/13			
B21:4/14			
B21:4/15			
B21:6/0			FLTA DP START
B22:0/0	BFILT_FR	Global	UNIT B FAST RINSE
B22:0/1	BFILT_OFFLINE	Global	B FORCED OFFLINE
B22:0/2	BFILT_REGN	Global	UNIT B FILTER IN REGENERATION
B22:0/3			CB2 ALARM OSR STORAGE
B22:0/4			CB2 ALARM OSR OUTPUT
B22:0/5			WELL PUMP AUX BIT
B22:0/6			MANUAL CONTROL VISIBILITY BIT
B22:0/7			
B22:0/8	B_AUTO_MAN_PB	Global	B AUTO/MAN PUSHBUTTON
B22:0/9			FILT B MANUAL ADVANCE PB FROM OIT
B22:0/10			FILT B ADVANCE REGN PB OSR FROM OIT ONLY
B22:0/11			FILT B /ADVANCE REGN PB OSR FROM OIT ONLY
B22:0/12			FILT B ADVANCE REGN PB OSR FROM OIT ONLY
B22:0/13	BFILT_BW	Global	UNIT B FILTER BACKWASH
B22:0/14	B_ON_OFFLINE_PB	Global	B ONLINE / OFFLINE PUSHBUTTON
B22:0/15			B ON/OFF OSR1
B22:1/0	B_MAN_REGEN_PB	Global	B MANUAL REGEN PUSHBUTTON
B22:1/1			FILT B MANUAL START PB-START ONLY NOT ADVANCE
B22:1/2	BFILT_PULSE	Global	UNIT B FILTER PULSE MASTER CTR
B22:1/3			FILTER B REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DLYY-M
B22:1/4	BFILT_SERVICE	Global	UNIT B FILTER IN SERVICE
B22:1/5	B_ABORT_PB	Global	B REGEN ABORT PB

Address/Symbol Database

Address	Symbol	Scope	Description
B22:1/6			
B22:1/7			UNIT B HIGH FLOW OSR
B22:1/8			UNIT B HIGH FLOW OSR
B22:1/9			UNIT B HIGH FLOW ALARM MSG
B22:1/10			
B22:1/11			UNIT B LO FLOW OSR
B22:1/12			UNIT B LO FLOW OSR
B22:1/13			UNIT B LO FLOW ALARM MSG
B22:1/14			
B22:1/15			B ON/OFF OSR2
B22:2/0			OSR
B22:2/1			OSR
B22:2/2			OSR
B22:2/3			OSR
B22:2/4			OSR
B22:2/5			OSR
B22:2/6			OSR
B22:2/7			OSR
B22:2/8			OSR
B22:2/9			OSR
B22:2/10			OSR
B22:2/11			OSR
B22:2/12			OSR
B22:2/13			OSR
B22:2/14			OSR
B22:2/15			OSR
B22:3/0			FLOW SENSOR B PULS OSR
B22:3/1			FLOW SENSOR B PULS OSR
B22:3/2			
B22:3/3			FILTER B REGN START BY TOTAL BATCH OR INTERVAL WITH TIME OF DAY DEI
B22:3/4			FILTER B REGN START BY TOTAL BATCH OR INTERVAL WITH TIME OF DAY DEI
B22:3/5			FILT B RAW INLET GRAPHIC DISPLAY
B22:3/6			FILT B SERVICE OUTLET/BW INLET TO OTHER TANK GRAPHIC DISPLAY
B22:3/7			FILT B BW OUT GRAPHIC DISPLAY
B22:3/8			FILT B RINSE OUT GRAPHIC DISPLAY
B22:3/9			B SERVICE THOUSANDS > 1 MILLION
B22:3/10			UNIT B ONGOING SERVICE FLOW CTR IS GREATER THAN 1 MILLION OSR
B22:3/11			
B22:3/12			OSR FLOW MATH
B22:3/13			OSR FLOW MATH2
B22:3/14			
B22:3/15			
B22:4/0			
B22:4/1			FILT B BW IN GRAPHIC DISPLAY
B22:4/2			
B22:4/3			
B22:4/4			
B22:4/5			
B22:4/6			
B22:4/7			
B22:4/8			
B22:4/9			
B22:4/10			
B22:4/11			
B22:4/12			
B22:4/13			
B22:4/14			
B22:4/15			
B22:5/0			
B22:5/1			
B22:5/2			
B22:5/3			
B22:5/4			
B22:5/5			
B22:5/6			
B22:6/0			FLTB DP START
B23:0/0	CFILT_FR	Global	UNIT C FAST RINSE
B23:0/1	CFILT_OFFLINE	Global	C FORCED OFFLINE
B23:0/2			
B23:0/3			FILT C DP STARTREGN MEM IN CASE OF REGN LO OR TREATED /BW SUPLYTANK
B23:0/4			
B23:0/5			

Address/Symbol Database

Address	Symbol	Scope	Description
B23:0/6			C WENT INTO REGN DUE TO HI DP MEMORY SEAL
B23:0/7			FILTER C DP WITH TIME OF DAY DELAY IMMED OSR START
B23:0/8			FILTER C DP WITH TIME OF DAY DELAY IMMED OSR START
B23:0/9			FILT C /ADVANCE REGN PB FROM OIT ONLY
B23:0/10			FILT C ADVANCE REGN PB OSR FROM OIT ONLY OSR
B23:0/11			FILT C ADVANCE REGN PB OSR FROM OIT ONLY OSR
B23:0/12			FILT C ADVANCE REGN PB OSR FROM OIT ONLY
B23:0/13	CFILT_BW	Global	UNIT C FILTER BACKWASH
B23:0/14	CFILT_SETTLE	Global	FILTER C IN SETTLE
B23:0/15			FILT C RUN INTERVAL REGN START MEM.+ AND BW WASTETANK IS NOT LOW EN
B23:1/0			FILT C DP AFTER LVL OR REGN LOCKOUT TO AVOID TC START OF BOTH FILT
B23:1/1			FILT C REGN START PB ONLY- NOT ADVANCE
B23:1/2	CFILT_PULSE	Global	UNIT C PULSE MASTER CTR
B23:1/3			FILTER C REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DLYY-M
B23:1/4	CFILT_SERVICE	Global	UNIT C FILTER IN SERVICE
B23:1/5			FILT C END REGN PB
B23:1/6			RESET FILT C SERVICE/REGN GALLON TOT ON A NEW SERVICE RUN
B23:1/7			UNIT C HIGH FLOW OSR
B23:1/8			UNIT C HIGH FLOW OSR
B23:1/9			UNIT C HIGH FLOW ALARM MSG
B23:1/10			UNIT C HIGH FLOW ALARM FOR HORN
B23:1/11			UNIT C LO FLOW OSR
B23:1/12			UNIT C LO FLOW OSR
B23:1/13			UNIT C LO FLOW ALARM MSG
B23:1/14			UNIT C LO FLOW ALARM FOR HORN
B23:1/15			FILT C RUN INTERVAL REGN START MEM.+ AND BW WASTETANK IS NOT LOW EN
B23:2/0			FILT C DP START REGN MEMORY IN CASE OF REGN LO OR BW SUPPLYTANK NOT
B23:2/1			FILT C DP START REGN MEMORY IN CASE OF REGN LO OR BW SUPPLYTANK NOT
B23:2/2			UNIT C DP START REQUESTED AND TREATED WTR/BW SUPPLY TNK IS NOT HI E
B23:2/3			UNIT C DP START REQUESTED AND BW SUPPLY TNK IS NOT HI ENOUGH ALM MS
B23:2/4			FILT C RUN INTERVAL REGN START MEM.+ AND BW WASTETANK IS NOT LOW EN
B23:2/5			FILT C DP STARTREGN MEM IN CASE OF WASTE NOT TANK LOW ENOUGH TO STA
B23:2/6			FILT C DP STARTREGN MEM IN CASE OF WASTE NOT TANK LOW ENOUGH TO STA
B23:2/7			FILT C DP STARTREGN MEM IN CASE OF WASTE NOT TANK LOW ENOUGH TO STA
B23:2/8			C BATCH REGN START MEMORY AND TREATED WATER/BW SUPPLY TANK IS HI OS
B23:2/9			C BATCH REGN START MEMORY AND TREATED WATER/BW SUPPLY TANK IS HI OS
B23:2/10			C BATCH REGN START MEM.TREATED/BW SUPPLY TANK IS NOT HI ENOUGH ALM
B23:2/11			C BATCH REGN START MEM.TREATED/BW SUPPLY TANK IS NOT HI ENOUGH ALM
B23:2/12			FILT C RUN INTERVAL REGN START MEMORY AND BW SUPPLYTANK IS NOTHI EN
B23:2/13			FILT C RUN INTERVAL REGN START MEMORY AND BW SUPPLYTANK IS NOTHI EN
B23:2/14			C RUN TIME ELAPSED DONE START MEMORY AND BW TANK IS HI OR REGN LOCK
B23:2/15			FILT C RUN INTERVAL REGN START MEM.+TREA TED/BW SUPPLYTANK IS NOTHI
B23:3/0			FLOW SENSOR C PULS OSR
B23:3/1			FLOW SENSOR C PULS OSR
B23:3/2			RESET FILT C SERVICE/REGN GALLON TOT ON A NEW SERVICE RUN
B23:3/3			FILTER C REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DELAY
B23:3/4			FILTER C REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DELAY
B23:3/5			FILT C RAW IN GRAPHIC DISPLAY
B23:3/6			FILT C SERVICE OUTLET/BW INLET TO OTHER TANK GRAPHIC DISPLAY
B23:3/7			FILT C BW OUT GRAPHIC DISPLAY
B23:3/8			FILT C RINSE OUT GRAPHIC DISPLAY
B23:3/9			UNIT C ONGOING SERVICE FLOW CTR IS GREATER THAN 1 MILLION OSR
B23:3/10			UNIT C ONGOING SERVICE FLOW CTR IS GREATER THAN 1 MILLION OSR
B23:3/11			
B23:3/12			C BATCH REGN START MEMORY& BW WASTE TANK IS NOT LO ENOUGH TO ACCEPT
B23:3/13			C BATCH REGN START MEMORY& BW WASTE TANK IS NOT LO ENOUGH TO ACCEPT
B23:3/14			C BATCH REGN START MEMORY& BW WASTE TANK IS NOT LO ENOUGH TO ACCEPT
B23:3/15			
B23:4/0			FILT C DP STARTREGN MEM IN CASE OF WASTE NOT TANK LOW ENOUGH TO STA
B23:15/0	CARBAFILTREGN	Global	CARBON FILTER IN REGENERATION
B24:0/0	DFILT_FR	Global	UNIT D FAST RINSE
B24:0/1	DFILT_OFFLINE	Global	D FORCED OFFLINE
B24:0/2			
B24:0/3			FILT D DP STARTREGN MEM IN CASE OF REGN LO OR TREATED /BW SUPLYTANK
B24:0/4			D BATCH REGN START MEMORY& TREATED/BW SUPPLY TANK IS NOT HI ENOUGH
B24:0/5			FILT D RUN INTERVAL REGNSTART MEMORY AND BW SUPPLY TANK IS NOT HI E
B24:0/6			D WENT INTO REGN DUE TO HI DP MEMORY SEAL
B24:0/7			FILTER D DP WITH TIME OF DAY DELAY IMMED OSR START
B24:0/8			FILTER D DP WITH TIME OF DAY DELAY IMMED OSR START
B24:0/9			FILT D /ADVANCE REGN PB FROM OIT ONLY
B24:0/10			FILT D ADVANCE REGN PB OSR FROM OIT ONLY OSR
B24:0/11			FILT D ADVANCE REGN PB OSR FROM OIT ONLY OSR

Address/Symbol Database

Address	Symbol	Scope	Description
B24:0/12			FILT D ADVANCE REGN PB OSR FROM OIT ONLY
B24:0/13	DFILT_BW	Global	UNIT D IN BACKWASH
B24:0/14	DFILT_SETTLE	Global	UNIT D IN SETTLE
B24:1/0			FILT D DP AFTER LVL OR REGN LOCKOUT TO AVOID TC START OF BOTH FIL
B24:1/1			FILT D REGN START PB ONLY- NOT ADVANCE
B24:1/2	DFILT_PULSE	Global	UNIT D PULSE MASTER CTR
B24:1/3			FILTER D REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DLYY-M
B24:1/4	DFILT_SERVICE	Global	UNIT D FILTER IN SERVICE
B24:1/5			FILT D END REGN PB
B24:1/7			UNIT D HIGH FLOW OSR
B24:1/8			UNIT D HIGH FLOW OSR
B24:1/9			UNIT D HIGH FLOW ALARM MSG
B24:1/10			UNIT D HIGH FLOW ALARM FOR HORN
B24:1/11			UNIT D LO FLOW OSR
B24:1/12			UNIT D LO FLOW OSR
B24:1/13			UNIT D LO FLOW ALARM MSG
B24:1/14			UNIT D LO FLOW ALARM FOR HORN
B24:2/5			FILT D DP STARTREGN MEM IN CASE OF WASTE NOT TANK LOW ENOUGH TO STA
B24:2/8			D BATCH REGN START MEMORY AND TREATED WATER/BW SUPPLY TANK IS HI OS
B24:2/9			D BATCH REGN START MEMORY AND TREATED WATER/BW SUPPLY TANK IS HI OS
B24:2/10			D BATCH REGN START MEM.TREATED/BW SUPPLY TANK IS NOT HI ENOUGH ALM
B24:2/11			D BATCH REGN START MEM.TREATED/BW SUPPLY TANK IS NOT HI ENOUGH ALM
B24:2/12			FILT D RUN INTERVAL REGNSTART MEMORY AND BW SUPPLY TANK IS NOT HI E
B24:2/13			FILT D RUN INTERVAL REGNSTART MEMORY AND BW SUPPLY TANK IS NOT HI E
B24:2/14			FILT D RUN INTERVAL REGNSTART MEMORY AND BW SUPPLY TANK IS NOT HI E
B24:2/15			FILT D RUN INTERVAL REGN START MEM.+TREA TED/BW SUPPLYTANK IS NOTHI
B24:3/3			FILTER D REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DELAY
B24:3/4			FILTER D REGN START TOTAL BATCH OR INTERVAL WITH TIME OF DAY DELAY
B24:3/11			D BATCH REGN START MEMORY& BW WASTE TANK IS NOT LO ENOUGH TO ACCEPT
B24:3/15			FILT D RUN INTERVAL REGN START MEM.+ AND BW WASTETANK IS NOT LOW EN
B221:0/0			
C5:0			FILTER A 60 MINUTE COUNTER FOR TIME INTERVAL
C5:0/DN			
C5:1			COMMON FILTER CYCLE TIME COUNTER
C5:1.PRE			FILTER MINUTES COUNTER
C5:1/DN			
C5:2			FILTER A HOURS IN SERVICE COUNTER FOR INTERVAL RUN PRESET FROM N34
C5:2.PRE			
C5:2.ACC			
C5:2/DN			
C5:3			FLOW SENSOR A PULS CTR
C5:3.ACC			
C5:3/DN			
C5:4			FILTER B 60 MINUTE COUNTER FOR TIME INTERVAL
C5:4.ACC			
C5:5			FLOW SENSOR B PULS CTR
C5:6			FILTER C 60 MINUTE COUNTER FOR TIME INTERVAL
C5:7			FILTER B HOURS IN SERVICE COUNTER FOR INTERVAL RUN PRESET FROM N34
C5:7/DN			
C5:8			A_REGENS
C5:8/DN			
C5:9			B_REGENS
C5:9.PRE			
C5:9.ACC			
C5:9/DN			
C5:10			
C5:10/DN			
C5:11			FILT A MASTER STEP CTR 0=SERV,1=BW,2=FR
C5:11.ACC			A FILTER COUNTER
C5:12			FILT B MASTER STEP CTR 0=SERV,1=BW,2=FR
C5:12.ACC			FILTER B COUNTER
C5:13			
C5:14			
C5:15			
C5:16			DUMP VALVE OPEN CTR PRE FROM N7:7
C5:16/DN			
C5:17			
C5:17.PRE			
C5:17/DN			
C5:18			
C5:18.PRE			
C5:18/DN			

Address/Symbol Database

Address	Symbol	Scope	Description
C5:19			
C5:20			
C5:21			DUMP VALVE OPEN DELAY AFTER A FR PRESET FROM
C5:21.ACC			
C5:21/DN			
C5:22			
C5:22.PRE			
C5:23			
C5:24			
C5:24/DN			
C5:25			
C5:26			
C5:27			
C5:28			
C5:29			
C10:0			
C10:1			
C15:1			FILTER B HOUR COUNTER FOR RUN INTERVAL LOGIC
C15:10			FILTER B DP MINUTES CTR TO INITIATE REGN ENTRY FROM N17:15
C15:10.ACC			
F8:0			A BATCH MILLIONS FOR DISPLAY
F8:1			A BATCH THOUSANDS FOR DISPLAY
F8:2			A SERVICE THOUSANDS
F8:3			B BATCH THOUSANDS FOR DISPLAY
F8:4			B BATCH MILLIONS FOR DISPLAY
F8:5			VESSEL B INLET PRESSURE SCALED VALUE
F8:6			B BATCH THOUSANDS
F8:7			A BATCH THOUSANDS
F8:8			VESSEL B OUTLET PRESSURE SCALED VALUE
F8:9			BATCH ENTRY FOR BATCH START
F8:10			UNIT B GALLONS PER SECOND
F8:11			VESSEL B ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
F8:12			B SERVICE THOUSANDS
F8:13			
F8:14	A_CAP_REM	Global	
F8:15	B_CAP_REM	Global	
F8:16	T_GAL	Global	
F8:17			
F8:18			
F8:19			
F8:20			TEMP # FOR PULSE MATH-PULSE DURATION/1000 ENTERED FROM N34:37
F8:21			1/F8:20 PULSE MATH FOR FREQUENCY MATH
F8:22			
F8:23			
F8:24			
F8:25			CHEM1 GALLONS PER PULSE
F8:26			B K FACTOR
F8:27			
F8:28			
F8:29			
F8:30			A PULSES PER SECOND
F8:31			A PULSES PER MINUTE
F8:32			B PULSES PER SECOND
F8:33			B PULSES PER MINUTE
F8:34			A GALLONS PER SECOND
F8:35	A_REG_THOU	Global	
F8:36	A_REG_MIL	Global	
F8:37	B_REG_THOU	Global	
F8:38	B_REG_MIL	Global	
F8:39			TANK HI
F8:40			TANK LO
F8:41			
F8:42			
F8:43			
F8:44			
F8:45			A BATCH MILLIONS
F8:46			A SERVICE MILLIONS
F8:47			B SERVICE MILLIONS
F8:48			TEMP STORAGE OF A+B REGN WATER FLOW TOTAL FOR MATH FOR DISPLAY
F8:49			B BATCH MILLIONS
F8:50			TEMP ADD OF A+B ONGOING SER FLOW TOT FOR SYSTEM TOT THOUSANDS CALC
F8:51			

Address/Symbol Database

Address	Symbol	Scope	Description
F8:52			TEMP STORAGE OF A+B SERVICE AND REGN WATER FLOW TOTAL FOR MATH FOR
F8:53			
F8:54			
F8:55			
F8:56			
F8:57			
F8:58			
F8:59			
F8:60			
F8:61			WELL PUMP HI LEVEL TURN OFF ENTRY
F8:62			WELL PUMP ENABLE LEVEL ENTRY
F8:63			
F8:64			
F8:65			
F8:66			A+B SERVICE THOUSANDS
F8:67			GALLONS PER 1/10 SECOND FOR CHEMPUMP2 MATH
F8:68			C GALLONS PER 1/10TH SECOND FOR PTO MATH
F8:69			P2 LOGIC FOR CL TOTAL INLET GALL PER 1/10TH SECOND FOR CHEM PULSINC
F8:70			
F8:71			P-2 CL INLET TOTAL GALLONS FOR CHEM PACING
F8:72			TEMP STORAGE OF A+B SERV ONGOING FLOW TOT MILLIONS FIELD
F8:73			
F8:74			
F8:76			A+B SERVICE MILLIONS
F8:77			A+B SERVICE THOUSANDS
F8:78			TEMP ADD OF A+B+C MILLIONS FIELD WHEN THOUS FIELD IS GRT 1 MILL AND
F8:79			
F8:80			
F8:81			
F8:82			BACKWASH RECYCLE/WASTE TANK LEVEL DISPLAY
F8:83			TEMP ADD OF A+B THOUSANDS FIELD
F8:84			TEMP ADD OF A+B THOUSANDS FIELD
F8:85			TEMP ADD A+B MILLIONS FIELD
F8:86			TEMP ADD OF A+B MILLIONS FIELD
F8:87			TEMP ADD OF A+B THOUS FIELD
F8:88			TEMP STORAGE OF A+B THOUSANDS FIELD
F8:89			GAL TOTAL ENTRY FROM PV TO STROKE PUMP
F8:91			NAOCL LOGIC FOR TOTAL INLET GALL PER 1/10TH SECOND FOR CHEM PULSINC
F8:92			CHEMPUMP2 LOGIC FOR TOTAL INLET GALL PER 1/10TH SECOND FOR CHEM PUI
F8:93			NAOH LOGIC FOR TOTAL INLET GALL PER 1/10TH SECOND FOR CHEM PULSING
F8:94			POLYMER LOGIC FOR TOTAL INLET GALL PER 1/10TH SECOND FOR CHEM PULSI
F8:95			
F8:96			
F8:101			
F8:102			CHEMPUMP2 GALLONS PER 1/10 SECOND TOTALIZER FOR PUMP PACING LOGIC
F8:103			NAOH A+B GALLONS PER 1/10 SECOND TOTALIZER FOR NAOH PUMP PACING LOC
F8:104			POLYMER A+B GALLONS PER 1/10 SECOND TOTALIZER FOR NAOH PUMP PACING
F8:105			
F8:106			
F8:107			
F8:117			VESSEL A ACTUAL DIFFERENTIAL PRESSURE INLET - OUTLET
F8:118	DP_SETPOINT	Global	PRESSURE DIFFERENTIAL SETPOINT
F8:119			VESSEL A INLET PRESSURE SCALED VALUE
F8:120			VESSEL A OUTLET PRESSURE SCALED VALUE
F8:121			GPM SETPOINT FOR CHEMPUMP1
F8:122			GPM SETPOINT FOR CHEMPUMP2
F8:123			GALLON TOTAL ENTRY FROM PV TO STROKE NAOH PUMP
F8:124			NAOH A+B GALLONS PER 1/10 SECOND TOTALIZER FOR POLYMER PUMP PACING
I:0.0			
I:0/0	FE_101A_IN	Global	FLOW SENSOR A PULSE INPUT
I:0/1			SPARE
I:0/2			SPARE
I:0/3			SPARE
I:0/4	FE_101B_IN	Global	FLOW SENSOR B PULSE INPUT
I:0/5	SPARE	Global	
I:0/6			
I:0/7			SPARE
I:0/8	FE_101C_IN	Global	FLOW SENSOR C PULSE INPUT
I:0/9			SPARE
I:0/10			SPARE
I:0/11			A CIRCUIT BREAKER 120VAC
I:0/12			B CIRCUIT BREAKER 120VAC

Address/Symbol Database

Address	Symbol	Scope	Description
I:0/13			
I:0/14			WELL PUMP AUX INPUT
I:0/15			
I:0/16			
I:0/17			
I:0.4			
I:0.5			
I:0.6			
I:1.0			UNIT A FLOW 4-20=0-150GPM
I:1/13	DPS2	Global	FILT DP OPEN ON HIGH
I:1.1			UNIT B FLOW 4-20=0-150GPM
I:2.0			VESSEL A INLET PRESSURE SCALED VALUE
I:2.1			VESSEL A OUTLET PRESSURE SCALED VALUE
I:2.2			VESSEL B INLET PRESSURE SCALED VALUE
I:2.3			VESSEL B OUTLET PRESSURE SCALED VALUE
I:3.0			RECYCLE TANK LEVEL TRANSMITTER 4-20
N7:0			TOTAL SECONDS LEFT IN CYCLE (FOR PV)
N7:1			HI FLOW DELAY ENTRY FOR ALM
N7:2			LOW FLOW DELAY FOR ALARM
N7:3			DUMP VALVE DELAY TO OPEN AFTER FR ENTRY FROM PV
N7:4			FILTER BACKWASH PRE FROM OIT
N7:5			DP REGEN STATUS 0=NOT IN CYCLE 1=REGEN A 2=REGEN B
N7:6			FILTER FAST RINSE ENTRY FROM OIT
N7:7			DUMP VALVE OPEN PRESET FROM PV
N7:8			
N7:9			
N7:10			BACKWASH RECYCLE/WASTE TANK LEVEL DISPLAY
N7:11			BW RECYCLE TANK HIGH LEVEL ALM ENTRY
N7:12			DELAY IN MIN
N7:13			MINUTES LEFT IN DUMP VALVE OPEN DISPLAY
N7:14			DUMP VALVE AUTO MANUAL CONTROL SELEC 0=AUTO 1=MANUAL
N7:15			FILTER A,B DP DELAY ENTRY IN MINUTES FROM OIT
N7:16			FILTER REGENERATION SECONDS LEFT IN REGN CYCLE DISPALY ON OIT
N7:17			REGN. MINUTES - 1
N7:18			FILTER MINUTES - 1 LEFT DISPALY ON OIT
N7:19			MINUTES LEFT IN DELAY DISPLAY
N7:20			DUMP VALVE HOA 0=AUTO 1=CLOSE 2=OPEN
N7:21			014A STATUS
N7:22			FILTER A UNIT A 0=SERV 1=BW,3=FR,6=OFFLINE
N7:23			FILTER B UNIT A 0=SERV 1=BW,3=FR,6=OFFLINE
N7:24			015A STATUS
N7:25			011B STATUS
N7:26			013B STATUS
N7:27			012B STATUS
N7:28			014B STATUS
N7:29			015B STATUS
N7:30			HORN STATUS
N7:31			REGEN STATUS STATUS
N7:32			LOW LEVEL ENTRY TO ENABLE A BW START.IF LVL IS NOT LOW THEN A BW MI
N7:33			DELAY AFTER A REGN END TO ALLOW SETTLING IN TANK B4 STARTING RECLAI
N7:34			BW STORAGE/RECYCLE TANK LVL LOW TO STOP ITS PUMP
N7:35			WELL PUMP LOCKOUT STATUS 0=RUN 1=LOCKOUT
N7:36			
N7:37			CHEM PUMP2 STATUS 0=MAN 1=AUTO
N7:38	PT_100_IN_	Global	
N7:39	PT_102_IN_	Global	
N7:40			PID AUTO.MANUAL MAINTAINED BUTTON 0=AUTO 1=MANUAL
N7:41	A_REGENZ	Global	
N7:42	B_REGENZ	Global	
N7:43			LEVEL IN BW WASTE TANK TO PREVENT A REGN START- TANK IS TOO HIGH TO
N7:44			
N7:45			
N7:47			SETTLE TIME REMAINING FOR PV
N7:51			
N7:52			PULSE OUTPUT LENGTH IN MILLISECS FOR CHEMPUMP2 PRESET
N7:53			
N7:54			
N7:55			
N7:56			
N7:57			
N7:58			
N7:59			

Address/Symbol Database

Address	Symbol	Scope	Description
N7:60			
N7:61			RESET TERM ENTRY FROM PV
N7:62			RATE TERM ENTRY FROM PV
N7:63			
N7:64			
N7:65			
N17:0			FILTER B DP DELAY LEFT IN MINUTES DISPLAY ON OIT
N17:3			WEDNESDAY CHOICE FROM OIT 0=NO 1=CHOSEN
N17:7			FRIDAY CHOICE FROM OIT 0=NO 1=CHOSEN
N17:9			SATURDAY CHOICE FROM OIT 0=NO 1=CHOSEN
N17:20			DAY OF THE WEEK FOR FILTER O-6=SUN-SAT
N34:0			FILTER B MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:1			FILTER B HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:2			A+B FLOW GPM
N34:3			FILT INTERVAL IN DAYS TO START A REGN
N34:4			
N34:5			
N34:6	BGPM	Global	UNIT B FLOW RATE
N34:7			B HRS LEFT TO REGEN
N34:8			
N34:9			A+B GPM FOR MATH
N34:10			
N34:11	A_DAYS_LEFT	Global	
N34:12	B_DAYS_LEFT	Global	
N34:13			FILTER A MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:14	AGPM	Global	FILTER A FLOW RATE
N34:15			REGEN TRIGGER ON DP FOR PV
N34:16			REGEN TRIGGER ON VOL FOR PV
N34:17			REGEN TRIGGER ON TIME FOR PV
N34:18			HOUR INPUT FROM PV
N34:19			MINUTE INPUT FROM PV
N34:20			BW PUMP HOA 1=HAND 0=OFF 2=AUTO
N34:21			FILT A VALVE 11 A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:22			FILT A VALVE 14A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:23			
N34:24			FILT A VALVE 15A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:25			FILT B VALVE 14B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:26			FILTER A HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:27			ENTRY FROM OIT TO ALLOW SETTLIN OF BW TANK BEFORE URNING BW PUMP
N34:28			
N34:29			
N34:30			DISPLAY OF CURRRENT REAL TIME HOURS COPIED FROM RTC
N34:31			FILT B VALVE 11B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:32			B FILT OUT/BWSUP 12B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:33			FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:34			C VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
N34:35			
N34:36			DISPLAY OF CURRRENT REAL TIME MINUTES COPIED FROM RTC
N34:37			FILTER C HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:38			FILTER C MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:39			A HRS LEFT TO REGEN
N34:40			UNIT A HIGH FLOW RATE SETPOINT FROM OIT
N34:41			UNIT A LOW FLOW RATE SETPOINT FROM OIT
N34:42			UNIT B HIGH FLOW RATE SETPOINT FROM OIT
N34:43			UNIT B LOW FLOW ALARM SETPOINT FROM OIT
N34:44			
N34:45			FILTER B AUTO/MANUAL ADVANCE SELECT 0=AUTO 1=MANUAL
N34:46			FILTER A TIME OF DAY TO START SELECT FOR BATCH START MODE
N34:47			FILTER B TIME OF DAY TO START SELECT FOR BATCH START MODE
N34:48			A VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
N34:49			B VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
N34:50	C_MINUS_A	Global	
N34:51			C HRS LEFT TO REGEN
N34:52			FILT A INTERVAL IN DAYS TO START A REGN
N34:53			C_DAYS_LEFT
N34:54			
N34:55			
N34:56			FILTER C HOUR OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:57			FILTER C MIN OF DAY TO START SELECT EITHER DUE TO BATCH OR ELAPSED
N34:58	DP_REGEN	Global	DP_REGEN
N34:59	TIME_REGEN	Global	TIME_REGEN
N34:60	VOLUME_REGEN	Global	VOLUME_REGEN

Address/Symbol Database

Address	Symbol	Scope	Description
N34:61			DP DELAY IN SECS FROM HMI
N34:62			TANK HI SCALE
N34:63			TANK LO SCALE
N34:64			COMMON BW/FR IN VALVES AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
N34:66			P2 CL PUMP DURATION OF PULSE ENTRY IN MILLSECONDS
N34:67			P2 CL CHEMICAL FEED PUMP OFF AUTO SELECT 0=OFF 1=AUTO
N34:68			HI DP AND REGN START IS NOT DP ALM ENABLE SELECT
N34:69			COMM BW/FR IN CNTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:70			FILT A VALVE 12A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:71	A_TIME_SUM	Global	
N34:72	B_TIME_SUM	Global	
N34:73	A_MINUS_B	Global	
N34:74			
N34:75			FILT B VALVE 15B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:76			
N34:77			
N34:78			
N34:79			COMMON BACKWASH INLET AUTO/MANUAL CONTROL 0=AUTO 1=MANUAL
N34:80			BW RECYCLE PUMP 0=AUTO 1=FORCE ON 2=OFFLINE
N34:81			COMMON BW IN VLV BV115 CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:82			
N34:83			WELL PUMP CONTROL 1=HAND 0=OFF 2=AUTO
N34:84			
N34:85			
N34:86			
N34:87			
N34:88			
N34:89			
N34:90			FILT A VALVE 13A CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:91			FILT B VALVE 13B CONTROL 0=AUTO 1=FORCE CLOSED 2=FORCE ON
N34:111			
N34:112			
N34:113			
N34:114			
N35:0			
N37:7			
O:0.0			
O:0/0			
O:0/1	MV_011A_EN	Global	A RAW WATER INLET VALVE
O:0/2	MV_012A_EN	Global	A SERVICE OUTLET VALVE
O:0/3	MV_013A_EN	Global	A BACKWASH INLET VALVE
O:0/4	MV_014A_EN	Global	A BACKWASH OUTLET VALVE
O:0/5	MV_015A_EN	Global	A RINSE OUTLET VALVE
O:0/6	MV_011B_EN	Global	B RAW WATER INLET VALVE
O:0/7	MV_012B_EN	Global	B SERVICE OUTLET VALVE
O:0/8	MV_013B_EN	Global	B BACKWASH INLET VALVE
O:0/9	MV_014B_EN	Global	B BACKWASH OUTLET VALVE
O:0/10	MV_015B_EN	Global	B RINSE OUTLET VALVE
O:0/11			SPARE
O:0/12			
O:0/13			SPARE
O:0/14			SPARE
O:0/15			SPARE
O:1/0			
O:1/1			BACKWASH RECYCLE PUMP OUT
O:1/2			NaCl PUMP OUT
O:1/3			
O:1/4			
O:1/5			WELL PUMP CALL
O:1/6			ALARM HORN OUT
O:1/7			
O:1/8			
O:1/9			
O:1/10			
O:1/11			
O:1/12			
O:2.0			TOTALIZED FLOW RATE IN GPM OUTPUT TO SERVICE WATER INLET FEED THROT
O:2/0			FILTER A REGN INLET VALVE FV11A ENG TO OPEN
O:2/1			FILTER B REGN INLET VALVE FV12B
O:2/2			FILTER C REGN INLET VALVE 12C -ENG TO OPEN
O:2/3			FILTER REGN WATER INLET VALVE FV12D-ENG TO OPEN
O:3.0			CO2 PROPORTINAL VALVE MORE CO2=LESS PH

Address/Symbol Database

Address	Symbol	Scope	Description
PD9:0/AM			PID AUTO/MANUAL BIT 1=MANUAL 0=AUTO
PD9:0.SPS			CO2SETPOINT TO MAINTAIN FOR CO2 FEED
PD9:0.MAXS			CO2 PID MAX ANALOG VALUE FOR NUMERIC RANGE 0-16363
PD9:0.MINS			CO2 PID MIN ANALOG VALUE FOR NUMERIC RANGE 0-16363
PTO:0			
PTO:0/DN	PTO		
PTO:0.OF			
PTO:0/3			
PTO:1/DN			
PTO:10/DN			
Q14:0			
Q15:0			
Q3:0			
Q4:0			
Q7:0			
RTC:0.DAY			
RTC:0.DOW			
S:0			Arithmetic Flags
S:0/0			Processor Arithmetic Carry Flag
S:0/1			Processor Arithmetic Underflow/ Overflow Flag
S:0/2			Processor Arithmetic Zero Flag
S:0/3			Processor Arithmetic Sign Flag
S:1			Processor Mode Status/ Control
S:1/0			Processor Mode Bit 0
S:1/1			Processor Mode Bit 1
S:1/2			Processor Mode Bit 2
S:1/3			Processor Mode Bit 3
S:1/4			Processor Mode Bit 4
S:1/5			Forces Enabled
S:1/6			Forces Present
S:1/7			Comms Active
S:1/8			Fault Override at Powerup
S:1/9			Startup Protection Fault
S:1/10			Load Memory Module on Memory Error
S:1/11			Load Memory Module Always
S:1/12			Load Memory Module and RUN
S:1/13			Major Error Halted
S:1/14			Access Denied
S:1/15			First Pass
S:2/0			STI Pending
S:2/1			STI Enabled
S:2/2			STI Executing
S:2/3			Index Addressing File Range
S:2/4			Saved with Debug Single Step
S:2/5			DH-485 Incoming Command Pending
S:2/6			DH-485 Message Reply Pending
S:2/7			DH-485 Outgoing Message Command Pending
S:2/15			Comms Servicing Selection
S:3			Current Scan Time/ Watchdog Scan Time
S:4			Time Base
S:5/0			Overflow Trap
S:5/2			Control Register Error
S:5/3			Major Err Detected Executing UserFault Routine
S:5/4			M0-M1 Referenced on Disabled Slot
S:5/8			Memory Module Boot
S:5/9			Memory Module Password Mismatch
S:5/10			STI Overflow
S:5/11			Battery Low
S:6			Major Error Fault Code
S:7			Suspend Code
S:8			Suspend File
S:9			Active Nodes
S:10			Active Nodes
S:11			I/O Slot Enables
S:12			I/O Slot Enables
S:13			Math Register
S:14			Math Register
S:15			Node Address/ Baud Rate
S:16			Debug Single Step Rung
S:17			Debug Single Step File
S:18			Debug Single Step Breakpoint Rung
S:19			Debug Single Step Breakpoint File

Address/Symbol Database

Address	Symbol	Scope	Description
S:20			Debug Fault/ Powerdown Rung
S:21			Debug Fault/ Powerdown File
S:22			Maximum Observed Scan Time
S:23			Average Scan Time
S:24			Index Register
S:25			I/O Interrupt Pending
S:26			I/O Interrupt Pending
S:27			I/O Interrupt Enabled
S:28			I/O Interrupt Enabled
S:29			User Fault Routine File Number
S:30			STI Setpoint
S:31			STI File Number
S:32			I/O Interrupt Executing
S:33			Extended Proc Status Control Word
S:33/0			Incoming Command Pending
S:33/1			Message Reply Pending
S:33/2			Outgoing Message Command Pending
S:33/3			Selection Status User/DF1
S:33/4			Communicat Active
S:33/5			Communicat Servicing Selection
S:33/6			Message Servicing Selection Channel 0
S:33/7			Message Servicing Selection Channel 1
S:33/8			Interrupt Latency Control Flag
S:33/9			Scan Toggle Flag
S:33/10			Discrete Input Interrupt Reconfigur Flag
S:33/11			Online Edit Status
S:33/12			Online Edit Status
S:33/13			Scan Time Timebase Selection
S:33/14			DTR Control Bit
S:33/15			DTR Force Bit
S:34			Pass-thru Disabled
S:34/0			Pass-Thru Disabled Flag
S:34/1			DH+ Active Node Table Enable Flag
S:34/2			Floating Point Math Flag Disable,Fl
S:35			Last 1 ms Scan Time
S:36			Extended Minor Error Bits
S:36/8			DII Lost
S:36/9			STI Lost
S:36/10			Memory Module Data File Overwrite Protection
S:37			Clock Calendar Year
S:38			Clock Calendar Month
S:39			Clock Calendar Day
S:40			Clock Calendar Hours
S:41			Clock Calendar Minutes
S:42			Clock Calendar Seconds
S:43			STI Interrupt Time
S:44			I/O Event Interrupt Time
S:45			DII Interrupt Time
S:46			Discrete Input Interrupt- File Number
S:47			Discrete Input Interrupt- Slot Number
S:48			Discrete Input Interrupt- Bit Mask
S:49			Discrete Input Interrupt- Compare Value
S:50			Processor Catalog Number
S:51			Discrete Input Interrupt- Return Number
S:52			Discrete Input Interrupt- Accumulat
S:53			Reserved/ Clock Calendar Day of the Week
S:55			Last DII Scan Time
S:56			Maximum Observed DII Scan Time
S:57			Operating System Catalog Number
S:58			Operating System Series
S:59			Operating System FRN
S:61			Processor Series
S:62			Processor Revision
S:63			User Program Type
S:64			User Program Functional Index
S:65			User RAM Size
S:66			Flash EEPROM Size
S:67			Channel 0 Active Nodes
S:68			Channel 0 Active Nodes
S:69			Channel 0 Active Nodes
S:70			Channel 0 Active Nodes
S:71			Channel 0 Active Nodes

Address/Symbol Database

Address	Symbol	Scope	Description
S:72			Channel 0 Active Nodes
S:73			Channel 0 Active Nodes
S:74			Channel 0 Active Nodes
S:75			Channel 0 Active Nodes
S:76			Channel 0 Active Nodes
S:77			Channel 0 Active Nodes
S:78			Channel 0 Active Nodes
S:79			Channel 0 Active Nodes
S:80			Channel 0 Active Nodes
S:81			Channel 0 Active Nodes
S:82			Channel 0 Active Nodes
S:83			DH+ Active Nodes
S:84			DH+ Active Nodes
S:85			DH+ Active Nodes
S:86			DH+ Active Nodes
T4:0			FILT A FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:0/DN			
T4:1			DELAY BW FEED PUMP ON TILL THE BW INLET VALVE OPENS FIRST
T4:1.PRE			
T4:1/DN			
T4:2			FILT C FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:2.PRE			
T4:2/DN			
T4:2/TT			
T4:3			
T4:3/DN			
T4:3/TT			
T4:4			POWER UP DELAY TO ALLOW TIME FOR SCREN TO BOOTUP
T4:4/DN			
T4:5			UNIT A FLOW ALARM DELAY
T4:5/DN			
T4:6			
T4:7			
T4:8			UNIT B FLOW RATE ALARM DELAY
T4:9			
T4:9/DN			
T4:10			FILT B FILTER 60 SECOND PULSER FOR RUN INTERVAL
T4:11			RESET FLOW TOTAL DELAY FOR OP INTENT
T4:11/DN			
T4:12			INTERVESSEL DP REGEN TIMER 5 MIN
T4:12/DN			
T4:12/TT			
T4:13			DELAY FOR SETTING DP ALARM STATUS
T4:13/DN			
T4:14	A_REGEN_COUNTER	Global	
T4:14/DN			
T4:15	B_REGEN_COUNTER	Global	
T4:16	SHOW_REGENS	Global	
T4:16/DN			
T4:17			
T4:18			
T4:19			LEVEL IN BW SUPPLY TANK FALLS BELOW PERMISSIVE BW START LEVEL-
T4:19/DN			
T4:20			BW RECYCL TANK HI LEVEL DELAY FOR ALM
T4:20/DN			
T4:21			PULSE OUTPUT LENGTH FOR NAOH PUMP PRESET FROM N7:53
T4:21/DN			
T4:22			TREATED WATER/BW SUPPLY STORAGE TANK HI LVL DELAY FOR ALM
T4:22/DN			
T4:23			PULSE OUTPUT LENGTH FOR POLYMER PUMP PRESET FROM N7:54
T4:23/DN			
T4:23/TT			
T4:24			B DRIVE MULTIPORT VALVE TIMER
T4:24/TT			
T4:25			VALVE A HOMING DELAY FRO ALM
T4:25/DN			
T4:26			
T4:26/DN			
T4:27			A DELAY BETWEEN PULSES TO ALLOW VALVE TO MOVE
T4:27/DN			
T4:27/TT			
T4:28			

Address/Symbol Database

Address	Symbol	Scope	Description
T4:28/DN			
T4:28/TT			
T4:29			VALVE C DELAY IN BETWEEN PB ADVANCES TO ALLOW VALVE TO TURN
T4:30			VALVE D DELAY IN BETWEEN PB ADVANCES TO ALLOW VALVE TO TURN
T4:31			LEVEL OK TO RUN BW FEED PUMP DELAY
T4:31/DN			
T4:32			VALVE B HOMING DELAY FRO ALM
T4:32/DN			
T4:33			HI LEVEL STOP WELL PUMP DELAY
T4:33/DN			
T4:34			TREATED WATER STORAGE TANK/BW SUPPLY WATER AT WELL PUMP RE-START LE
T4:34/DN			
T4:35			VALVE C HOMING DELAY FRO ALM
T4:35/DN			
T4:36			NOT IN DP START MODE FILTER SYSTEM DP DELAY ONE MINUTE TIMER
T4:36/DN			
T4:37			ENABLE A REGN START BW RECYCLE/WASTE TANK IS AT LOW DELAY
T4:37/DN			
T4:38			DELAY AFTER A REGN TO ALLOW BW RECYCLE/STORAGE TANK TO SETTLE B4 TU
T4:38.PRE			
T4:38/DN			
T4:39			BW STORAGE/RECYCLE TANK LEVEL LOW TO DISABLE ITS PUMP
T4:39/DN			
T4:40			
T4:40/DN			
T4:41			
T4:41/DN			
T4:42			ONE SECOND PULSER FOR FLOW TOTAL ADDITION
T4:42/DN			
T4:43			FILT B DP DELAY TO AVOID SIMUTANEOUS REGN STARTS DUE TO DP IF TC DI
T4:43/DN			
T4:44			
T4:44/DN			
T4:45			
T4:46			FILTER B REGN START BY BATCH OR INTERVAL WITH TIME OF DAY DELAY
T4:47			
T4:48			
T4:49			FILT B DELAY AFTER LOCKOUT TO AVOID DP START OF BOTH FILTER AT SAME
T4:50			
T4:51			
T4:52			FILTER A REGN START BY BATCH OR INTERVAL WITH TIME OF DAY DELAY
T4:52/DN			
T4:53			
T4:53/DN			
T4:54			ONE MINUTE PULSER FOR MASTER CYCLE COUNTER
T4:54/DN			
T4:55			BW RECYCLE/WASTE TANK LEVEL OK DELAY TO ALLOW ITS PMP TO BE ON
T4:55/DN			
T4:56			DUMP VALVE DELAY IN SECONDS-ENTRY FROM PV N7:3
T4:56/DN			
T4:57			DUMP VALVE OPEN TIME PERIOD MINUTE PULSER
T4:57.PRE			
T4:57/DN			
T4:57/TT			
T4:58			
T4:58/DN			
T4:59			
T4:59/DN			
T4:60			
T4:60/DN			
T4:61			
T4:62			C5:16 DONE
T4:62/DN			
U:3			REGEN
U:4			FLOW
U:5			ALARMS
U:6			TIME INTERVAL
U:7			A VESSEL
U:8			ANALOG IN
U:9			PRESSURE
U:10			B VESSEL
U:11			DIGITAL IN

Address/Symbol Database

Address	Symbol	Scope	Description
U:12			WELL PUMP
U:13			
U:14			
U:15			
U:16			DIGITAL OUTS
U:17			REGEN A
U:18			REGEN B
U:19			RECYCLE

Instruction Comment Database

Address	Instruction	Description

Symbol Group Database

Group_Name Description